



Joint Venture Exploration Market Opportunity Business Feasibility



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Acronyms and Abbreviations

ADNOC	Abu Dhabi National Oil Company
AED	UAE Dirham
Axiom / Axiom Global / AGOT	Axiom Global Oil & Gas Trading DMCC
B24VLSFO	Biofuel Blended Very Low Sulphur Fuel Oil
BDN	Bunker Delivery Note
CAGR	Compounded Annual Growth Rate
CBM	Cubic Meters
CEO	Chief Executive Officer
CFO	Chief Finance Officer
DED	Department of Economic Development
DMCA	Dubai Maritime City Authority
DMCC	Dubai Multi Commodities Centre
DSCR	Debt Service Coverage Ratio
DWT	Dead Weight Tonnage
EBIT	Earnings Before Interest and Taxes
EBITDA	Earnings Before Interest, Taxes, Depreciation, Amortization
EU	European Union
FOIZ	Fujairah Oil Industry Zone
FOTT	Fujairah Oil Tanker Terminals
FTA	Federal Transport Authority
FUJCON	Fujairah Bunkering & Fuel Oil Forum
FZE	Free Zone Enterprise
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
GHG	Green House Gas
HSFO	High Sulphur Fuel Oil
IMEC	India-Middle East- Europe Economic Corridor
IMO	International Maritime Organization
IOC ME	IOC Middle East FZE
IOC	Indian Oil Corporation
IRR	Internal Rate of Return
KSA	Kingdom of Saudi Arabia
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
LSFO	Low Sulphur Fuel Oil
LSMGO	Low Sulphur Marine Gas Oil
M / Mt	Meters
MARPOL	International Convention for Prevention of Pollution from Ships
MASS	Maritime Autonomous Surface Ships
ME	Middle East
MENA	Middle East & North Africa
MFA	Miscellaneous Fixed Assets
MGO	Marine Gas Oil
MMTPA	Million Metric Tonnes Per Annum
MT	Metric Tonnes
MTPA	Metric Tonnes Per Annum
NPV	Net Present Value
O&G	Oil & Gas
P&M	Plant & Machinery
PAT	Profit After Tax
SWOT	Strengths, Weaknesses, Opportunities & Threats
TMT	Thousand Metric Tonnes
UAE	United Arab Emirates
UNCTAD	United Nations Conference on Trade and Development
USD Mn	One Million United States Dollar

USD	United States Dollar
VLCC	Very Large Crude Carriers
VLSFO	Very Low Sulphur Fuel Oil



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Date: 13 April 2026



Executive Summary

Demand supply scenario

#	Key factors	Description	Current impact on the JV
1	Demand scenario	Historic: Downtrend The port's bunker fuel demand has declined by 12.8% CAGR from 2013-2021 respectively	Positive - Declining historical demand puts pressure on existing and new start-ups. However, given the market has already declined and had its share of downfall citing the political and economic challenges - The forecast is now more toward the upside, presenting opportunities for new ventures. - Historically, the Qatar diplomatic crisis was the key factor behind the historical decline in the total sales at Fujairah port
		Current: Demand continues to stabilize with a reduction in the declining rate. The Market declined by 1.5% and 7.2% in 2022 and 2023 respectively	Positive - Stabilized demand not only presents opportunities but also instills confidence in new players and market entrants - Decline in demand in the year 2023 was primarily due to short-term crisis created due to geopolitical tensions in the MED region
		Forecast: Positive forecast with near about 3.7% CAGR over the next 10 years	Positive A modest increase in demand ensures a growing bunkering business momentum at the port Key factors behind the growth in demand at Fujairah port are: - Growth in maritime sea-trade - Growth of new ships/vessel categories - Rise in oil and gas exploration activities leading to the growth of tanker fleet in the MENA region - Traffic growth expected due to India-Europe Economic Corridor Given the projected demand growth, there is an opportunity for existing players to expand their barge fleets and for new players to enter the Fujairah market. To meet the rising demand, there is a scope to add 2-3 barges annually until 2027, with this number increasing to 3-4 barges per year from 2028 to 2033. These barges should have a capacity ranging between 10-15 (Thousand Metric Tonnes) TMT per month
2	Supply	Historical: Growth from 8.8 MMT in 2013 to 15.9MMT in 2021 at a CAGR of 7.67%	Positive Given the availability of bunker oil at a locally competitive rate compared to global alternatives. Nevertheless, this advantage may be offset by heightened competition resulting from an influx of new players aiming to capitalize on the abundant supply and market opportunity
		Current: Continued increase from 26MMT to 28.2MMT with growth of 8.46%	

		Future: Supplies will continue to increase at a drastically slower rate at a 2023-2033 CAGR of 0.62%	<u>Neutral to Positive</u> - The impact of increasing supply has already been accounted for, and there should not be any major impact - Future supplies are expected to increase support from UAE-based ADNOC refineries. Also, Oman, Kuwait, and Qatar are expected to be the key suppliers of LSFO and VLSFO in the near future
3	Demand supply gap	Oversupplied	<u>Positive</u> - The assurance of affordable oil availability has a positive impact on profitability. Further challenge of heightened competition is also offset by a low higher barrier to entry for competitors due to high regulation and huge investments - LSFO which contributes 70-75% of the total bunker fuel sales at Fujairah port is being sourced from Vitol Bunkers, Montfort Trading FZE, ADNOC refineries in UAE. Also, approximately ~ 150 TMT per month is being supplied from Qatar Petroleum Corporation (QPC) refinery. Oman and Kuwait refineries is also expected to supply LSFO by 2026 and 2nd half of 2024 respectively

Fuel grade mix trends

#	Key factors	Description	Impact on the JV
1	LSFO demand shifted to HSFO post-IMO 2020 regulations	Shift in fuel grade use pattern from HSFO to LSFO HSFO: LSFO demand mix is estimated change from 81%:18% in 2017 to 24%:71% in 2023 and is expected to reach a mix of 17%:78% in 2033	<u>Positive</u> Given Indian Oil's capabilities in LSFO, the shift in fuel grade represents a promising opportunity to explore in the near future
2	HSFO demand to be stable and to settle at 15-17% during the 2023-2030 period	- Larger ships will be able to achieve profitability in two years post-installation of scrubbers - Usage of scrubbers expected to increase for a few large vessel owners	<u>Positive</u> Continued HSFO demand will ensure business in this fuel category as well, driven by VLCC and Capesize vessels fitted with scrubbers

Opportunity analysis

#	Area	Opportunity	Description
1	Strategic location	Right strategic location for new bunkering business	This route sees 80% of ME oil traffic, and around 35-40% of ships transiting the Strait of Hormuz make a stop at Fujairah port. This presents an opportunity to capitalize on the addressable and untapped demand from approximately 20,000 large ships using this route

			- The bunker fuel market in the ME and Africa is expected to grow at a CAGR of more than 12% during the forecast period 2024-2029 - No other ports in the nearby areas will be able to give serious competition to Fujairah port due to issues like an unstable economy, weak infrastructure, and the production capabilities of LSFO. A few examples are below: - Oman do not have their own Bunker fuel (LSFO) production capabilities - High prices of Bunker fuel in India which is USD 60-70 costlier than Fujairah - Bankruptcy in Sri Lanka - Weak economies in Pakistan and Afghanistan - Saudi Arabia will primarily cater to the demand originating in the Red Sea region
		Rise in global maritime transport	- According to UNCTAD, global maritime trade is projected to increase at a CAGR of 2.2% from 2024 to 2028 - Maritime transport, in alignment with the expectations of the United Nations, foresees a consistent expansion in trade volumes. Projections indicate a growth of 2.4% Y-o-Y growth in 2023, with an average increase of 2.1% over the next five years (2024-2028)
		Oil & gas offshore exploration activities picking up post-COVID-19 creating sustained demand for bunker oil for tanker fleets	Estimated for 2023 is a global Y-o-Y surge of over 20% in spending on offshore oil and gas exploration, propelled by significant development initiatives in the ME, Africa and the east Mediterranean region. This surge in the tanker fleet is expected to sustain the demand for bunker fuel at Fujairah port
2	Sustained demand of bunkering oil grades	Opportunity across LSFO fuel oil due to regulatory changes and lower cost implication. Abundant supply in the ME region	- The global demand for VLSFO is forecasted to achieve a CAGR of 19% from 2022 to 2032. Despite the majority of sales being attributed to LSFO, there is a sustained demand for the product in Fujairah. This scenario presents an opportunity for new players to enter the market - Fujairah's proximity to complex Asia and ME based refineries will help it to source relatively lower cost LSFO in comparison to bunkering ports based out in the European and West region
		HSFO is here to stay with continued demand backed by scrubber-fitted ships	- The number of vessels fitted with exhaust gas cleaning systems has increased multi-folds from ~117 in 2014 to about 4,379 ships worldwide. As such, the demand for HSFO will continue and not deplete all together, providing opportunities for existing and new players for continued revenue - Overall, HSFO will still remain the fuel of choice for large vessel owners fitted with scrubbers (VLCC & Capesize vessels)

3	Favorable geopolitical scenario	Fujairah port gains traction with accepting imports of west origin crude and fuel oil products, as Europe and the US sanction ban on Russian imports	West region continues to be the top exporter of fuel oil to Fujairah port and is utilizing Fujairah storage capacity at an optimum level
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Customer Insights

The primary customer segments for bunkering fuel are Small ships, Charters, Ferries and **VLCC** with their application in container and bulk carriers, oil, chemical and gas tankers, general cargo ships and fishing vessels. Container, bulk carriers and oil tankers constitute 60-70% of the total bunker fuel consumption globally and at Fujairah port.

At Fujairah port, approximately 45-50% of demand originates from charterers with remaining share originating directly from ship owners and bunker fuel traders.

Competition

Fujairah market exhibits a semi-fragmented landscape, with large and mid-size players controlling 85-90% of the market share

The bunker fuel market in Fujairah exhibits semi-fragmentation, with top 6 players accounting for more than two-thirds of the total market share. In general, the landscape is primarily dominated by large and mid-size players, accounting for around 85-90% of the market share. The top 6 players account for ~70% of the market share. Smaller players collectively hold the remaining 10-15% of the market. Overall, there are 13 licensed and compliant bunker fuel suppliers in Fujairah. Key players include Akron Trade and Transport, Al Arabia Bunkering, Montfort Trading FZE, Oil Marketing and Trading International, Vitol Bunkers among others.

In the last 4-5 years, Fujairah's market has seen some shift in terms of new players evolving as market leaders. In 2023, players like Akron Trade and Transport, Al Arabia Bunkering, Montfort Trading FZE, Oil Marketing and Trading International, Vitol Bunkers and have gained market share and control significant share of bunkering fuel business at Fujairah.

Suppliers realized the shortage of barges at Fujairah post demand getting stabilized in 2021 and started increasing their fleet size to meet the untapped potential demand

As demand stabilized in 2021, vessel operators encountered extended waiting periods of 7-8 days for receiving bunkers at Fujairah. It was during this period that suppliers identified a shortage of barges as a key issue hindering their ability to meet the growing and untapped potential demand. With vessel operators prioritizing voyage efficiency and meeting customer deadlines, some began redirecting towards nearby ports such as Khorfakkan, Yanbu, Jeddah, Muscat, and Egyptian ports.

Since 2021, the suppliers realized the potential untapped demand, which was getting diverted towards adjacent ports due to longer waiting times and a shortage of barges with Fujairah-based suppliers. Furthermore, following the implementation of IMO 2020 regulations, Fujairah possesses a strategic advantage in offering competitively priced LSFO compared to European and Russian ports. This advantage stems from its proximity to state-of-the-art refineries in the ME and Asia region. Consequently, suppliers have begun expanding their fleets of barges, resulting in approximately 40 barges and 13 licensed and compliant bunker fuel suppliers at Fujairah in the year 2023.

All major suppliers operating in the Fujairah bunker fuel market engage in diverse activities across the value chain, which carries its strategic significance in terms of cost efficiency, customer retention, and real-time decision-making and strong customer relationships.

Why would IOC ME survive the competition

- **Capacity for incorporating additional barges:** There is a potential to integrate 2-3 barges of 10-15 TMT each annually until 2027, followed by 3-4 barges from 2028 to 2033, in anticipation of stabilizing demand and future growth.
- **Synergistic strength of IOC ME and AGOT positions the joint venture favourably to seize market opportunities:**
 - Robust financial prowess and adept management of working capital and customer credit terms
 - Streamlined costs through innovative process implementation
 - Augmented marketing, sales, and research and development capabilities
 - Improved access to innovative and digital technologies
 - Axiom's deep understanding of local markets complemented by IOC ME's extensive capabilities and network

The most common strategic focus area for the local suppliers of Fujairah and their action-impact analysis is presented below:

	Focus Area	Strategy	Level of Impact
1	Reduction of operational expense	Develop captive capabilities such as owning barges, ship-building companies, storage terminals and/or refineries	High
		Long terms relationships with customers help them to decide their supplying capacity and fixed assets	High
		Long-term contracts with bunker fuel producers in neighbouring locations help the suppliers to mitigate any potential risk due to crude oil price fluctuations	High
		Adoption of cloud-based collaboration tools for better communication between the stakeholders involved in the supply chain	Medium to High
2	Customer retention	Bring in more transparency through the adoption of innovative technologies and digital solutions (Internet of things, Blockchain, etc.), in terms of fuel quantity and quality which helps in winning customer's trust	High
		Agency services	Medium to High
		Ship repair and maintenance	High
		Marine and bunker surveys	Medium to High
		Crew training and services	Medium to High
		Sharing market and technical knowledge related to engine specifications, fuel consumption reduction techniques and environment-friendly maritime practices	Medium to High
3	Tapping potential unmet demand	Expansion of operations and existence of their barges at adjacent ports helps in optimum utilization of its supplying capabilities	High

	Adoption of innovative and digital tools for better demand estimation	High
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Market intelligence - Porter's 5 forces

Bargaining power of suppliers (Medium-to-High)

The bargaining power of bunker fuel suppliers is significant, given the challenging process of becoming a registered and profitable supplier, characterized by highly regulated and complex licensing procedures. Moreover, suppliers must make substantial investments in capital-intensive fixed assets such as barges, with lengthy payback periods, particularly considering the narrow profit margins in the bunkering sector.

Since 2021, demand has stabilized at Fujairah port, and the anticipated market expansion in Fujairah, driven by planned increases in oil and gas production in the UAE and KSA, alongside rising offshore oil and gas expenditures globally, and consistent growth in global maritime trade, will further strengthen the bargaining position of suppliers who have established themselves as mature and robust players in the Fujairah bunker fuel market.

Bargaining power of buyers (Medium)

Since 2021, vessel operators at Fujairah port have faced prolonged waiting times due to escalating oil and gas exploration activities and the expanding maritime trade in the ME region. In response to the potential surge in demand, neighbouring ports in the UAE, including Khorfakkan as well as bunkering ports in KSA, Qatar, Bahrain, and Egypt, have been enhancing their port infrastructure to facilitate efficient bunkering operations. Additionally, major ports like Singapore and Rotterdam pose formidable competition to the bunkering industry in the ME by offering competitive prices and superior services.

The bargaining power of buyers, namely vessel operators and owners, is expected to be moderate. This is because demand for bunker fuels is anticipated to remain stable, allowing buyers to consider multiple suppliers, both domestically and globally, thus exerting a reasonable degree of bargaining power.

Threat of neighboring ports (High)

The Port of Fujairah, ranked as the world's 3rd-largest bunkering hub, is encountering heightened competition in supplying marine fuels across the region. New contenders from KSA, Qatar, Bahrain, and Egypt are entering the market, intensifying the competition among domestic players and eroding Fujairah's position as a leading bunkering hub. Furthermore, the emergence of alternative refueling options at strategic locations throughout the ME region, coupled with a recent surge in bunkering activity beyond the port limits off Fujairah's coast, may exert additional downward pressure on Fujairah's market dominance.

The threat of new entrants (Low)

Due to the complex regulatory framework and licensing process and high investment in infrastructure and technology, threat of new entrants or bunker fuel suppliers is low.

Threat of new sustainable alternatives (High)

The pressing need for decarbonization within the maritime sector, driven by the carbon emission reduction targets outlined by the U.N.'s IMO, is prompting the industry to transition to alternative fuels like LNG, Biofuels, and Methanol. According to a survey conducted among bunker fuel suppliers at the FUJCON forum, Methanol and LNG are projected to dominate the market share of alternative bunker fuels by 2050. Nevertheless, it is anticipated that Methanol supplier capabilities could be introduced into the Fujairah market by 2030.

Project Location

Conclusively, the Fujairah region occupies an exceptionally strategic position as a pivotal hub for oil production and maritime activities. Fujairah port, boasting a well-developed infrastructure, ranks as the 5th busiest port in the UAE. Furthermore, it stands out as the 3rd largest bunkering port worldwide, supported by a robust bunkering ecosystem. Following a decade of declining sales, bunkering oil transactions have stabilized, indicating positive

growth prospects over the next decade. Overall, Fujairah port represents a promising opportunity for new entrants like **IOC ME** & Axiom Global JV, as detailed below.

#	Key factors	Description	Impact on the JV
1	Strategic geographic location	Situated within the Gulf of Oman, this location boasts strategic access to crucial shipping routes like the Strait of Hormuz, rendering it a pivotal transit point for crude oil and petroleum products. Its advantages encompass: • Close proximity to oil production centers • Strategic positioning as a maritime hub • Access to prominent markets across the ME, Asia, Africa, and Europe	Positive The strategic geographical placement of the port will guarantee a steady demand for bunkering services, as the number of transit ships in the area continues to rise. • Modest growth in global maritime trade • Rise in tanker fleets due to increased offshore oil and gas spending
2	Bunkering port ranking	Ranked #3 among all the ports across the globe in bunkering space, after Singapore and ARA region (Antwerp, Rotterdam & Amsterdam)	Positive A prominent global ranking in the bunkering sector reflects a well-established bunkering ecosystem, offering promising opportunities for a Joint Venture (JV)
3	Strong port infrastructure for bunkering	Well established bunkering business infrastructure ecosystem • 2 Refineries producing LSFO at Fujairah • Number of barges for LSFO/HSFO/MGO: ~40 (As of 2023) • Oil Berths: 9 Main Berths / 14 Wing Berths • Bunker Barge Berths (Low Sulphur oil tankers): 4 Berths • Capacity of Anchor positions at one time: 133 • VLCC Jetty since 2016	Positive The robust bunkering infrastructure places Fujairah port players in a more advantageous position compared to neighboring ports like Khorfakkan and others
4	Lower visible threats from other ports	No near time serious competition from neighboring ports. For example, Oman port and India ports struggle with underdeveloped infrastructure and high product cost.	Positive The minimal threats from both existing and emerging nearby ports enhance the likelihood of success for any new operations at Fujairah port
5	LNG Bunkering at Fujairah will give tough competition to Qatar's LNG bunkering capabilities	ADNOC's LNG plant expected to complete between 2026 and 2028 The facility will include two 4.8 million MMTPA trains which will raise ADNOC's LNG production capacity to 15.6 MMTPA	Positive The LNG bunkering infrastructure at Fujairah port will attract more customers which are currently being diverted to Qatar
6	The port is planning to develop infrastructure for	As per the online poll at the FUJCON forum, Methanol and LNG are expected to take the top market share of alternative bunker fuels at Fujairah by 2050	Positive IOC ME can utilize its alternative fuel capability to support the JV in adding and retaining more customers

	Methanol, Ammonia and LPG		
7	Advanced mass flow meters to be mandatory in future	Fujairah port planning to make usage of mass flow meters compulsory in coming years along with regulatory backup from the port authorities	Positive Implementation of mass flow meters will make bunkering operations more transparent and will attract more customers at the port

Regulatory and risk framework

UAE regulatory framework maintains high standards and compliance with international maritime laws and practices

The UAE's regulatory framework in the bunkering sector is geared towards maintaining high standards of operation, focusing on safety, environmental protection, and quality assurance.

The recent drive towards establishing unified bunkering standards, led by the Dubai Maritime City Authority (DMCA), underscores the commitment to enhancing the quality of bunkers and enforcing stringent regulations that minimize environmental footprints and enhance safety in the industry. The key regulations in the UAE Bunker fuel market are The International Convention for the Prevention of Pollution from Ships (MARPOL) Annex VI, IMO 2020 Sulphur Cap, IMO safety standards, Bunker delivery agreements, ISO standards, Fuel testing and inspection-related standards, regulations related to waste disposal, Maritime health, safety and security standards.

The UAE presents a favourable investment climate for the bunkering industry, characterized by political stability, economic diversity, a strong legal framework, commitment to sustainability, and a strategic geographical location.

UAE's country risk and investment climate analysis from a Bunkering perspective is presented below in the tabular format:

Country risk and investment climate analysis	
Factors	Level of risk
Political stability and governance	Medium to low
Economic factors and market dynamics	Low
Legal and regulatory framework	Low
Environmental compliance and sustainability	Low
Geopolitical risk and global economic trends	Medium to low

Potential investors and JV partners in UAE Bunkering market can leverage the country's political and economic stability, strategic location and well-developed infrastructure at Fujairah port. They should also navigate the country's stringent regulatory environment and should start aligning their businesses towards sustainable and environment-friendly practices. They should also look for international JV partners that align with their strategic objectives and complement the UAE's market dynamics. The local players should look for partnerships that can enable the implementation of innovative and digital solutions, encouraging transparent processes related to

bunker fuel quantity and quality. They should also look for international partners that can implement smart bunkering solutions and can further streamline and digitize the bunkering process, reducing their operational expenses.

Assessment of proposed JV Partner - Axiom Global

Financial performance, Assets and Liabilities

- AGOT was incorporated in April 2019, and hence gradually commenced its business in 2019. Thus, effectively, its first full year of operations was 2020.
- It is noted that AGOT achieved a top line (operating turnover) of USD 3.6 Mn in 2019 and USD 74.6 Mn in 2020, which further increased to a level of USD 393.6 Mn. This demonstrates the steep positive leaps y-o-y AGOT has been able to achieve in a short span of 4 years, which also is due to a relatively higher increase in product prices in 2022.
- Along with scaling up the business growth, AGOT has been able to improve its profit margins too in 2022, as compared to the previous years.
- Also, it is pertinent to note that the years 2 and 3 (2020 and 2021) have been the COVID-19 years, which impacted global industrial operations and logistics, and likewise has impacted the Indian region and AGOT's operations. Amid these volatile challenging years too, AGOT has been able to maintain growth through and beyond these years, which is a positive sign of having strong understanding and good insights of their core focus business areas and the geography.
- Coupled with this, 2020 also had brought in the new regime (IMO 2020 regulations) with change in Bunker fuel compliance requirements for sulphur content permissible limits being reduced. This actually proved to be a positive turn for AGOT as Indian refineries are well poised to provide the low sulphur fuels
- The relative stretched profit margins in 2021, despite the considerable growth in top line, is evident of the same; but for the year 2022, AGOT achieved gross profit margin of 4.35% and PAT margin of 3.41%.
- Majority of operating revenues are from trading (in the range of 96% to 99%), with a very marginal income from consultancy services, forming the balance share of revenues.

SWOT of Axiom Global

Strengths

- AGOT - promoted by a highly experienced professional of Oil & Gas industry,
- Supported by suitably experienced, capable team leaders, trade analysts.
- Senior management / team has good understanding of IOCL (worked with IOCL over two decades, heading various business units and operations and India and Sri Lanka)
- Has existing business relationship with IOCL through procurement and trade of bunkers in the South Asia region.
- AGOT has demonstrated its performance through its high growth in a short span, in the focus region of east coast of UAE (in the Fujairah - Khorfakkan ports)
- Good understanding across the value chain.
- Has developed a platform, with capability to grow in future, adapting to the changing bunker consumption patterns regionally / globally.

S

Weaknesses

- Due to its fast paced growth in last two years, AGOT experiences intermittent shortfalls in working capital, with limited banking limits and support. This in a way, restricts to some extent, its future plans and growth potential.
- AGOT is relatively a young firm of 4 years, and hence although with experienced professionals, would be required to pass through few more years to develop its brand and elevate to the levels of leading players in this region.

W

- The Middle East region is expected to observe very dynamic economic / business environment in O&G and other capital intensive sectors, post the recent past volatilities.
- Massive national level development programmes by few leading countries in ME - expected to lead to considerable increase in domestic and regional transportation requirements, which in turn is expected to lead to renewed increasing demand for bunkers.
- The changing domestic business scenario is expected to turn the local and regional bunker / lubricants market more attractive. Thus, shall provide more opportunities to cater to in the local / regional markets.
- In view of the JV partners' experience and knowhow in the South Asian region, this would help in tapping the growing opportunities in these regions also.

O

Opportunities

- Few other countries – Qatar, Kingdom of Saudi Arabia (KSA), Egypt & Oman are also envisaging enhancing / developing bunkering ports, although would require time to develop and catch up with Fujairah.
 - The Fujairah port presently has few well established players in the bunkering business operating on a larger scale, as compared to present AGOT's levels, and hence it has would have to compete directly with these players, if it has to grow its business.
- Mitigation measures:**
- It is essential firstly to have presence at the established bunkering hub – Fujairah, to strengthen / stabilize its position and increase its bunkering volumes, with help of additional funds. Study the regional competing bunker ports more closely.
 - Further, as the local / regional market demand is expected to grow in near future, AGOT can aim to have a share of this pie, along with partial contribution coming in as a share from existing overall supplies from Fujairah.

T

Threats

Comparative assessment of different business models

Assessment parameters	Captive development	Joint Venture	Acquisition
Present experience of the Company / Group in the proposed line of business	✓	✓✓✓	✓✓✓
Company's know-how about the focus geography, in relation to the proposed products and business activities.	✓	✓✓✓	✓✓✓
Required skill sets - status of in-house manpower availability or time and cost required to develop team of skilled manpower.	✓	✓✓	✓✓✓
Preparedness of the Company to develop these teams	✓	✓✓✓	✓✓✓
Relative scale of investments required	✓✓	✓✓✓	✓
Gestation time for implementing the plan and commencing operations	✓	✓✓✓	✓✓
Knowledge and liaison capabilities for confirming compliance to regulatory and local laws	✓	✓✓✓	✓✓✓
Level of freehand in planning, decision making during project and operations stage.	✓✓✓	✓✓	✓✓✓
Management control over the proposed business plan and realizations / returns from it	✓✓✓	✓	✓✓✓
Exit route options and ease for the same.	✓	✓✓✓	✓

It is noted that the JV route would be more preferred one, in view of the complimenting advantages, key strengths each of the JV partner brings in.

IOC ME's Rationale for this Project

Axiom Global:

In view of AGOT's past performance, its senior leaders' experience in UAE, presence in Fujairah region, in-depth understanding of local markets, its past association and existing business relationship with IOC / IOCME, the synergies with IOC, indicate it to be a preferred Partner for the proposed JV Company.

JV Route:

As explained above, looking into pros and cons of alternative business models (captive expansion, acquisition, JV), looking into the scenario of proposed JV partners - aims and goals, preparedness, complimenting expertise areas and knowhow about products and markets, time required for venturing / establishing presence in new markets, exit options, their synergies, cost of implementation and operations, sharing of authority and responsibilities, risks and rewards, it can be noted that JV option indicates to be a balanced and advantageous route -

Bunkering:

Bunkering is a core and integral part of the shipping industry / value chain and operations. Considering the growing business activities in South Asia, ME, Africa, and the industrial activity dynamics within ME countries, it is pertinent that the demand for bunkers would be increasing in future, irrespective of the short / midterm volatilities and transitions in the product formats. It also blends well as an extension to IOC's other allied products

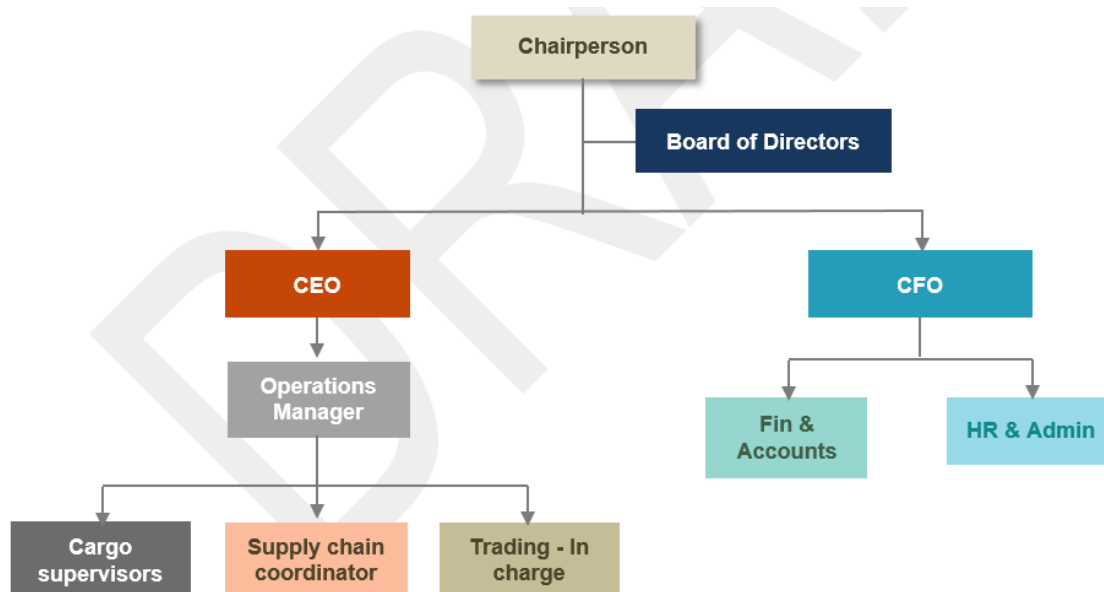
Fujairah:

On one hand it is the 3rd largest bunkering hub presently, but considering the volumes, it would be preferable / attractive location for any new entrant into bunkering, as compared to Singapore and Rotterdam (handling much higher volumes). IOCME already has presence in UAE; well-studied geography, region by AGOT with existing business. UAE, KSA growth plans indicate good local / regional opportunities. Other ports plan for bunkering, so it's a good time, to enter the leading hub in this region, as to have time to develop strong foothold in this region.

Its location is strategic, in terms of route options for vessels, as compared to other proposed bunker ports.

Company structure

The JV Company shall have the senior management positions such as Chairman, CEO, CFO, shared among the partners and these would be on rotation basis after a period of every three years.



Mode of Bunkering

Storage & logistics

For the storage and logistics in Bunkering, there are multiple options / configurations through which these can be arranged and it depends on the present position and overall goal & strategy of the project proponents. The storage & logistics infrastructure requirements can be satisfied / addressed through the below mentioned alternative routes.

Short duration contracts / on-demand services

Bunkering traders operating on a small scale and not planning for capital outlay allocations towards these facilities, owing to relatively low volumes of business. The storage and logistics facilities may be arranged on the basis of on-demand approach / model, wherein the equipment

Deployment on lease basis

For players who are relatively established, engaged in considerable higher level of volumes of transactions, and catering to multiple clients and through repeat orders, but still has no plans for huge investments towards captive in-house facilities, the storage & logistics facilities can be secured by having the optimal no. of barges and in land logistics support on mid to long term lease

Captive / in-house development of facilities

For players, who are engaged in activities across multiple nodes of value chain, engaged in refining, operating on a much larger / global scale, and certain of contributing towards a considerable share of trade / business in a given geography, they shall propose having secured storage and logistics facilities within their complete / higher control

Key guiding parameters of storage & logistics for this Project

Location	The focus area for the proposed venture is Fujairah Port
Products	VLSFO / LSFO, HSFO & MGO
Capacities	Starting with 0.72 MMT / year aiming to grow to 1.8 MMT / year in 4 years
Storage	To manage the bunker transit and supplies through barges – 3 barges each of 10,000 MT capacity

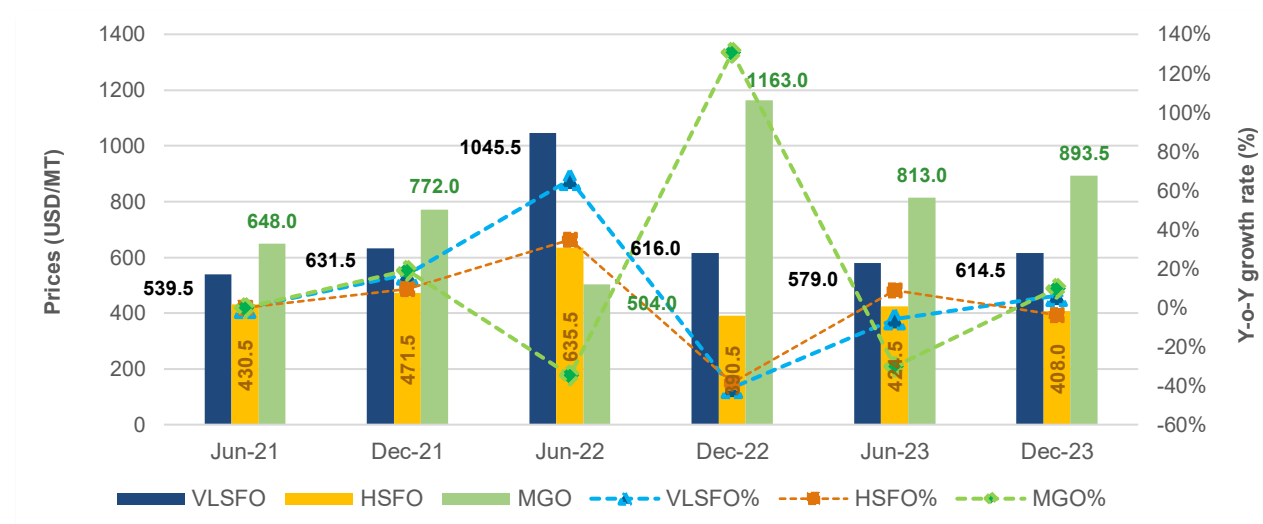
Sales Plan and Procurement of Bunkers

The JV Company shall engage largely into VLSFO, HSFO and MGO, and aims to reach of peak of 150,000 MT sales per month. While HSFO and MGO are available domestically to address the volumes envisaged by the JV, the domestic output of VLSFO is limited. Hence majority of procurement of HSFO and MGO can be supported locally, but for VLSFO almost 25-30% of procurement is envisaged to be imported.

Proposed Sales mix and sourcing Plan

Sales Mix	Sourcing Plan		
- VLSFO: 60-65% - HSFO: 30-35% - MGO: 05-10%	Supply Source	VLSFO	HSFO
	Domestic	70-75%	85-90%
	Imported	25-30%	10-15%
			MGO
			95-100%
			0-5%

Bunker recent pricing trend



Bunker fuel is a derivative of crude and hence has close correlation to **crude oil price trends, refining costs and its volatilities**. Bunker prices also depend on the availability of the product of different grades in the market.

Also, any major geopolitical upheavals / downturn events regionally or globally impact the prices of crude and bunkers, being closely linked to key macroeconomic indicators globally. In the year 2022, the crude and bunker prices were volatile due to the scenario of gradual return to normalcy post COVID-19 and the Ukraine-Russia war and remained volatile throughout 2022.

Proposed Capital Expenditure

The capital investment for the bunkering business through the proposed JV Company, is estimated to be about USD 12.5 Mn. The actual capital investment shall be quite low, about 3.7 Million, covering the company incorporation, bunkering license expenses, and few office and operations support costs for a small tenure of 2-3 months. The majority share of capital outlay about USD 8.7 million would be towards margin money of

Working Capital (WC) finance. Usually the WC is secured through a lien on the vessels to whom the bunker is supplied. Thus, the bunker supplier would have right on these vessels in event of need of any recoveries.

Projected Operating finance performance

The proposed JV shall largely deal with VLSFO, HSFO and MGO. The JV envisages targeting about 1.8 MMTPA of sales in a span of four years, commencing with volumes around 0.72 MMTPA in 1st year. Further, from 5th year onwards a modest 3% growth has been considered.

Product	Product Mix	Peak Sales Volumes (MTPA)
VLSFO	65%	1,170,000
HSFO	30%	540,000
MGO	05%	90,000
Total		1,800,000

It is further proposed that gradually the JV Company shall discontinue HSFO and increase share of VLSFO, in view of the then prevailing market trends.

For the operating expenses, it is proposed to have 3 barges of 10,000 Tonnes capacity on lease. The other expenses include captive consumption of bunker, insurance, wharfage, license fee, manpower costs, management fees and other administrative costs.

The projected P&L statement is as shown below.

(USD Mn unless specified)						
Operating costs	1	2	3	4	5	6
Revenues						
Sales Volumes (MTPA)						
VLSFO	429,600	540,000	780,000	1,170,000	1,297,800	1,336,734
HSFO	250,600	314,650	359,750	539,000	371,740	381,862
MGO	35,800	44,950	59,917	89,833	184,870	190,931
Total	716,000	899,600	1,199,667	1,798,833	1,854,410	1,909,527
Sales Value						
VLSFO	259	332	489	749	847	890
HSFO	107	137	160	245	172	180
MGO	33	42	57	87	182	192
Total	399	511	706	1,080	1,201	1,262
Expenses						
Procurement costs						
Domestic						
VLSFO	168.74	224.16	304.86	320.29	336.50	353.52
HSFO	49.80	66.16	89.98	94.53	99.31	104.34
MGO	27.83	36.25	49.30	75.42	158.48	166.50
Imports						
VLSFO	73.52	97.67	132.83	139.55	146.62	154.04
HSFO	50.39	66.94	91.04	95.64	100.48	105.57
MGO	1.55	2.02	2.74	4.20	8.82	9.26
Total procurement cost	371.83	484.28	669.02	1,023.61	1,135.03	1,192.46
Barge Hire charges	10.95	11.17	11.39	11.62	11.85	12.09
Bunker for barges	1.10	1.12	1.14	1.16	1.19	1.21
Marine Insurance	0.04	0.04	0.04	0.04	0.04	0.04
Wharfage costs	0.27	0.28	0.28	0.29	0.29	0.30
agency charges	0.04	0.04	0.04	0.04	0.04	0.04
testing costs	0.12	0.12	0.12	0.13	0.13	0.13
license fee	0.50	0.51	0.52	0.53	0.54	0.55
Office & Admin	0.12	0.12	0.12	0.13	0.13	0.13

BD & Marketing manpower expenses	0.12	0.12	0.12	0.13	0.13	0.13
Mgmt. Expenses	0.84	0.88	0.92	0.97	1.02	1.07
	0.30	0.31	0.31	0.32	0.32	0.33
Total Other Expenses	14.39	14.70	15.02	15.35	15.69	16.03
Total Operating Costs	386.2	499.0	684.0	1,039.0	1,150.7	1,208.5
Operating Profit	12.5	12.1	22.0	41.0	50.5	53.5
Margin (%)	3.1%	2.4%	3.1%	3.8%	4.2%	4.2%
Depreciation	0.0	0.1	0.1	0.1	0.1	0.1
Interest - LT Loan	0.0	0.0	0.0	0.0	0.0	0.0
Interest – WC	2.1	2.7	3.7	5.7	6.3	6.6
Sub Total	2.1	2.8	3.8	5.8	6.4	6.7
Total Expenses	388.3	501.8	687.9	1044.7	1157.1	1215.2
Profit Before Tax	10.4	9.3	18.2	35.2	44.1	46.8
Tax	0.9	0.8	1.6	3.2	4.0	4.2
Profit After Tax (PAT)	9.5	8.5	16.6	32.1	40.1	42.6
PAT Margin (%)	2%	2%	2%	3%	3%	3%
Dividend	1.4	1.3	2.5	4.8	6.0	6.4
Retained Earnings	8.1	7.2	14.1	27.3	34.1	36.2
Margin (%)	2%	1%	2%	3%	3%	3%

Based on these estimates of projected performance, the key performance indicators of this project are as below:

- Average EBITDA Margin (%) : 4.43%
- Average PAT Margin (%) : 3.55%
- Net Present Value (NPV) : USD 238.3 Mn.
- Internal Rate of Return (IRR) : 372.9 %
- Debt Service Coverage Ratio (DSCR) : 7.0

Key Ratios

Parameters	1	2	3	4	5	6	7
Profitability Ratio							
Gross Profit Margin	7%	5%	5%	5%	6%	6%	6%
Operating Profit Margin	3%	2%	3%	4%	4%	4%	5%
EBIT Margin	3%	2%	3%	4%	4%	4%	5%
Net Profit Margin	2%	2%	2%	3%	3%	3%	4%
Y-o-Y growth							
Growth in revenues		28%	38%	53%	11%	5%	17%
Growth in Net profits		-11%	96%	93%	25%	6%	34%
Liquidity ratios							
Current Ratio	1.3	1.5	1.7	1.9	2.2	2.6	2.9
Quick Ratio	1.2	1.4	1.6	1.8	2.1	2.5	2.8
Other key Ratios							
Inventory Turnover Ratio	12.2	12.2	12.2	12.2	12.2	12.2	12.2
Asset Turnover Ratio	11.6	10.1	8.7	8.0	6.7	5.7	5.1
Return on Assets	24%	12%	16%	19%	19%	16%	17%
Return on Net Worth (RONW)	39.3%	55.0%	70.2%	81.9%	87.9%	91.0%	93.4%
Return on equity (ROE)	39.3%	25.9%	33.7%	39.5%	33.0%	26.0%	25.9%
Interest Coverage Ratio	6.0	4.5	5.9	7.2	8.0	8.1	9.2
Debt Service Coverage Ratio	5.5	4.2	5.5	6.7	7.4	7.4	8.4

The reduction in the key ratios from year 1 to year 2 are due to the cost increment considered. From year 2 onwards the ratios indicate a gradual improvement over the tenure of project.

Sensitivity Analysis

Year		Base Case	Varying Revenue	Varying Pricing	Varying Margin
			Decline by 5%	Decrease by 5%	Decrease by 3%
Income					
EBITDA Margin	%	4.43%	4.36%	4.36%	1.45%
PAT Margin	%	3.55%	3.48%	3.48%	0.84%
NPV	USD Mn	238.3	221.77	222.05	48.67
IRR	%	373%	288.83%	288.83%	34.50%
WC	USD Mn	94.46	89.74	89.74	94.63
CR	ratio	3.71	3.69	3.7	1.64
Inventory ratio	ratio	12.17	12.17	12.2	12.2
Return on Assets (RoA)	%	14%	13%	14%	5%

It is noted that the decrease in revenues or pricing has marginal impact on the project's overall performance, as the major operating cost is that of product procurement, which is defined by a spread margin.

A decrease in spread margin has significant impact as an inherent attribute of any commodity trading sector. This would be a high-volume low margin trading business activity and the Company can sustain a decrease of up to 3% in spread margin, beyond which such variations and impacts have to be passed on to the customers.

Recommendations and Conclusions

Considering the key business parameters – markets, products, geographies, competition, opportunities, relevance and synergies to the Client's existing businesses, below are the key findings and conclusions.

The continued relevance of bunkering in the evolving shipping industry: Bunkering is an integral part and one of the core operating components of the shipping industry. This scenario and fact would remain relevant in future too, although the product forms and dynamics may undergo change.

Expansion of IOC ME into the bunkering business - A seamless value chain extension: IOC ME, along with its parent company IOC is already engaged in production, marketing and sales of Oil & Gas product with a wide range of products and also has a physical presence in select global markets, selling other O&G products and lubricants. Hence it can be seen as a seamless and suitable extension of the value chain, for IOC ME to focus on the Bunkering business.

Unlocking opportunities in bunkering - Leveraging presence and group strengths for long-term growth: Bunkering is a high value and volumes business as compared to lubricants and hence although competitive, indicates opportunities to the Client, to leverage its presence, group strengths to increase its top line and overall returns to the Company over the long term.

Strategic entry into bunkering - Capitalizing on Fujairah's potential as a balanced and optimal hub:

IOC ME has a presence in UAE, for quite a long duration now, and is the same country where the 3rd largest bunkering hub is situated- at Fujairah. Hence, from the perspective of a new entrant, Fujairah appears to be an optimized and balanced option to enter into the bunkering business; as it would have relatively less competition as compared to the top two largest bunkering hubs, but still holds the 3rd position globally and thus shall have an upside to offer in the medium to long term. Also, this geography being more familiar to the Client.

Partnership with Axiom Global - Accelerating entry into bunkering with established expertise: Axiom Global has suitable experience in the UAE bunkering business having presence in the Fujairah region, with its presence in Khorfakkan, and thus can be a suitable JV Partner with IOC ME, for the proposed venture. The JV route shall help IOC ME to have an entry and presence into the bunkering segment within a very short span and shall have a live operating platform to kick-start its operations as to ramp up much faster.

Synergistic partnership potential - Leveraging past collaborations and shared expertise between Axiom Global and IOC ME: Axiom Global has past and present business relationship with IOC, through purchasing of bunker products for their South Asia markets and also trading IOC bunker products in this region. The Senior Management of AXIOM Global have past career experience spanning over two decades with IOC, and this indicates good synergies in terms of overall business strategies, management and operational philosophies, corporate cultural and ethos.

Regional growth prospects - Capitalizing on economic and industrial development in the Middle East and Africa for bunkering business opportunities: The Middle East and African markets indicate quite a string of economic, infrastructure and industrial development programmes, which would fuel the overall business and transportation activity in this geography. Thus, this geography indicates an uptrend in local and regional demand for marine transportation, indicating that the bunkering business may see more traction from local / regional demand in future, apart from global demand. Therefore, there are possibilities of the demand dynamics being more regional / local centric as compared to the past trends, which may offer better opportunities to the Client to leverage its presence.

Strategic approach - Entry into bunkering through Fujairah JV with Axiom Global, embracing an asset-light model for optimized exploration and growth

Thus, it would be suitable for IOC ME to look towards the bunkering business – with an entry through the Fujairah region as a JV with Axiom Global, with a balanced, asset light business model. This model would allow IOC ME to have a presence and explore this industry and the geography in more depth, with a more optimized / lower risk economically and operationally. This would help IOC ME to devise future strategies in terms of time to address the product assortment and gaps if any at this location, other multiple locations options it has to tap to increase its presence, if required, and to allocate future capital outlay allocation to start developing a stronger foothold.

1 Introduction

1.1 Study background

IOC Middle East FZE ('IOC ME' or 'Client' or 'Company'), is a private company setup as a Free Zone Establishment (FZE), incorporated on 19th April 2006, at Jebel Ali Free Zone, as per the UAE Law No. 2 of 1986 and Jebel Ali Companies Implementing Regulations 2016. It is being promoted by Indian Oil Corporation Limited (IOC), based out of India, with a 100% shareholding.

IOC ME with its primary interests in the Oil & Gas (O&G) sector and allied products / industries, is exploring an opportunity of entering into the bunkering business in the Middle East region, and thus had proposed to undertake a study to assess the potential for the same. Also, simultaneously, it had identified another entity - Axiom Global Oil & Gas Trading DMCC (AGOT), which is present in bunkering business and thus are exploring suitable options of collaborative opportunities with AGOT.

Thus, with Fujairah as the focus location, IOC ME had envisaged a study to assess the local bunkering market, including competitive landscape, regulatory framework and other local issues; an overview assessment of AGOT, as a prospective partner, to assess the options for entering this industry through tie-up, and also assess the feasibility of this proposed business.

Based on this aim, IOC ME proposed the study to be taken up in two phases, covering broadly the following aspects:

Phase I: Focusing on the UAE market assessment, industry value chain, dynamics, competitive landscape, for the bunker products and regulatory environment, focus location review, procurement plan, assessment of proposed partner, for bunkering business.

Phase II: Assess the overall JV trends in this industry / region, advantages and disadvantages of JV, corporate structure for the proposed JV, a feasibility analysis based on financial projections analysis.

In view of this, IOC ME appointed CRIF Gulf DWC LLC, UAE (CRIF), to carry out the proposed study and present the findings. This is the Draft Report being submitted to Client.

1.2 Client Profile

IOC ME was setup in UAE as an FZE, in Jebel Ali, in year 2006, with an aim to focus on the Middle East and North Africa regions, and explore opportunities to grow their business and diversify into new product segments and sub-segments.

1.2.1 Business Objective

IOC ME's defined business objectives are:

- Manufacturing and marketing of lubricants
- Trading of Base Oil from and to IOC based on availability & requirement
- Other business opportunities in downstream oil sector
- Investment in any company in any overseas Country for development of new business activities required for expansion of IOC ME

1.2.2 Business activities

Trade Licensees

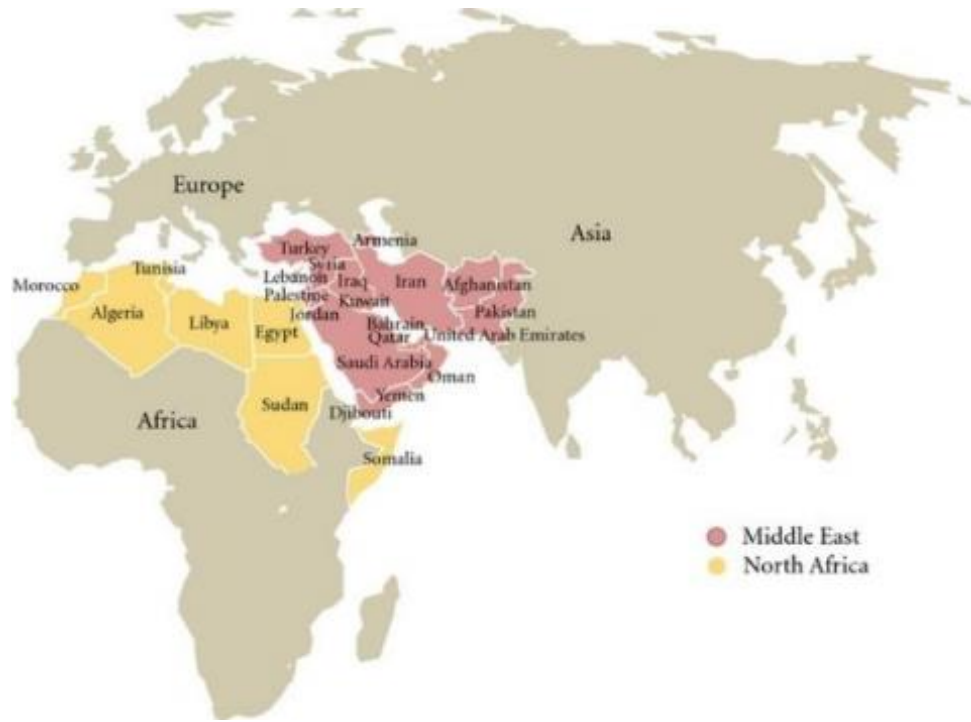
IOC ME's presently has secured trade license for the following products:

- Lubricants and Grease trading
- Trading of Crude Oil abroad
- Trading of Refined Oil products abroad
- Tar and Asphalt trading

- Petrochemicals trading

Geography

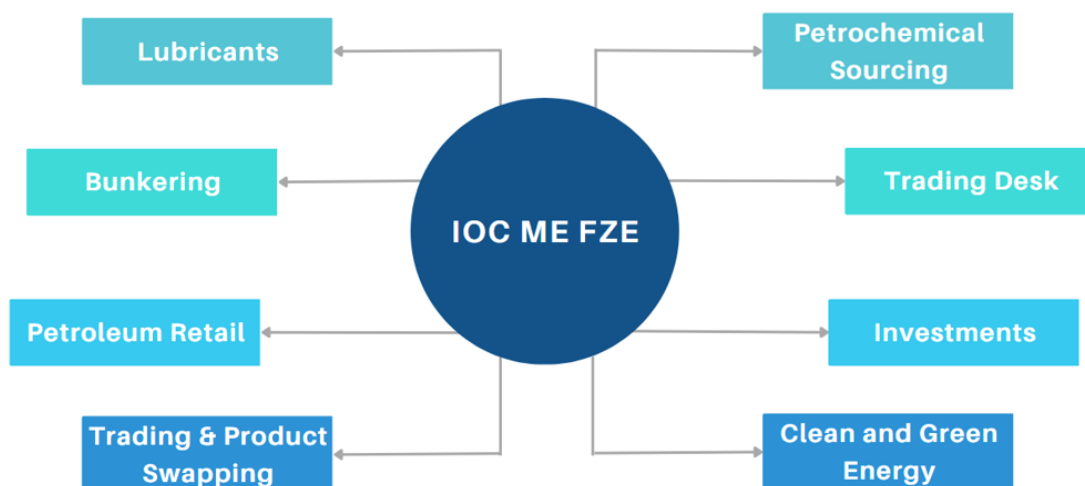
Figure 1.1: IOC ME's present focus geographies



Future Plans

In the long term, IOC ME envisages to have structured and active verticals, with aimed dedicated business focus and penetration in each of the following areas.

Figure 1.2: IOC ME's present and future proposed business portfolio



2 Industry overview and Market assessment

2.1 Overview of focus location

For the proposed bunkering business, **IOC ME** has identified Fujairah port as the focus area, to venture into this segment. Overview of Fujairah Port is as below.

Fujairah Port

Deep water port with large cargo handling facilities: Situated in the UAE and serving as the sole port in the country along the Indian Ocean, the Port of Fujairah ranks as the 5th busiest port in the UAE. Boasting a deep-water harbor, the port is capable of accommodating large vessels, and it features a cargo handling facility. With a capacity to manage 100+ million tons of cargo annually, the port handles diverse shipments, including containers; break bulk, general cargo, and vehicles. As a consequence, Fujairah gained prominence as a strategic location for bunkering activities.¹ [Link](#)

Significantly contributes to the UAE economy accounting for a-tenth of country's GDP: The Port of Fujairah plays a crucial role in driving the UAE's economic growth and serves as a vital contributor to the country's overall economic prosperity. The port significantly contributes to the UAE's economy, accounting for 10% of the country's GDP and employing over 15,000 people.

It represents a substantial source of revenue for the UAE government and is a major employer, significantly impacting the nation's workforce. Beyond its direct economic contributions, the port is integral to supporting the country's strategic objectives.

2.1.1 Strategic location

- Trade corridor between east and the west economies and trade hub of GCC with Indian sub-continent, Red sea and Eastern Africa
- Significant oil route with entry point of Strait of Hormuz as 80% of ME oil passes through this location
- Strategic position outside the Strait of Hormuz, a chokepoint where daily oil flow accounts for 21% of global petroleum liquids consumption
- Transit point for crude oil and refined products
- Only UAE port located on the Indian Ocean, making it ideally placed to serve the growing markets of Asia
- Only multipurpose port located on the eastern seaboard of the UAE, about 70 nautical miles from the Strait of Hormuz

Figure 2.1: Overview of Oil & Gas network in UAE

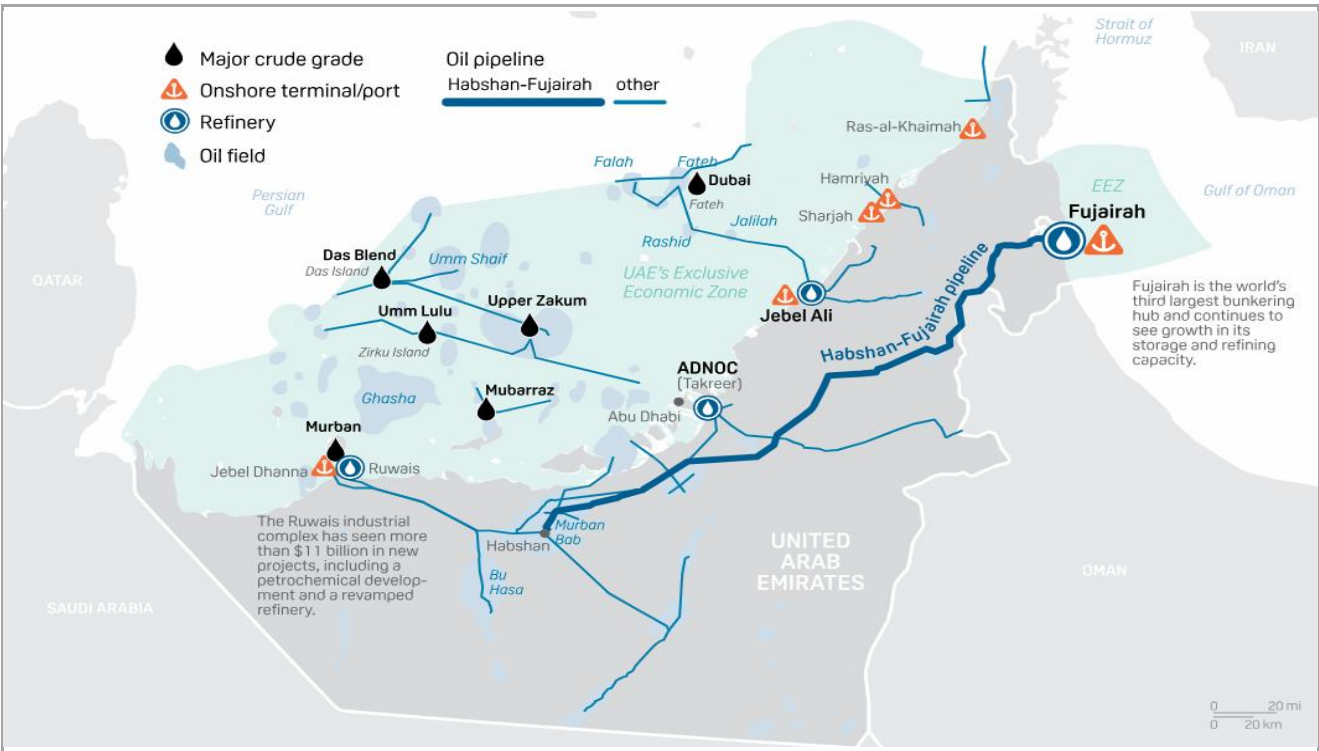
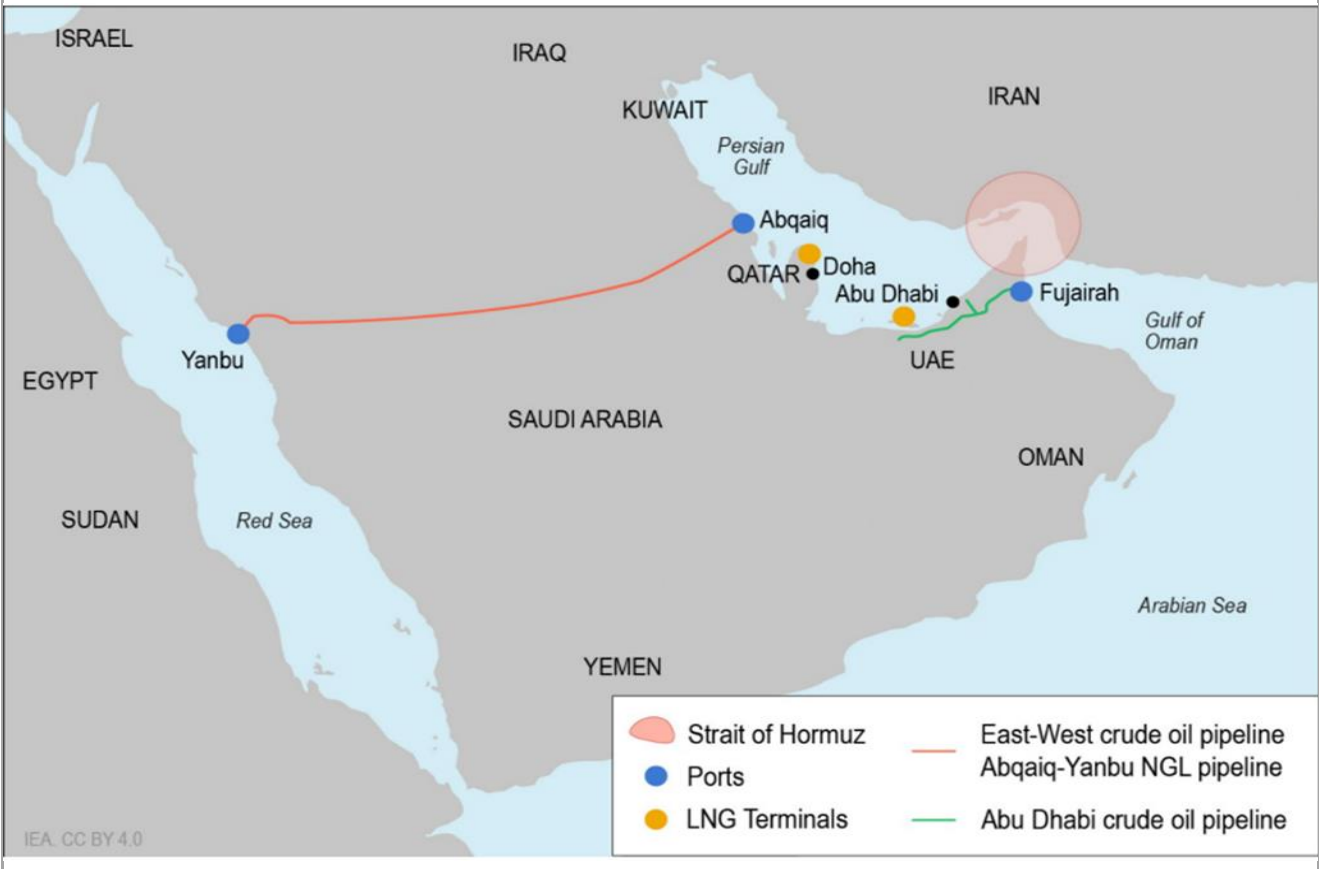


Figure 2.2: Fujairah- Strategic Location



The Fujairah anchorage has the capacity to host over 100 vessels simultaneously, and its role as a bunkering hub is significantly bolstered by its efficient provision of comprehensive maritime services. These services encompass ship supplies, ship repair, spare parts, inspection services, and serve as a convenient hub for crew changeovers.

2.1.2 Global Significance – Among the World’s Top 5 Bunkering Ports

Total global bunker output as of 2022 for the top 17 supply countries and regions were approximately 141 MMT. Top 17 bunkering ports globally accounted for 61% of the global market of 217 MMT in the year 2022. The global bunker fuel market declined by ~ 1.7% in 2023, with approximately 213 MMT sales. Fujairah port is currently at 3rd spot in terms of bunker fuel with ~12,500 vessels calling at the port’s offshore anchorage in 2022. Fujairah has an advantage in terms of demand coming from tanker fleet as 80% of ME oil passes through this route.

Figure 2.3: Leading global bunkering hubs

Bunkering Port (Country)		Bunkering Fuel (Volume of Sales (in MMT) - 2023)	Y-o-Y growth % (2022-2023)
Global		213.0	(1.7%)
Singapore		51.8	8.1%
Rotterdam (Netherlands)		7.2	(26.8%)
Fujairah (UAE)		7.1	(7.8%)
Zhoushan (China)		7.0	16.3%
Antwerp (Belgium)		~4.5	18.4%

Source: Ship & Bunker, Maritime & Port Authority of Singapore, Port of Fujairah, Port of Rotterdam, Port of Antwerp, Ship& Bunker

Other key bunkering ports which fall under the top 10 list are Hong Kong Port, Panama Port, Gibraltar port and Los Angeles – Long Beach port

Figure 2.4: Middle East bunker demand -2023 [Link](#)

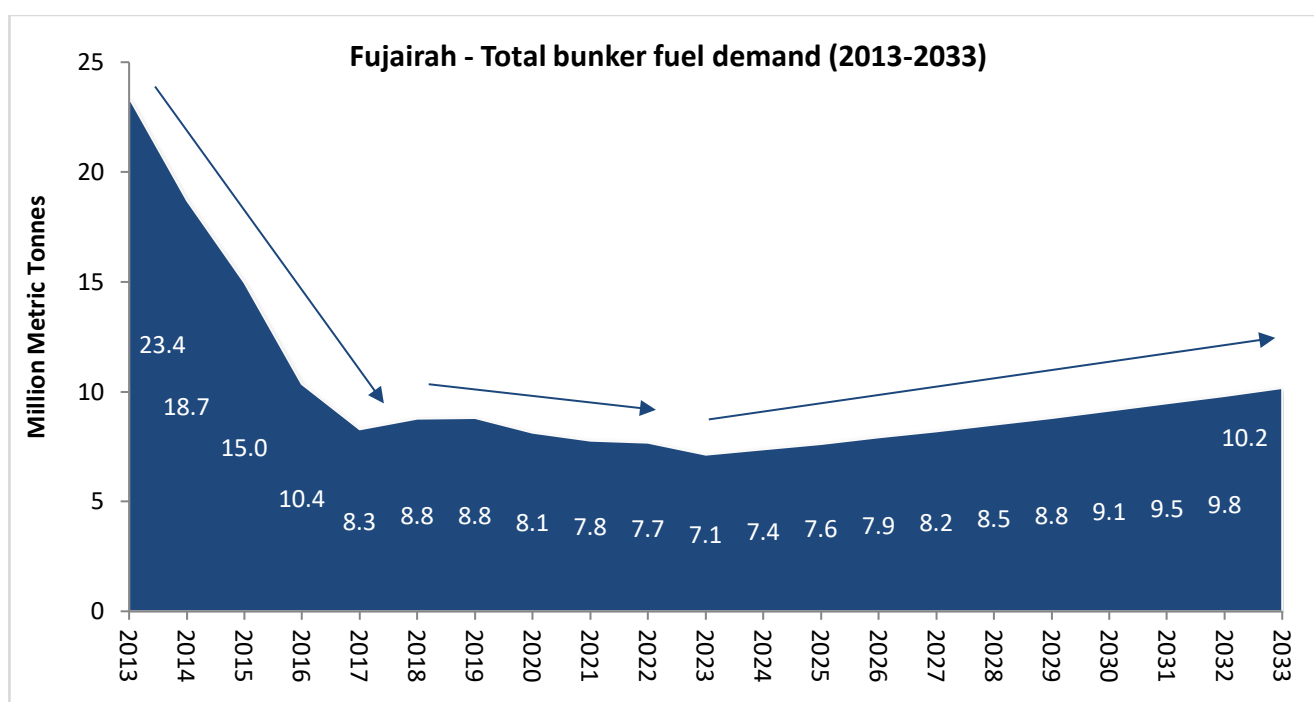
ME Bunkering locations	Volume of Sales (in MMT) - 2023
UAE (including Fujairah & Khorfakkan)	8.5-9.0
Saudi Arabia (East coast)	2.0-2.5
Saudi Arabia (West coast)	1.1-1.2
Qatar	0.8-0.9
Iran	0.6-0.7
Iraq	0.4-0.5
Egypt	0.5-0.6

ME Bunkering locations	Volume of Sales (in MMT) - 2023
Oman	0.4-0.5
Djibouti	0.3-0.4
Middle East (Total)	17-17.5

Source: Bunkerspot.com

2.2 Demand Analysis

Figure 2.5: Fujairah bunker demand



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

2.2.1 Historical scenario

Continued depletion in demand owing to Qatar diplomatic crisis^{2,3} [Link](#) [Link](#)

In the past, Fujairah recorded annual bunker sales of 23 MMT in 2013, establishing itself as the world's second-largest bunkering port following Singapore. Nevertheless, the port has experienced a continuous Y-o-Y decrease of 20-25% in bunker fuel sales since 2013. This decline is attributed to the Qatar diplomatic crisis that commenced in early 2017, which led to imposition of sanctions on Qatar by KSA, the UAE, Bahrain, and Egypt. This adversely affected the bunker fuel market at key UAE bunkering ports like Fujairah and Khor Fakkan.³ [Link](#) The coalition led by KSA accused Qatar of supporting militant groups throughout the ME region and forming alliances with its regional rival, Iran. Qatar denied these allegations, leading to Arab nations imposing restrictions on Qatar, which encompassed diplomatic ties, travel, and trade. These restrictions also extended to bunkering activities at Fujairah.

Impact of Qatar diplomatic crisis

Drop in bunker sales by 170,000 tonnes per month

Perpetually, few of the major bunkering suppliers at Fujairah such as Chemoil Middle East DMCC ceased physical supply in the UAE's Fujairah's port in early-2017. Several suppliers also reduced the number of their barges at the port.

Conversely, Qatar embarked on establishing its bunkering capacity with the goal of achieving self-sufficiency in the bunkering industry. For instance, the Qatar Fuel Company, or Woqod, planned to invest approximately USD 174 million in expansion projects within Qatar to meet domestic demands for various products, including bunker fuel. ⁴ [Link](#)

Modest increase in demand witnessed in 2018

Despite the ongoing impact of the Qatar situation on sales, traders observed that demand was influenced by higher bunker fuel prices at the port in comparison to those in Singapore. The majority of the demand originates from tankers, and their choice of bunkering location is often determined by price differentials and vessel routes. In 2018, total sales experienced a modest increase compared to 2017, likely attributed to competitive prices relative to Singapore. According to a bunkering supplier in Fujairah, bunker fuel prices at the port were more favorable, especially in the latter half of 2018, in contrast to other ports like Singapore. This led some customers to shift their marine fuel procurement to Fujairah. Bunker fuel sales at the Fujairah port were estimated to have reached at least 9 MMT in 2018, marking a recovery from the 20% decline in sales observed as of 2017, when sales dropped to around 8.5 MMT, as reported by S&P Platts. ⁵ [Link](#) [Link](#)

Fujairah stands out as a unique port when compared to Singapore and Rotterdam in terms of the types of vessels that frequent the port. While container and bulk traffic dominate demand in Singapore and Rotterdam, Fujairah has consistently relied more heavily on its tanker fleet.

In 2020, marine fuel sales at Fujairah faced challenges due to Covid-19, primarily influenced by the decrease in demand by oil tankers visiting the UAE's east coast port and production cuts by OPEC. In contrast, both Singapore and Rotterdam reported increased volumes for the same period. ⁶ [Link](#)

Current scenario

Decline remerged in 2022 and 2023 on back of global slump in the industry, followed by subdued macroeconomic sentiments

Marine fuel sales experienced a 1.5% Y-o-Y decrease to approximately 7.7 MMT in 2022, aligning with the overall decline in bunkering volumes across global ship refueling ports, influenced by a subdued macroeconomic sentiment. However, in 2023, the sales decline intensified to 7.2%, driven by heightened geopolitical tensions in the ME region. The port faced challenges from rising war insurance premiums, contributing to the downturn. Additionally, some of the demand was redirected to the neighboring port, KhorFakkan. ⁷ [Link](#)

2.2.2 Future Outlook

(Forward 10 years): Bunker oil demand at Fujairah port is poised for ~3.7% growth CAGR

Fujairah's demand is expected to grow at a CAGR of 3.7% during the period of 2023-2033 and it is anticipated to reach 10-11 MMT by 2033.

The accelerated exploration activities in deep offshore locations, the expansion of sea-borne trade, growth in number of ships/tankers and the overall global GDP growth rate are expected to propel Fujairah's demand growth. ⁸ [Link](#)

According to UNCTAD, global maritime trade is projected to increase at a CAGR of 2.2% from 2024 to 2028. ⁹ [Link](#)

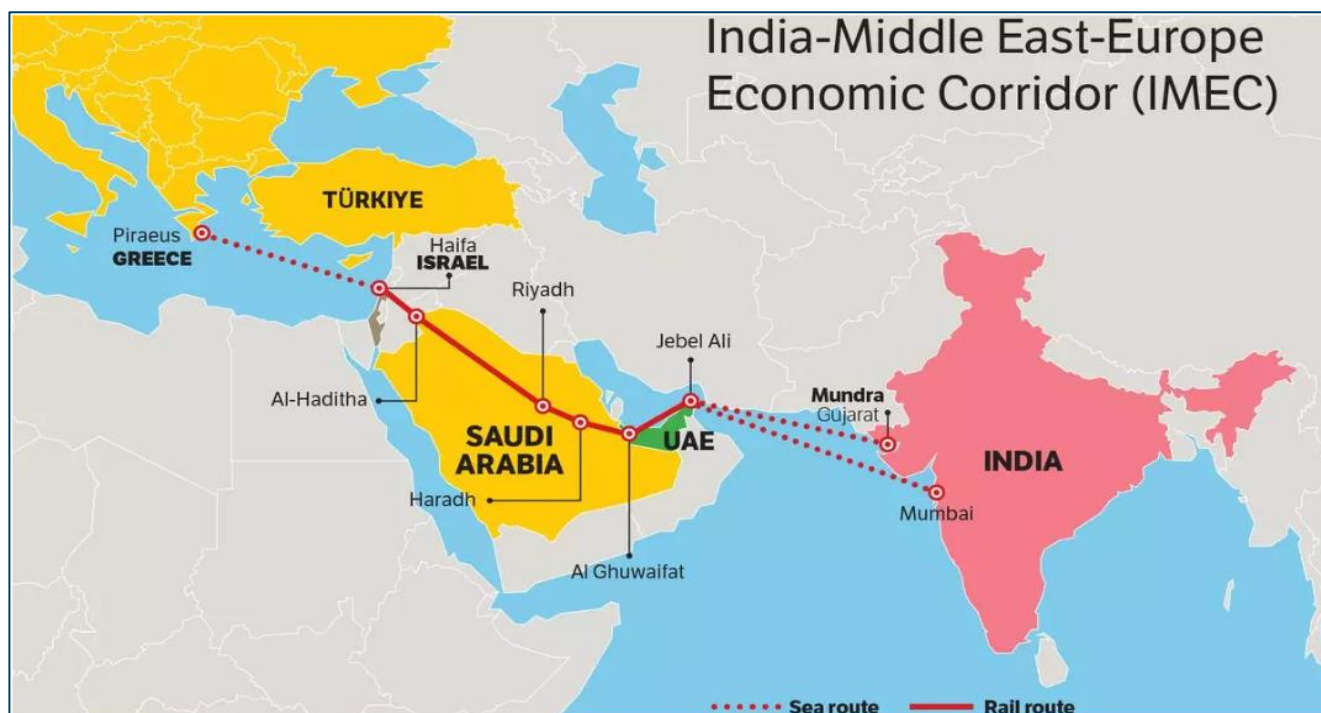
The increase in number of vessels/ships owing to the growing sea borne trade, rapid industrialization and commercialization in developing countries such as China, India, Vietnam and Philippines, exports as well as imports in these countries is rising significantly. The rising number of ship vessels, in turn, is resulting in the increased consumption of fuels used in ship bunkers. ¹⁰ [Link](#)

Table 2.1: Projected growth rates for different types of shipping vessels

Shipping / vessels global future growth rate (%)		
Ship Categories	CAGR	Forecast Period
Container	4.1%	2021-27
Bulk Carrier	4.1%	2024-29
Oil Tanker	2.5%	2020-29
General Cargo	7.0%	2022-31
Chemical Tanker	4.6%	2023-30

Also, the development of India- Middle East- Europe economic corridor (IMEC) is expected to increase vessel traffic passing Strait of Hormuz which will further support the bunker fuel demand growth at Fujairah port. IMEC will enable goods and services to transit to from, and between India, the UAE, Saudi Arabia, Jordan, Israel and Europe ¹¹ [Link](#)

Figure 2.6: India- Middle East-Europe Economic Corridor route



Source: Economic times

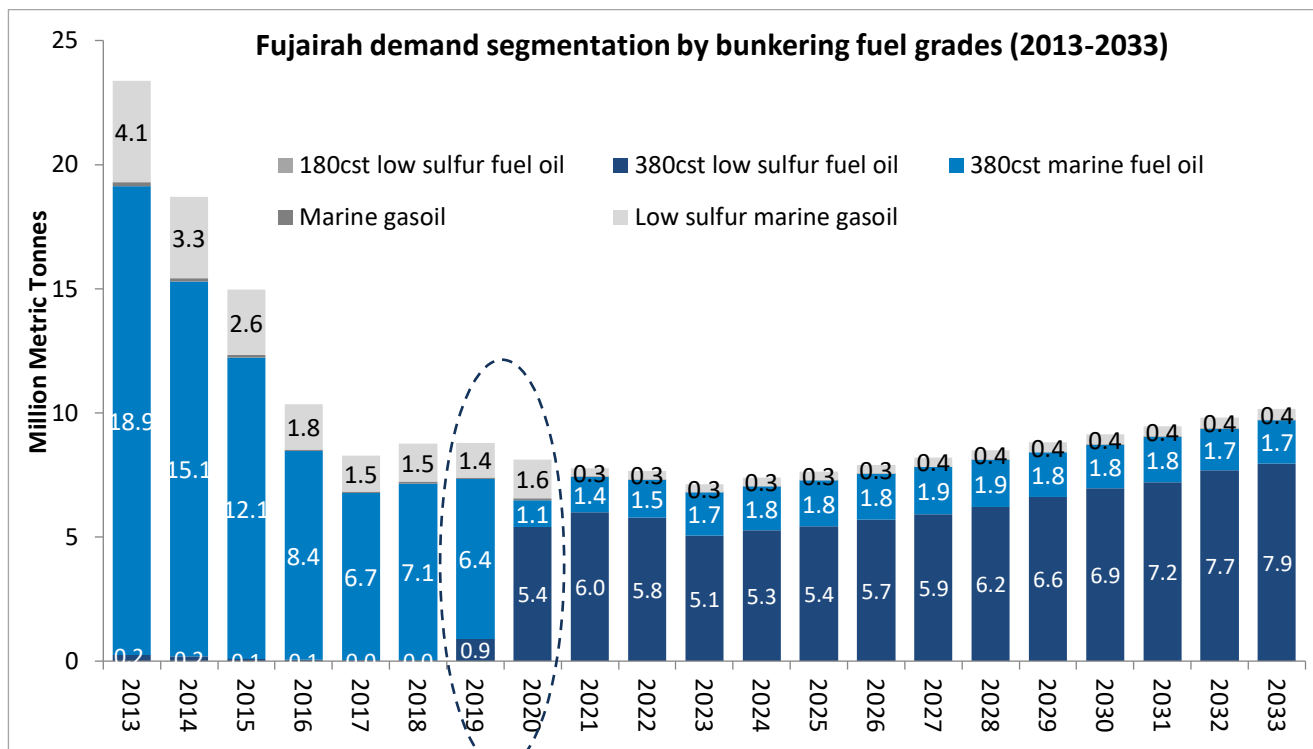
LNG Bunkering

ADNOC intends to establish LNG bunkering infrastructure at Fujairah port by 2026-2028. Currently in the design phase, the project aims to incorporate 2*4.8 (9.6) MMT /year trains. This initiative is a JV involving ADNOC, Mitsui & Co., BP, and Total as key stakeholders. Additionally, plans are underway to develop necessary storage and bunkering facilities for alternative fuels like Ammonia, Methanol, and LPG. According to an online poll conducted at the FUJCON forum, Methanol and LNG are anticipated to dominate the alternative fuels market at Fujairah by 2050, followed by Biofuels and Ammonia. Future plans also include the implementation of advanced mass flow meters, which will become mandatory during bunkering operations. These strategic measures collectively aim to enhance the port's appeal to customers and drive-up bunkering demand at Fujairah. ^{12 13 14} [Link](#) [Link](#) [Link](#)

Other factors driving the market are the rapid increase in the tanker fleet due to increased hydrocarbon resource development activities in offshore areas.

The expansion of exploration activities will be fueled by significant development projects in ME region, along with capacity expansions in Africa. Additionally, there is a resurgence of exploration and appraisal activities in emerging offshore regions such as India, and the East Mediterranean. ¹⁵ [Link](#)

Figure 2.7: Fujairah bunker demand - product and grade wise

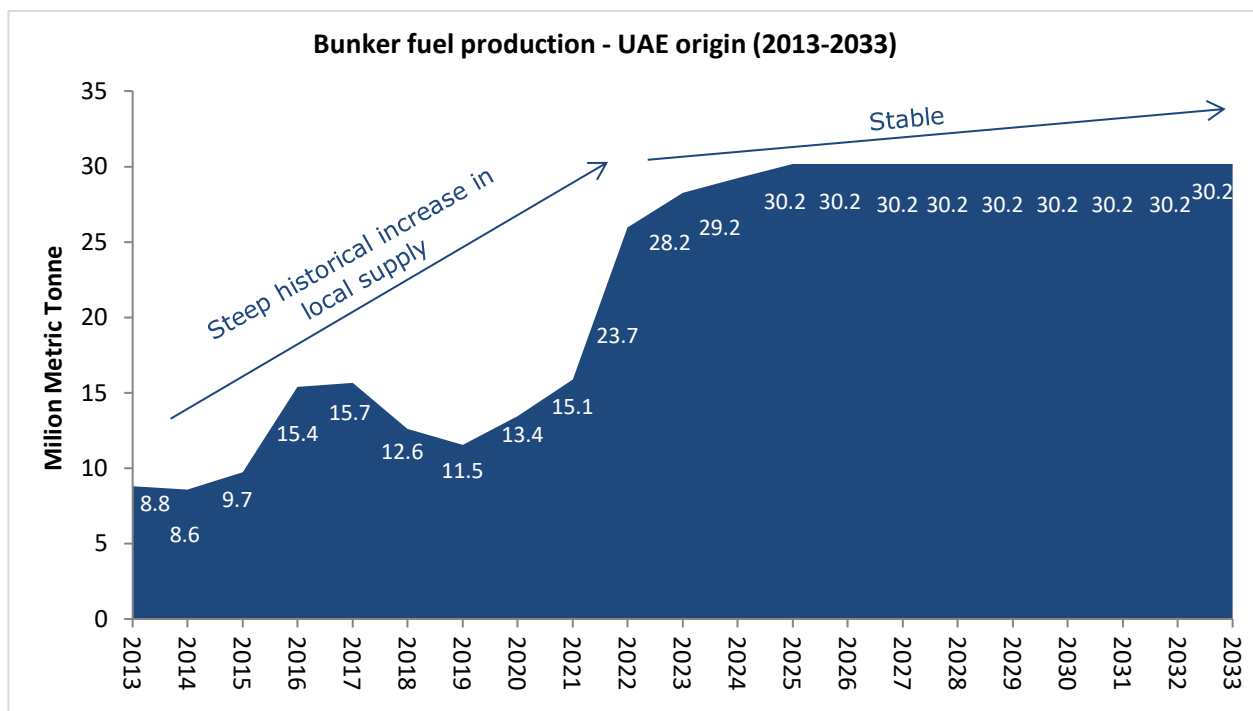


Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

HSFO sales share was ~70-80% during 2013-19 period, however post the implementation of IMO 2020, LSFO captured the HSFO market contributing ~65-75% share of total bunker fuel sales at Fujairah port during 2020-2023 period. It is anticipated that HSFO will continue to hold 17-24% share of total bunker fuel market during 2024-2033 period while LSFO sales will contribute approximately 70-80% of total bunker fuel sales. Marine gas oil (MGO) will continue contributing 4-5% of total bunker fuel sales.

2.3 Supply Analysis (Local Production- UAE Origin)

Figure 2.8: UAE total bunker production



Source: Statista, Reuters, D&B Analysis

The combined production of HSFO and MGO increased from 9.7 MMT to ~15-16 MMT due to refining capacity expansion in 2016 while it declined due to low refinery utilization in 2019-2020, due to COVID-19 and OPEC production cuts. However, there was an increase in production, reaching up to 13.5 MMT in 2020 due to the expansion of the Jebel Ali refinery.

The capacity of Ruwais refinery was doubled to 800,000 b/d, while Jebel Ali refining capacity reached 210,500 b/d in 2020.

The production (excluding exports) of Heavy Fuel Oil/Residual Fuel Oil amounted to approximately 2.7 MMT in 2013 reaching 4.4 MMT in 2023. ¹⁶ [Link](#)

In contrast, the production of LSFO commenced in 2022 and increased to about 8.8 MMT by 2024. This increase in LSFO production is facilitated by the operation of 4-5 refineries in Fujairah, UAE and other adjacent regions.

¹⁷ [Link](#) Few of these refineries are:

Table 2.2: Refineries producing and supplying LSFO at Fujairah

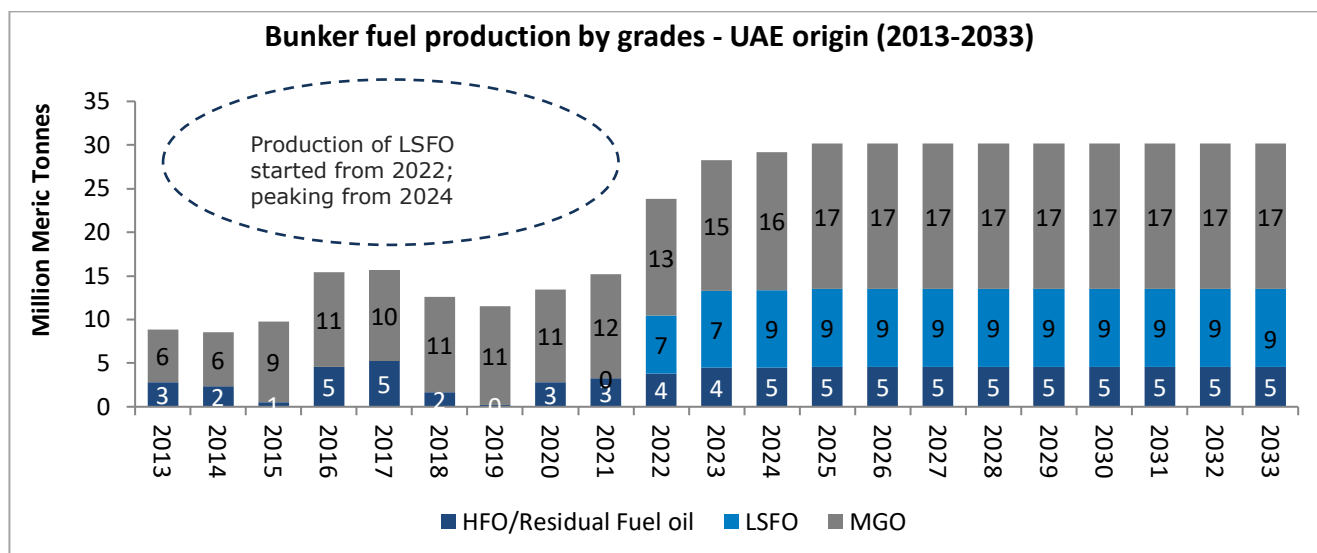
Name of Refinery	Location	Distance from Fujairah port (Kms)	LSFO Production Capacity (Tonnes per month)
Vitol VTTI refinery	Fujairah Oil Industry Zone (FOIZ)	Within the port premises	550,000 (Supplied at Fujairah port)
Montfort refinery (Formerly Uniper)	Fujairah Oil Industry Zone (FOIZ)	Within the port premises	
QPC Refinery	Umm Said (Mesaieed) and Ras Laffan	~850-900	447,000 (~ 150,000 per month supplied to Fujairah port)

ADNOC refineries	UAE based refineries	~450	Details not available
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Note: Brooge Energy is planning a modular refinery with capacity of 25,000 b/d LSFO in coming years; Kuwait's Al Zour refinery is also expected to supplement LSFO supply at Fujairah port in coming years

Sources: Fujairah port authority, Reuters

Figure 2.9: UAE Bunker production - product and grade wise



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group

Montfort and the trading firm Vitol manage refineries with capacities of 67,000 b/d and 82,000 b/d, respectively, collectively producing 550,000 MT per month of VLSFO. Oman based refinery is also expected to commission by year 2026 and it is expected to supply LSFO at Fujairah port. Also, Kuwait's Al Zour refinery is expected to become key supplier of LSFO at Fujairah port.

In 2022, the production of LSFO amounted to approximately 7 MMT, and it is projected to increase to around 9 MMT by 2033. Presently, there are no refineries in the UAE that produce HSFO in Fujairah. The bunkering market in the region depends on HSFO imports from Russia, Iran, and Singapore.

2.4 Demand Supply Gap

Oversupply of Bunker Oil

Over the years, the Supply demand gap across the marine/bunker oil segments became stable and gradually increased over the supply.

HSFO

HSFO had a head-to-head demand-supply match in the year around 2020, after which the supply increased and overtook the demand. It was due to the expansion of Jebel Ali refinery and planned expansion of refineries in UAE. Ecomar Energy and Brooge Refinery plans to add 42,000 b/d and 180,000 b/d by 2024, while Jebel Ali and Ruwais are also planning expansions, adding another 70,000 b/d and 150,000 b/d capacity by 2025.

In 2023, sales of HSFO reached approximately 1.8 MMT, indicating a 14.3% increase from the previous year. This development elevated the market share of high-sulfur bunkers in Fujairah to 24% in 2023, compared to 20% in 2022 (as per the available data by S&P Platts). This pattern corresponds to the growing global demand for HSFO, propelled by the increasing prevalence of vessels equipped with scrubbers. These vessels, which came into operation in the previous year, have the flexibility to refuel with HSFO, offering a more cost-effective alternative to LSFO. Following the adjustment to the global sulfur cap (<0.5%), the decline in HSFO's share of total bunker sales has ceased. Although its share remains considerably lower than pre-4Q2019 levels, the continual increase in scrubber-equipped ships has sustained the demand for HSFO.

LSFO

On the other hand, since the implementation of IMO 2020 and the increasing pressure and demand of using compliant fuels such as LSFO variants, the refining companies started ramping up LSFO production in 2021.

Refineries owned by Vitol, Montfort, and ADNOC-owned refineries started producing approximately 7 MMT per annum. Combining the capacities of all 3-4 refineries, there was an oversupply of ~ 1.5-2 MMT, when compared with the demand of ~ 5 MMT in 2023. Also, in the near term future, Oman and Kuwait based Al Zour refinery are expected to supplement LSFO and VLSFO at Fujairah port.

Also, it is anticipated that Fujairah will be able to match its demand for LSFO till 2030-31 based on the current refinery expansion plans in the UAE market and other adjacent countries such as Oman and Kuwait. Overall, there will be an oversupply of approximately ~3.5-4 MMT till 2027.

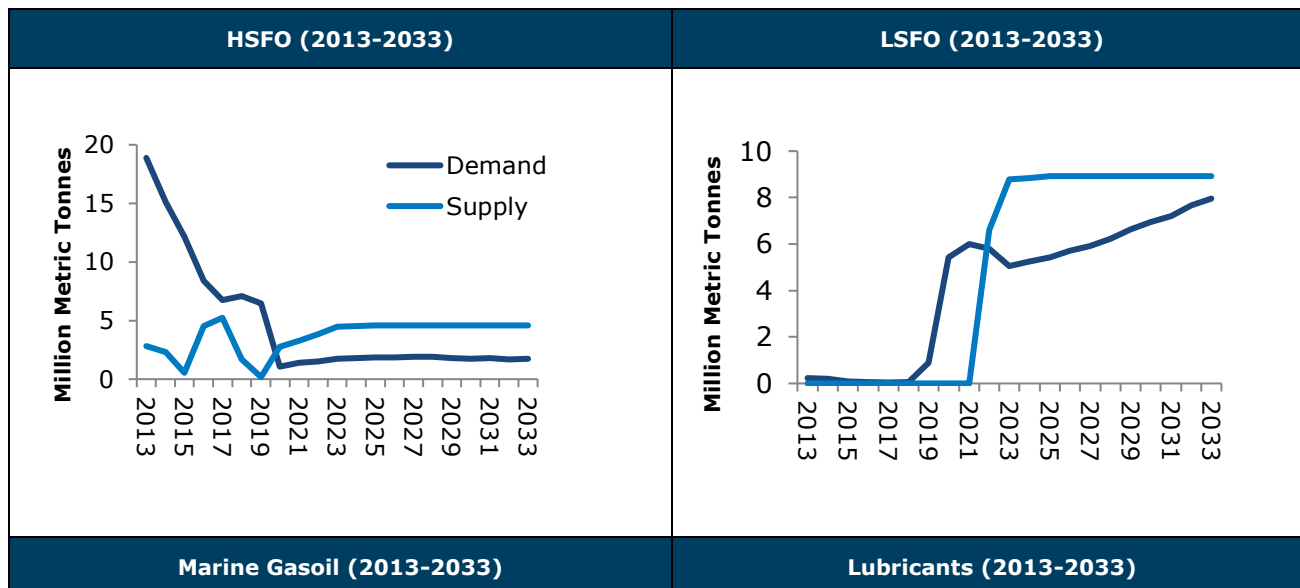
MGO

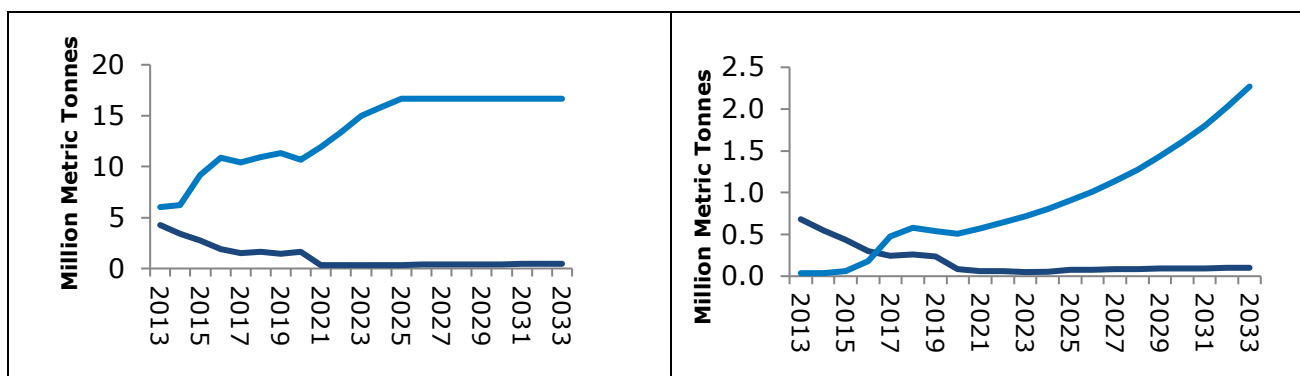
MGO demand was ~4 MMT in 2013 which declined to the level of approximately 0.5 MMT in 2023 along with the overall sales decline in Fujairah. It is anticipated that the gasoil demand will reach till 0.6 MMT in 2033. High prices of marine gasoil contribute towards its less consumption and in future as well it will constitute 4-5% of total Fujairah sales of bunker fuel. UAE's production of gasoil was ~6 MMT in 2013, which reached ~15 MMT in 2023. Also, it is projected that the gasoil production will reach around 17.5 MMT in 2033.

Lubricants

Lubricants sales was approximately 0.6 MMT in 2013 declining up to the level of 0.05 MMT in 2023 and is anticipated to reach 0.15 MMT in 2033. On the supply side, Lubricant production was ~ 0.7 MMT in 2023 and it is anticipated to reach 2.2 MMT in 2033.

Figure 2.8: Demand Supply scenario of Bunker and Lubricants in UAE



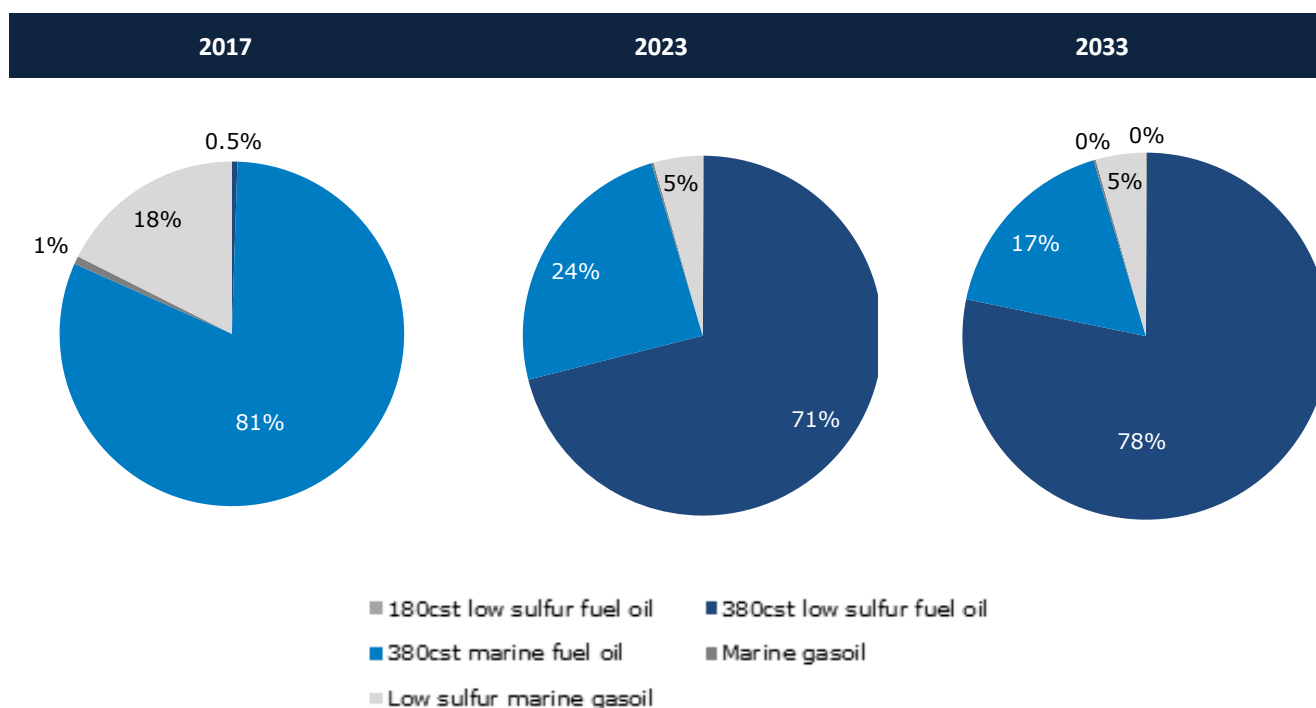


Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

2.5 Fuel grade wise market shares

Figure 2.9: Bunker fuel grade - Market share in UAE

Fuel grade mix % (2017-2033)



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

Drastic shift observed in current scenario vs. the historical scenario: The global market dynamics for bunker fuel sales have undergone a shift. In 2023, the market share of LSFO reached approximately 71% at Fujairah, contrasting with the 24% market share held by HSFO in total sales. Prior to the introduction of the IMO 2020 regulations, the market share of LSFO was less than 1% in 2017, while HSFO dominated with an 81% market share.

Also, it is expected that LSFO share is expected to reach ~ 78% of total sales in 2033 while HSFO sales will constitute 17% of total bunker fuel sales.

Trend development [Link](#)

In the 4Q2019, commercial maritime vessels began transitioning from high-Sulphur fuels to their low-Sulphur counterparts, anticipating the implementation of the IMO 2020 sulfur regulations.

For the 2020 IMO deadlines, four choices were available:

- **Switching from HSFO to MGO or distillates**

Switching to distillate fuels will mean a significant increase in fuel cost and may also require upgrading to a fuel treatment plant due to the significantly lower viscosity of the fuel. Fuel tanks previously used for HSFO must be carefully cleaned before bunkering MGO to avoid contamination and non-compliance problems.

- **Using VLSFO or compliant fuel blends (0.50% Sulphur)**

Low-Sulphur-compliant fuel blends are available in the market through a variety of products. New fuel blends may experience compatibility problems, which will make fuel handling very important for safe operation. Quality control when bunkering to ensure that on-spec fuel is received will be important.

- **Installing exhaust gas cleaning systems (scrubbers)**

HSFO continued to be an option after 2020. However, to be in compliance, it requires the installation of exhaust gas cleaning technology commonly known as SOX scrubbers. No changes will have to be made to the engines or fuel treatment plant, but the installation of a scrubber could be complex, especially for retrofits. In 2022, UAE tightened HSFO regulations at all UAE ports and they will now have to submit a copy of an International Air Pollution Prevention (IAPP) certificate, proving that the ship has a working scrubber.

- **Using LNG or other Sulphur-free fuels**

LNG is expected to gain a more favorable position as an alternative for marine fuel for complying with the global Sulphur cap. LNG as ship fuel is now a technically proven solution, and bunkering infrastructure is developing rapidly around the world. While conventional oil-based fuels will remain the main fuel option for most existing vessels, the commercial opportunities of LNG can be interesting for new ship building. Taking the leap to LNG should only be made based on the best possible information and a thorough analysis.

Also, there are currently a few vessels operating with Methanol as fuel, and several gas carriers have been ordered which use LPG as fuel.

2.6 Product wise Demand Supply scenario

2.6.1 LSFO

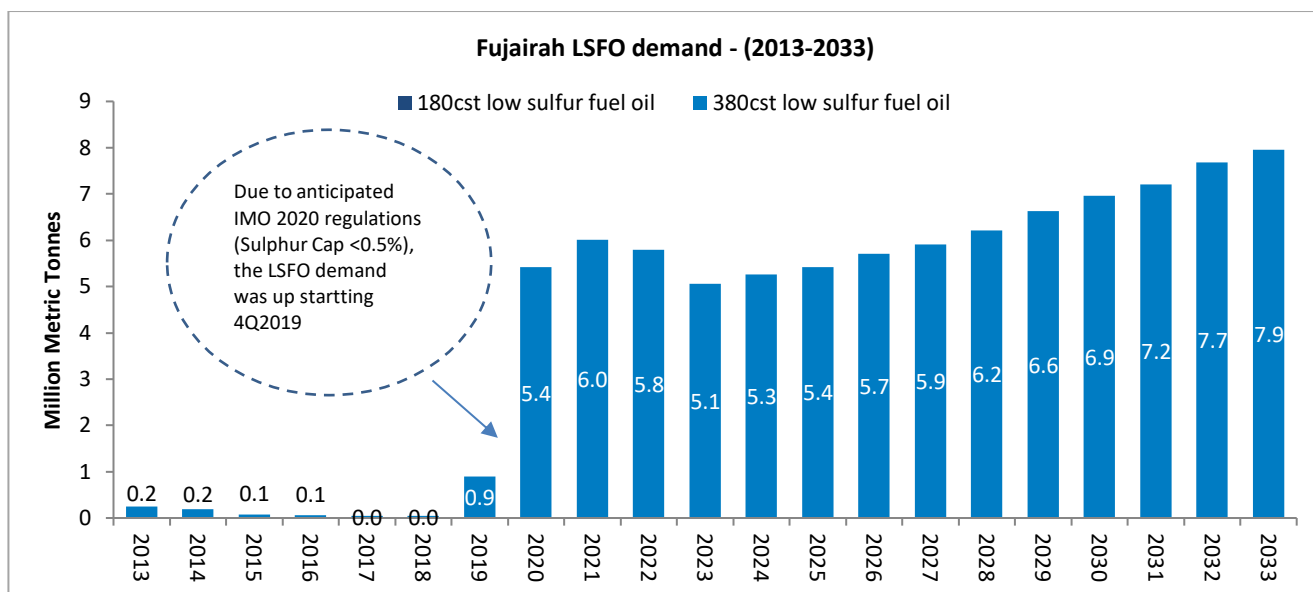
2.6.1.1 Demand

After the enforcement of the IMO 2020 regulations, there has been a notable increase in the demand for LSFO, with the objective of reducing sulfur emissions from maritime vessels. In 2022, the global market for LSFO was valued at USD 48.8 billion, with a projected growth to USD 139.28 billion by 2028. This indicates a CAGR of 19% from 2023 to 2028. [Link](#)

The notable increase in demand for LSFO is primarily linked to the implementation of IMO 2020 regulations, mandating a substantial reduction in sulfur emissions from maritime vessels. Since 2020, shipowners and operators have been transitioning from HSFO to LSFO, creating opportunities for refineries, suppliers, and traders. [Link](#)

In 2023, the demand for LSFO amounted to 5.1 MMT and is projected to reach 7.9 MMT by 2033.

Figure 2.10: LSFO Demand at Fujairah



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

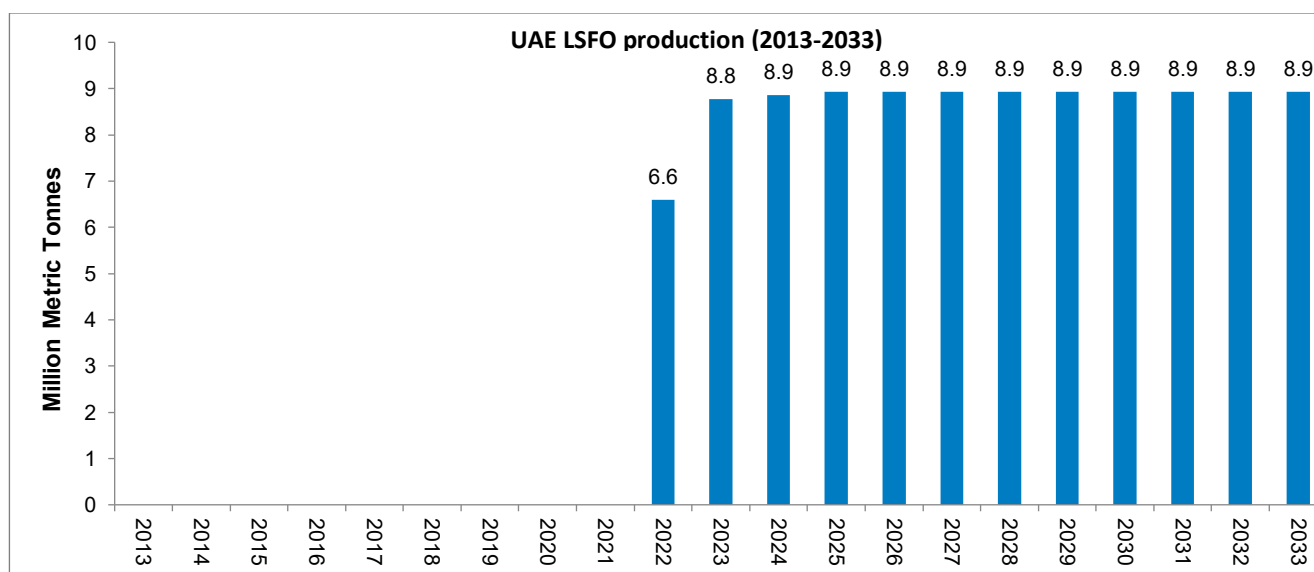
LSFO demand driver: The expansion of the industry is propelled by a growing emphasis on sustainability, the reduction of carbon emissions, and a heightened interest in alternative fuels and technologies. The demand is additionally stimulated by the increasing adoption of **low sulfur propulsion technologies through retrofitting initiatives**. Another contributing factor to the rising demand is the integration of **advanced technologies like AI, ML, and blockchain**, aimed at elevating product quality and efficiency. These technologies have played a crucial role in developing innovative products that outperform conventional alternatives in terms of effectiveness and efficiency.

Starting from early 2019, Fujairah has been supplying LSFO, facilitated by local production. Currently, Fujairah hosts four refineries with crude distillation units and modular refineries, all actively involved in the manufacturing of LSFO. Key contributors to this production in Fujairah include Montfort, Vitol, as well as Brooge Energy and Ecomar Energy, operating a modular refinery in Fujairah designed specifically for LSFO production.

2.6.1.2 LSFO (Local Production- UAE origin)

Refineries owned by Vitol, Montfort, and ADNOC owned refineries are the producers of LSFO in UAE. LSFO production was 6.6 MMT in 2022 and is expected to reach 8.9 MMT in 2033.

Figure 2.11: UAE LSFO Production (2013-2033)



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

Montfort has two 40,000 b/d distillation columns in Fujairah which have been designed to process low sulfur crude oils to produce ultra-LSFO. Montfort's Fujairah plant currently has an annual production capacity of about 3 million tonnes of LSFO with Sulphur content between 0.1% and 0.5%.

Montfort and trading firm Vitol operate 67,000 b/d and 82,000 b/d refineries, respectively, that produce a combined 550,000 MT/month of VLSFO.

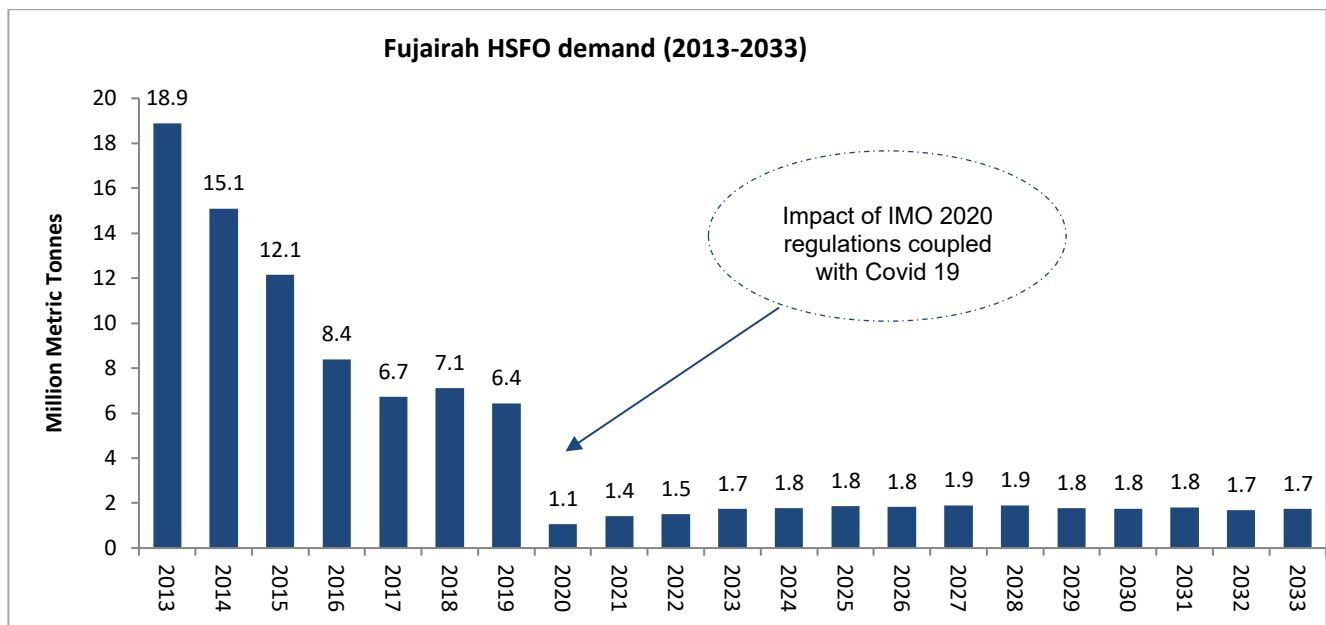
2.6.2 HSFO

2.6.2.1 Demand

In 2023, sales of HSFO reached approximately 1.81 MMT, indicating a 14.3% increase from the previous year. This development elevated the market share of high-sulfur bunkers in Fujairah to 24% in 2023, compared to 20% in 2022, as per the available data. It is expected that the HSFO sales will be stagnant and will be approximately 1.74 MMT by 2033.

Demand driver: This pattern corresponds to the growing global demand for HSFO, propelled by the increasing prevalence of vessels equipped with scrubbers. These vessels, which came into operation in the previous year, have the flexibility to refuel with HSFO, offering a more cost-effective alternative to LSFO. Following the adjustment to the global sulfur cap, the decline in HSFO's share of total bunker sales has ceased. Although its share remains considerably lower than pre-4Q2019 levels, the continual increase in scrubber-equipped ships has sustained the demand for HSFO. This demand is expected to endure until new solutions and future fuels gain widespread adoption, marking the industry's transition towards decarbonization. The number of ships equipped with scrubbers rose from 2,010 in January 2020 to 4,006 in April 2021, contributing to the heightened demand for HSFO. However, the pace of new builds with scrubbers and retrofit installations has slowed since reaching its peak in January 2020. ^{22 23} [Link Link](#)

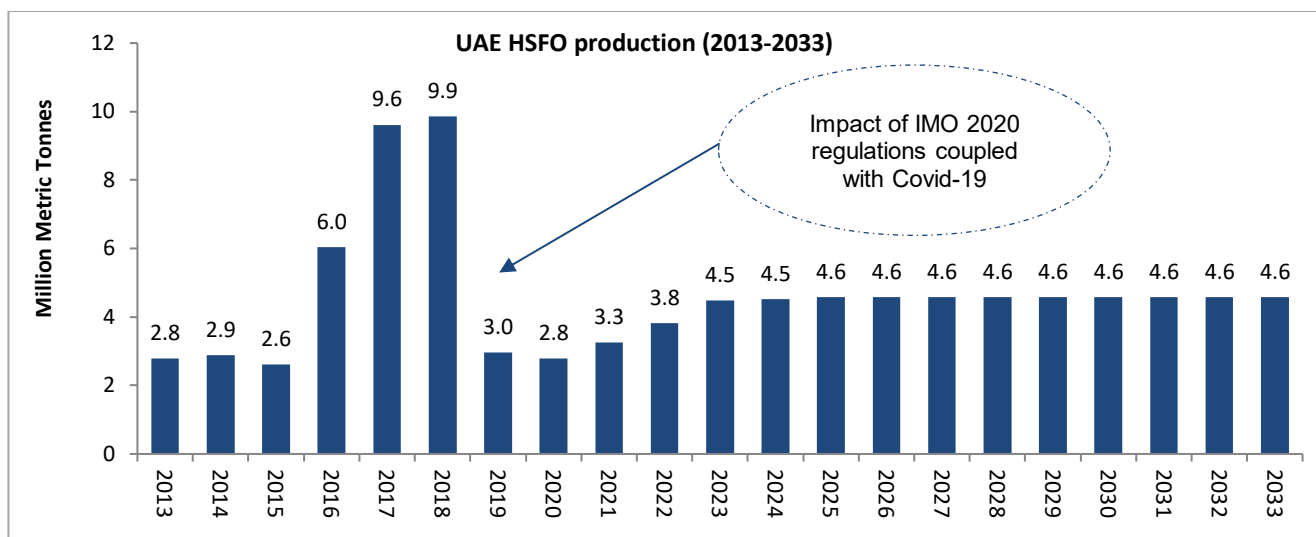
Figure 2.12: HSFO demand at Fujairah



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

2.6.2.2 HSFO (Local production – UAE origin)

Figure 2.13: UAE HSFO production



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

In the past, UAE-based refineries historically produced high-sulfur fuel oil, with production levels ranging from 9-10 MMT during the 2013-2015 period and reaching up to 9.6 MMT during the year 2017. However, production was impacted in 2019 due to the anticipated IMO 2020 regulations, compounded by a decline in demand resulting from the COVID-19 pandemic. Presently, none of the industry participants produce HSFO in Fujairah specifically, and the existing demand is met through imports from countries such as Kuwait, Iraq, Iran and Singapore.

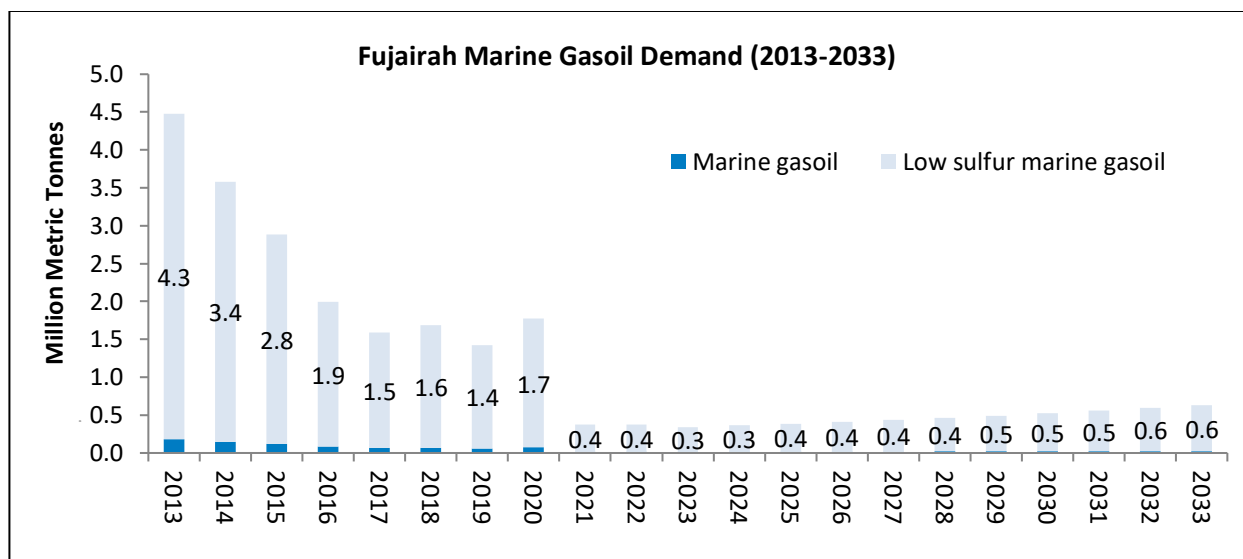
Anticipated refinery expansions in the UAE suggest that additional capacity for heavy fuel oil is likely to be introduced, with local production expected to reach approximately 4,000-5,000 TMT.

2.6.3 MGO

2.6.3.1 Demand

The demand for MGO was approximately 4.2 MMT in 2013, but it decreased to around 0.3 MMT in 2023. This decline was attributed to high prices and an overall reduction in sales at the Fujairah port, influenced by various factors including the Qatar dispute, pirate attacks in the Red Sea, and geopolitical tensions. It is anticipated that the demand for MGO will be to approximately 0.4 MMT by 2033.

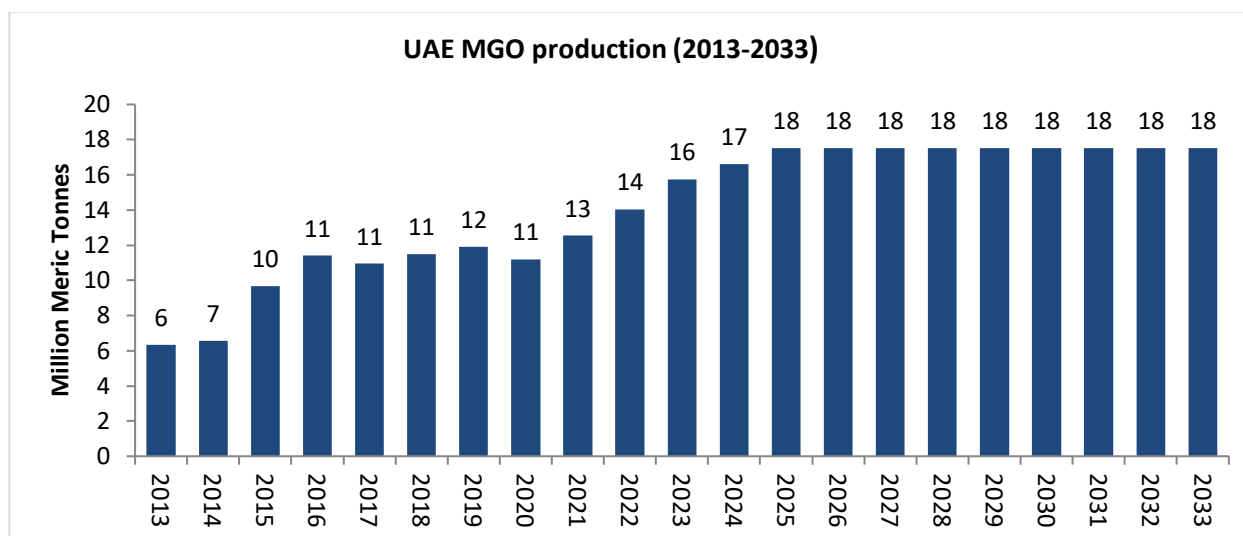
Figure 2.14: MGO demand at Fujairah



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

2.6.3.2 MGO (Overall local production- UAE origin)

Figure 2.15: UAE MGO production



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

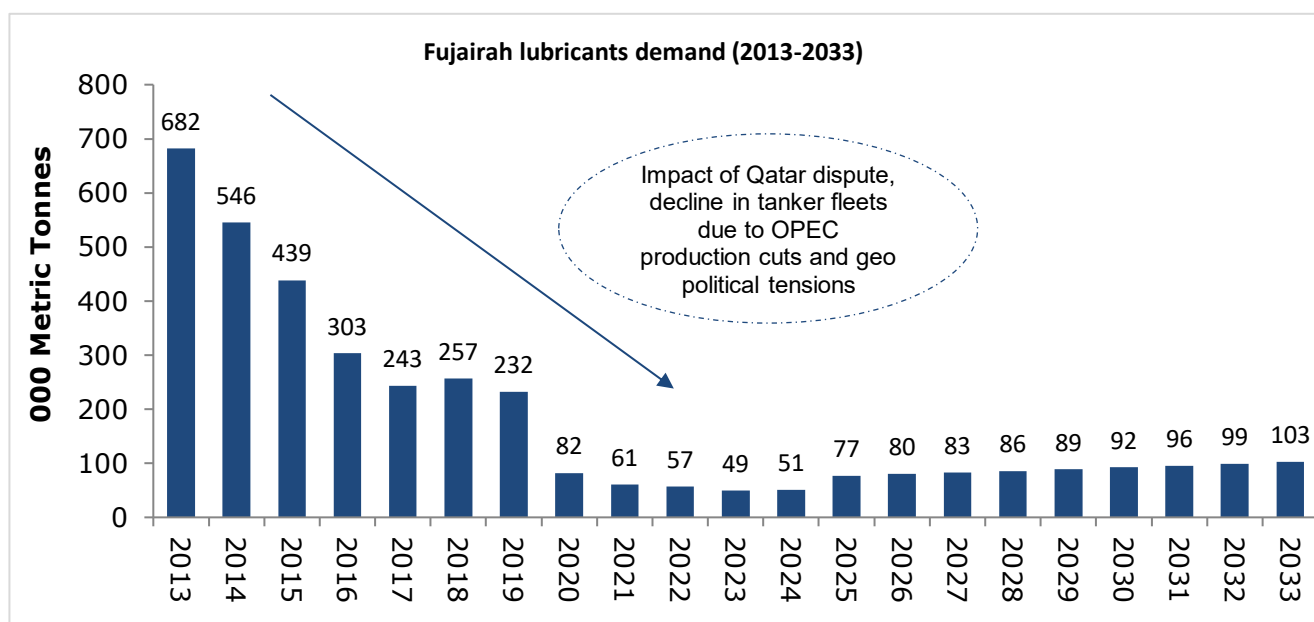
UAE based MGO production in 2013 was ~ 6 MMT which reached 16 MMT in 2023. It is anticipated that the production will reach 18 million tonnes in 2033.

2.6.4 Lubricants

2.6.4.1 Demand

The demand for lubricants was approximately 700 TMT in 2013 but decreased to around 50 TMT in 2023. This decline was attributed to high prices and an overall reduction in sales at the Fujairah port, influenced by various factors such as the Qatar dispute, pirate attacks in the Red Sea, and geopolitical tensions. It is anticipated that the demand for lubricants will increase to approximately 103 TMT by 2033.

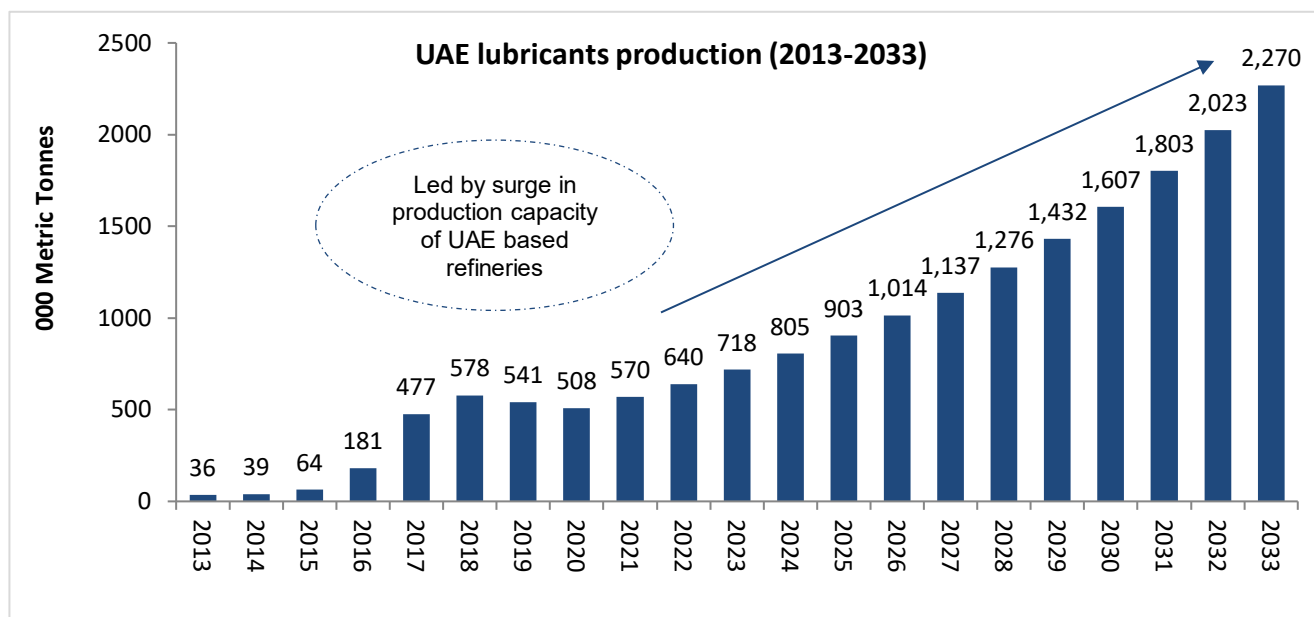
Figure 2.16: Lubricants demand at Fujairah



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

2.6.4.2 Lubricants production- UAE origin (2013-2033)

Figure 2.17: UAE lubricants production



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, D&B Analysis

UAE based lubricants production was ~ 0.03 MMT in 2013, which reached ~0.72 MMT in 2023. It is anticipated that the production will reach 2.2 MMT in 2033.

2.7 Opportunities for Bunkering Business in Fujairah

2.7.1 Opportunity from strategic locations benefits

Rise in maritime transport

Maritime transport and United Nations anticipate continuous growth in maritime trade volumes in the foreseeable future, with a projected increase of Y-o-Y 2.4 % in 2023 and an average of 2.1 % over the next five years. Although this signifies a deceleration compared to the average annual rate of around 3 % observed over the past four decades, it underscores the constraints on the concept of geo-economics fragmentation, at least in the short to medium term and in terms of volume.

The Strait of Hormuz plays a vital role in connecting the Persian Gulf to the Gulf of Oman, the Arabian Sea, and the Indian Ocean in the east. This maritime passage is of global strategic significance, serving as a crucial pathway for the international supply of oil. On a daily basis, an average of 14 oil tankers navigates through this water route, totaling over 5,000 oil tankers annually and facilitating the transportation of 16.5 to 17 million barrels of oil. In total 33,000 ships passed through this route out of which only 12,500 ships called at port for bunker fuel supply which is just 40% of the total addressable/potential demand.

Unfortunately, at present, there are no practical alternatives to this crucial maritime route, although there is a potential option of rerouting oil shipments through pipelines.

Oil & Gas offshore exploration activities picking up post Covid creating sustained demand for bunker oil for tanker fleets

The escalating hydrocarbon resource developments in offshore regions are contributing to the surge in bunker fuel demand. The rapid expansion of exploration activities, coupled with the establishment of new hydrocarbon reserves both offshore and onshore, has led to an increase in reserves. The rapid increase in the tanker fleet due to increased hydrocarbon resource development activities in offshore areas is expected to further boost the market growth for bunker fuel.

In 2023, there is a projected global increase of over 20% in spending on offshore oil and gas exploration, with expectations for this growth to persist into the next year as well. Schlumberger, the largest provider of oil services and equipment worldwide, anticipates a sustained period of activity, forecasting the conclusion of 65 lease rounds globally this year. Furthermore, numerous countries are expected to award leases through open-door policies during this period.

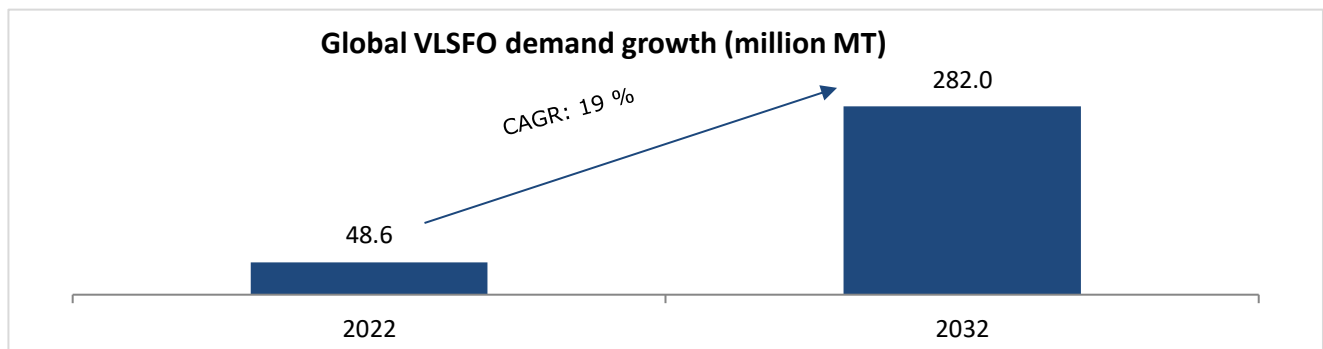
From 2022 to 2025, Schlumberger, the renowned oil services and equipment company, anticipates investment decisions exceeding USD 500 million, reflecting a substantial 90% surge compared to the period from 2016 to 2019 (pre-COVID period). This growth is poised to be fueled by major development initiatives in ME region, as well as capacity expansions in Africa. Additionally, there is a resurgence of exploration and appraisal activities in emerging offshore regions such as East Mediterranean. **With increase in the tanker fleet, Fujairah port will experience continued demand for bunker fuel.**

2.7.2 Opportunities from the sustained demand perspective

Opportunity across LSFO fuel oil due to regulatory changes and lower cost implication

With the International Maritime Organization's prohibition on ships utilizing fuels containing more than 0.5% sulfur in 2020, there has been a growing demand for LSFO. As of January 2024, the anticipated waiting period for refueling with LSFO from barges in Fujairah is expected to be within the range of seven to nine days. Limited immediate availabilities are projected throughout the month, indicating robust bunker demand for LSFO. Going forward, the demand and prices of LSFO is anticipated to be directly influenced by the relaxation of sulfur regulations in different countries.

Figure 2.18: Projected global growth of VLSFO



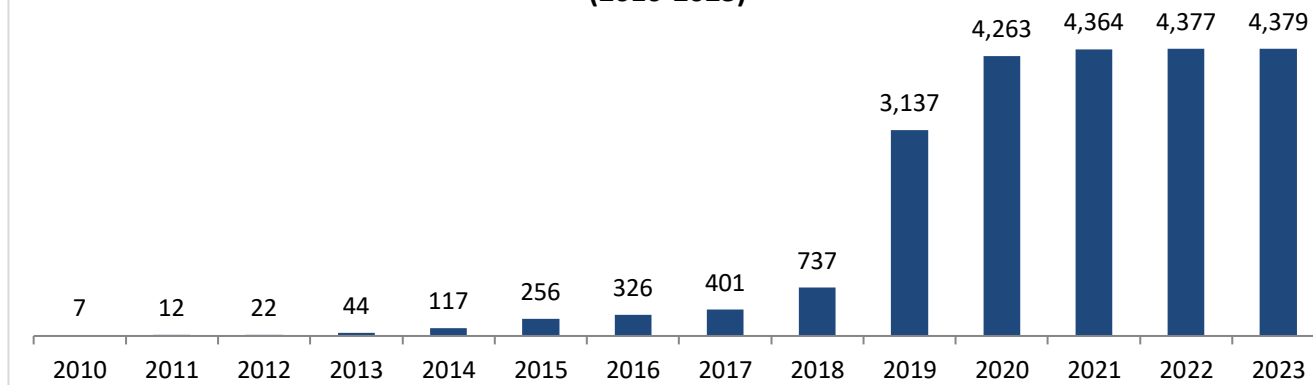
Source: Market data forecast

HSFO is here to stay with continued demand backed by scrubber fitted ships

The annual statement from the Ship and Bunkers 20 Index highlights that carriers equipped with scrubbers have benefited from the widening price gap between heavy fuel and low-sulfur bunker oil, coupled with the accessibility of inexpensive HSFO. Ships can install exhaust gas scrubbers at a cost of around USD 2 million-USD 3 million per ship. If the price spread between HSFO and VLSFO is wide enough, the fuel-cost savings cover the cost of the scrubber installations and generate profits thereafter.

Figure 2.19: Vessels fitted with EGCS globally

Number of vessels fitted with exhaust gas cleaning systems (EGCS) worldwide (2010-2023)



Source: S&P Platts, Reuters

Container vessels with a capacity of 8,000 twenty-foot equivalent units (TEUs) or greater prominently lead in the two primary mainline trades: Asia-Europe and Asia-U.S. As per Clarkson's data, 46% of container ships with a capacity of 8,000 TEUs or more have already been fitted with scrubbers, and additional installations are in progress. Among ships sized 12,000 TEUs or larger, 59% have already undergone scrubber installations

Given the massive amount of fuel consumed by larger ships on long-haul voyages, such spreads are creating huge earnings differentials in the spot market.

In the bulk commodity shipping sector, the responsibility for covering fuel costs in a spot voyage deal lies with the ship operator, not the cargo shipper. According to estimates from Clarkson Platou Securities, VLCCs that are ten years old, without scrubbers and utilizing VLSFO, earned an approximate daily equivalent of USD 16,200 per day in the spot market. Conversely, the same VLCC equipped with a scrubber, burning HSFO, incurred fuel expenses USD 22,000 lower per day, resulting in a net day rate of USD 38,400. With Clarkson's average breakeven for a 10-year-old VLCC set at USD 28,000 per day, a non-scrubber VLCC is still experiencing financial losses in the spot market. In contrast, the scrubber-equipped VLCC is earning 2.4 times more and is operating with a positive balance.

In the dry bulk shipping industry, a Capesize vessel (a bulk carrier with a capacity of approximately 180,000 deadweight tons) without a scrubber generated a daily equivalent of USD 22,000 in the spot market, as reported by Clarksons. Conversely, a Capesize equipped with a scrubber realized savings of USD 22,400 per day on fuel, resulting in net daily earnings of twice that amount: USD 44,400 per day

Considering the breakeven point of a 10-year-old Capesize set at USD 18,000 per day, a Capesize without a scrubber is making a modest profit, while a Capesize equipped with a scrubber is enjoying a substantial one.

Scrubber fitted VLCC's and average Capesize vessels utilizing HSFO earn almost double the profit and achieve breakeven over the ship cost in almost half the time, in comparison to without scrubber VLCC's and Capesize vessels utilizing LSFO

Table 2.3: Bulk vessels profitability – HSFO vs. LSFO

	VLCC		Capesize vessel (DWT: 180,000)	
	With scrubber (Utilizing HSFO)	Without scrubber (Utilizing LSFO)	With scrubber (Utilizing HSFO)	Without scrubber (Utilizing LSFO)
Breakeven period	10 years	10 years	10 years	10 years

Average breakeven (10-year period)	USD 28,000 per day	USD 28,000 per day	USD 18,000 per day	USD 18,000 per day
Net day rate (Daily profit)	38,400	16,200	44,400	22,000
Profitability level	High	Low	High	Medium
Comments	2.4 times more profit in relative to VLCC's utilizing LSFO	Incurring financial loss	2 times more profit in relative to VLCC's utilizing LSFO	Modest Profit

Source: S&P Platts, Reuters

As illustrated in the table above, VLCCs can achieve 2.4 times greater profits compared to those using LSFO as bunker fuel. Similarly, Capesize vessels with a deadweight tonnage (DWT) of 180,000 can realize twice the profit, whereas vessels utilizing LSFO earn a modest profit, for a break-even period of 10 years.

As such, the demand for HSFO will continue and not deplete all together, providing opportunities for existing and new players for continued revenue in the bunkering market.

2.7.3 Opportunity from geo political tensions

Fujairah port gains with accepting imports of crude and fuel oil products from Russia, as Europe and the US sanction ban on Russian imports

Post the ban imposed on the imports of Russian crude and oil products earlier this year, few traders at Fujairah have provided the storage and blending capacity to the western origin oil.

Western region continues to be top exporter of fuel oil to Fujairah port and is utilizing Fujairah storage capacity at optimum level. Robust storage demand is increasing revenues of operators such as Vopak, Vitol's VTTI, ENOC and ADNOC.

2.8 Growth Drivers of Bunkering Business

Cleaner fuel demand to rise (Methanol, Hydrogen and LNG) due to growing concerns regarding emissions reduction

Fujairah has now joined the league of Biofuel bunkering ports. The marine energy solutions provider KPI Ocean Connect, based in the UK, has supplied **B24 VLSFO** to two bulk carriers owned by the India-based company ArcelorMittal Shipping at the Port of Fujairah in the UAE. This marks the first-ever delivery of Biofuel stem at the Port of Fujairah, highlighting the growing availability of Biofuel bunkers on a global scale. The introduction of Biofuel bunkering in Fujairah, facilitated by KPI Ocean Connect, will open up new possibilities for vessel owners and operators to take steps in reducing the net greenhouse gas emissions during voyages.

The adoption of LSFO over high sulfur variants is being propelled by stringent environmental regulations.

Rapid globalization and increasing sea-borne trade

Rapid globalization and increasing seaborne trade boosting demand for bunker fuel for containers. Owing to the rapid industrialization and commercialization in developing countries such as China, India, Vietnam and the Philippines, exports as well as imports in these countries have risen significantly. To facilitate the movement of goods across various nations, the demand for shipping vessels is increasing considerably. The rising number of shipping vessels, in turn, is resulting in the increased consumption of fuels in shipping bunkers.

Digital transformation at Fujairah

In the past, the Port of Fujairah utilized an in-house software solution that proved incapable of managing customer complaints and streamlining supply chain operations. These intricate systems struggled to handle customer interactions and generate the necessary Management Information System (MIS) reports for reconciliation. Consequently, this had a detrimental effect on the port's performance and customer satisfaction.

Over the last few years, the authorities at the Fujairah port have accelerated their digitization efforts, actively pursued the automation of customer experiences and embraced digital transformation. As an example, the Port of Fujairah has opted for the Oracle Fusion Cloud Applications Suite to address finance, human resources, and supply chain needs, aiming to enhance insights, cut costs, boost employee engagement, and streamline global procurement processes. The port has successfully implemented Oracle Fusion Cloud Enterprise Resource Planning, Oracle Fusion Cloud Human Capital Management, and Oracle Fusion Cloud Supply Chain and Manufacturing.

Utilizing Oracle Fusion Applications, the port can leverage cloud technology to dismantle organizational silos, standardize processes, and consolidate financial, HR, and supply chain data within a unified cloud platform. The incorporation of advanced features like Artificial Intelligence (AI) and Machine Learning (ML), along with the implementation of innovative technologies, has empowered port authorities to enhance demand prediction, digitize supply chain operations, and achieve operational cost efficiencies.

2.9 Other Key Trends

Emergence of alternative refueling options and commitment of shipping industry to cleaner fuels

The rise of alternative refueling options at key locations in the ME region and the recent upswing in bunkering activities beyond the port limits off the coast of Fujairah could pose a competitive challenge in the future. For example, Minerva Bunkering and Aramco Trading have initiated collaborations since the end of 2021 to provide refueling services for vessels around KSA's Yanbu port.

Although Yanbu and Jeddah ports in KSA have a track record of supplying bunkers, they require additional time to solidify their position as formidable competitors to Fujairah. Bunkering operations in Oman are still in the developmental phase and may become more active in the extended period. The bunkering market in Oman is anticipated to experience gradual growth in the long term, with established strategies in place to develop essential infrastructure like storage facilities and barges. Under favorable demand conditions, Oman could reportedly sell up to approximately 20,000 MT of bunker fuels in coming future.

As per market participants surveyed by S&P Global, suppliers based at the nearby Sharjah port of Khor Fakkan are likely the most significant competitors for Fujairah's demand, alongside regional competition within the ME region. Some suppliers are exploring the acquisition of bunkering licenses around Khor Fakkan instead of Fujairah port, owing to less stringent regulatory oversight. Vessel operators are increasingly favoring more affordable bunker fuel options, particularly in the form of floating refueling alternatives.

Storage capacity expansions of clean fuels

Fujairah in the UAE needs to enhance investments in the storage and provision of new alternative fuels. Despite the imperative for maritime decarbonization, these markets have not gained substantial momentum at the world's third-largest bunkering hub.

While serving as a crucial and strategic bunkering port, Fujairah has fallen behind other hubs such as Singapore and Rotterdam in preparing for emerging alternative fuel markets like LNG, Biofuels, and Methanol.

The results of an online poll conducted at the FUJCON forum indicate that by 2050, Methanol and LNG are projected to dominate the market share of alternative bunker fuels in Fujairah. Biofuels and ammonia are expected to follow in terms of popularity.

As of December 2023, Vitol has initiated a sustainable fueling project, marking its first Biofuel deliveries in Fujairah, UAE. Additionally, KPI Ocean Connect, a UK-based marine energy solutions provider, has supplied B24 VLSFO to two bulk carriers owned by the India-based company ArcelorMittal Shipping at the Port of Fujairah, UAE.

Fuel reduction initiatives, prioritizing the adoption of efficient technologies

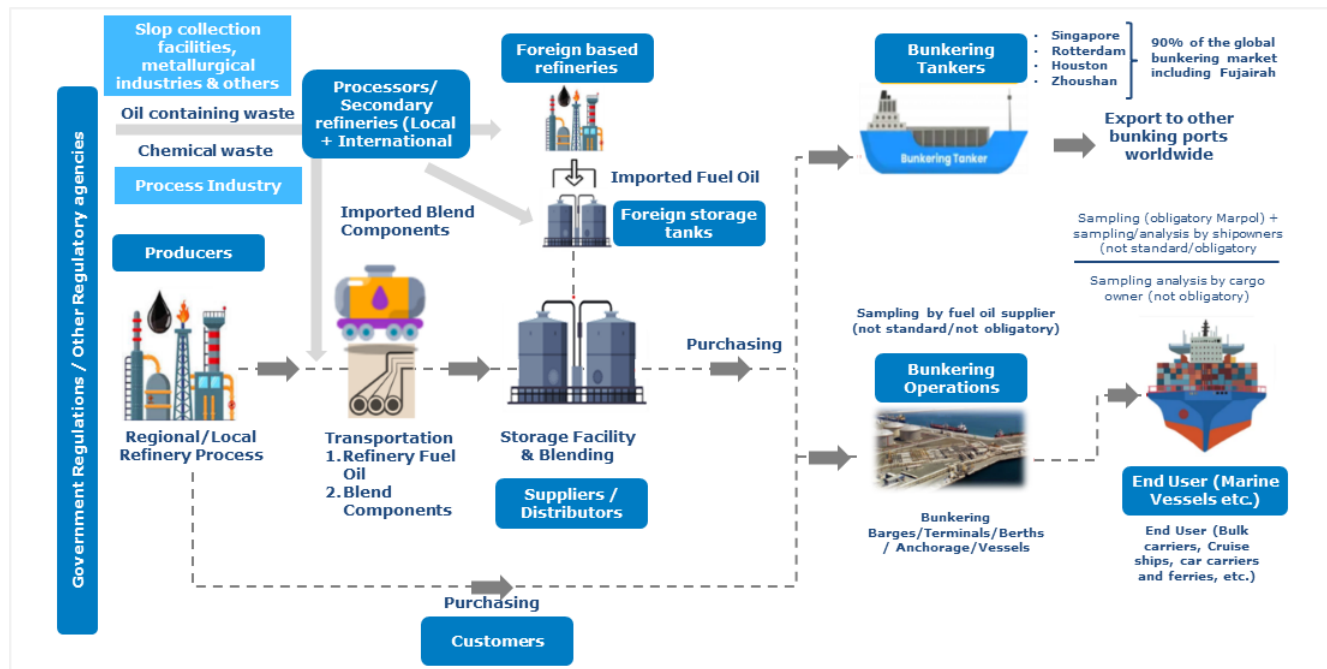
Large maritime vessels consume significant amounts of fuel and represent a primary cost factor for many shipping companies. Therefore, enhancing energy efficiency stands as the primary catalyst for innovation in the shipping industry.

The transition from conventional fuel-based propulsion systems to environmentally friendly hybrid propulsion is gaining momentum due to the efficiency and cost-effectiveness offered by hybrid systems. According to United Nations Conference on Trade and Development (UNCTAD)'s latest Review of Maritime Transport, there is a projected 2.4% growth in global maritime trade volume in 2023, rebounding from a 0.4% contraction in 2022. Maritime trade was estimated to experience a 2.4% Y-o-Y growth in 2023 and surpass 2.0% Y-o-Y growth between 2024 and 2028.

2.10 Value Chain Analysis

2.10.1 Value chain components

Figure 2.20: Overview of Bunkering value chain



Bunkering oil value chain consists of various nodes – including those of Refineries, Storage & Blending solution providers, Suppliers/brokers, Customers (ships and vessels), Government agencies, and other regulators.

Refineries are primarily large companies in the value chain and in the Fujairah port region; players like Montfort and Vitol are the major providers. Companies like GPS Bunkers, Vitol Bunkers, Minerva and Montfort are involved in the more than one node, and also active in the business of storage/blending and overall bunkering operations. **These players together account for 35-40% of total bunker fuel market share currently.**

2.10.2 Key nodes in bunkering supply chain

Role and concentration of key players and stakeholders

Table 2.4: Key nodes in bunkering supply chain

Key nodes in supply chain	Key actors / stakeholder roles in value chain			Concentration of actors
	Scenario 1	Scenario 2	Scenario 3	
Refinery	Bunker fuel producer 	Storage /Blending of Bunker fuel Bunker fuel supplier 	Refining, Storage/Blending and Supplier 	Low concentration (Specialized refineries for providing bunkering fuel)
Storage & Blending	Storage terminal operator			

	 			
Supplier/ Broker	Bunker fuel supplier   			Mid-level concentration (13 Licensed Bunker suppliers and 4 lube oil suppliers)
Government / other agencies	Specialized agency, Regulatory authorities   			Low number of agencies • The Ministry of Energy and Infrastructure • Ministry of Economy • Federal Maritime authority of UAE
Technology providers	Supporting smooth functioning of bunkering operations  			Low concentration (Limited players and discrete role)
End User (Bulk carriers, Cruise ships, Car carriers and ferries, etc.)	Consumption of Oil   			High concentration (Large number of ships for bunkering. UAE port of Fujairah Port of UAE is world's fourth largest bunkering)

Source: Fujairah port Authority, Supplier website

Note: The company logos mentioned in the above table is only for illustration purpose but not exhaustive

2.10.3 Roles by entities

Table 2.5: Role of key stakeholders

Key nodes in Supply Chain	Roles	Interdependencies & decision-making process
Refinery & blending	Refining crude oil and produce compliant fuel oils	Dependency on crude oil suppliers, oil traders, agents, port infrastructure
Storage terminal & blending	Provide storage, handling and blending requirements and accommodate oil products	Dependency on oil product suppliers, oil traders, agents, port infrastructure

Bunkering operations	Supplying marine gasoil and LSFO to vessels	Dependency on crude oil supplier, blender, storage terminal, barge supplier, and port infrastructure
Government / other agencies	Regulating Sulphur content in bunker fuels, Licensing bunker suppliers and other players in supply chain	Dependency on International Maritime Organization regulations for bunker fuel
Technology providers	Monitor and organize the vessel traffic in the port and make ship movements safe and more efficient	Influenced by drive towards cleaner fuels and improvement in processes, and productivity
End User (Bulk carriers, Cruise ships, car carriers and ferries, etc.)	Consumption of oil	Direct and indirect dependencies on all the actors in the value chain

Source: D&B Analysis

2.10.4 Value Chain Actors

Refinery & blending:

- **Vitol's FRCL (Fujairah Refinery Company Limited):** 82,000 b/d refining capacity; operated by VTTI; offers blended LSMGO and VLSFO
- **Montfort Refinery:** 65,000 b/d crude oil processing unit, Produce and supply IMO 2020-compliant LSFO; production facility and associated marketing business sell over 30 million barrels of LSFO to the shipping industry annually
- **ECOMAR:** Refinery's production slate currently includes LSFO, Capacity of the refinery will be expanded to 62,000 b/d from 22,000 b/d
- **Brooge Energy Refinery:** Operates a modular refinery with capacity of 25,000 b/d

Storage terminal & blending:

- **VTTI (Vitol Tank Terminal International) storage terminal:** Terminal offers over 1,176,600 KL storage capacity and 11 onsite jetties, 66 Tanks, Second-largest terminal operator at the port of Fujairah
- **Global Terminal Investments Limited (GTI Fujairah):** 1,200,000 Kilo Liters (KL) bulk liquid storage facility, first terminal in the Port to store ultra-LSFO, 14 tanks and, 6 jetty lines access to 9 main berths and 2 bunker barge berths
- **Minerva's Fujairah terminal:** Capable of storing 465,000 KL of fuel
- **GPS CHEMOIL:** Storage Capacity of 698,758 KL of storage capacity

Bunkering operations:

- **Vitol Bunkers:** Supply circa 1,800 vessels each year, Leading independent mover of bunker fuels in the region
- **Minerva Bunkering:** Three bunker barges to supply marine gasoil and LSFO to vessels
- **GPS Bunkers:** Fully-owned three double-hull bunker barges to vessels at anchorage and at berth in Fujairah and KhorFakkan, Deal IN – Intermediate Fuel Oil (IFO) – IFO 380 CST RMG and IFO 180 CST RME and Low Sulphur MGO (LSMGO)
- **Montfort Trading FZE:** Operates three bunker barges in the region, supply capacity of 100,000 mt/month, Offer all grades of bunkering fuels (VLSFO, HSFO, and LSMGO) from facilities in Fujairah

Regulatory agency

Port of Fujairah, Federal Maritime Authority of UAE, & Ministry of Energy: Identifies and regulate licensed bunkers suppliers and lube oil supplier

Technology providers

Endress+Hauser: Supplied Automated Overfill Prevention System (AOPS) to detects, indicates and prevents hazardous overfills of storage tanks

End users

- Marine vessels
- Shipping companies
- Tanker fleet owners

3 Customer Insights

3.1 Overview

The primary customer segments for bunkering fuel are Small ships, Charters, Ferries and VLSS with their application in container and bulk carriers, oil, chemical and gas tankers, general cargo ships and fishing vessels. Container, bulk carriers and oil tankers constitute 60-70% of the total bunker fuel consumption globally and at Fujairah port.

At Fujairah port, approximately 45-50% of demand originates from charterers with remaining share originating directly from ship owners and bunker fuel traders.

HSFO to remain fuel of choice for few large vessel owners

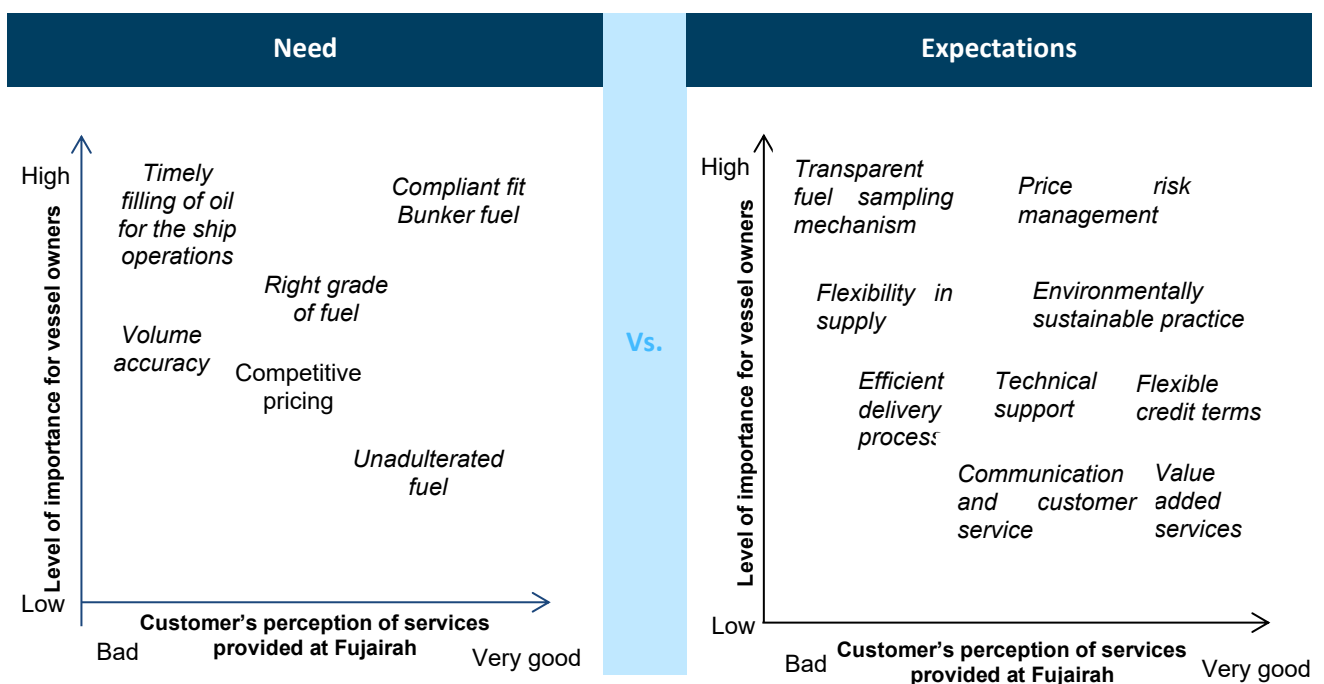
HSFO is still expected to be a fuel of choice for large vessel owners which consume more than 30 MT a day as they will be able to recover the amount of scrubber in 2-4 years through saving bunker fuel cost, as HSFO and LSFO have a price differential of ~USD150- USD250 per MT. Shipping freight charges Y-o-Y, growth rate has not been consistent and fluctuating due to oversupply of ships and sea-based freight movers in the market, demand getting affected due to restrictions imposed during Covid 19 and geo-political factors in the ME region. These market factors have put additional pressure on the large vessel owners to remain competitive in terms of saving operational and fuel expenses to remain profitable in relative to its competitors.

Rapid surge in LSFO demand at Fujairah port, post IMO 2020 regulations

Post IMO 2020, approximately 50% of the HSFO sales shifted towards LSFO in the last quarter of 2019, globally. Following the global trend, Fujairah experienced the rapid surge in LSFO demand post the IMO 2020 regulations were implemented, as the port authorities banned the usage of HSFO by the ships without scrubbers. Few end customers of vessel and cargo owners such as Amazon etc. pressurized the ship owners to use LSFO as a Bunker fuel as part of their sustainable and environmental practice.

Figure 3.1: Key insights from customer survey

What do customers want?

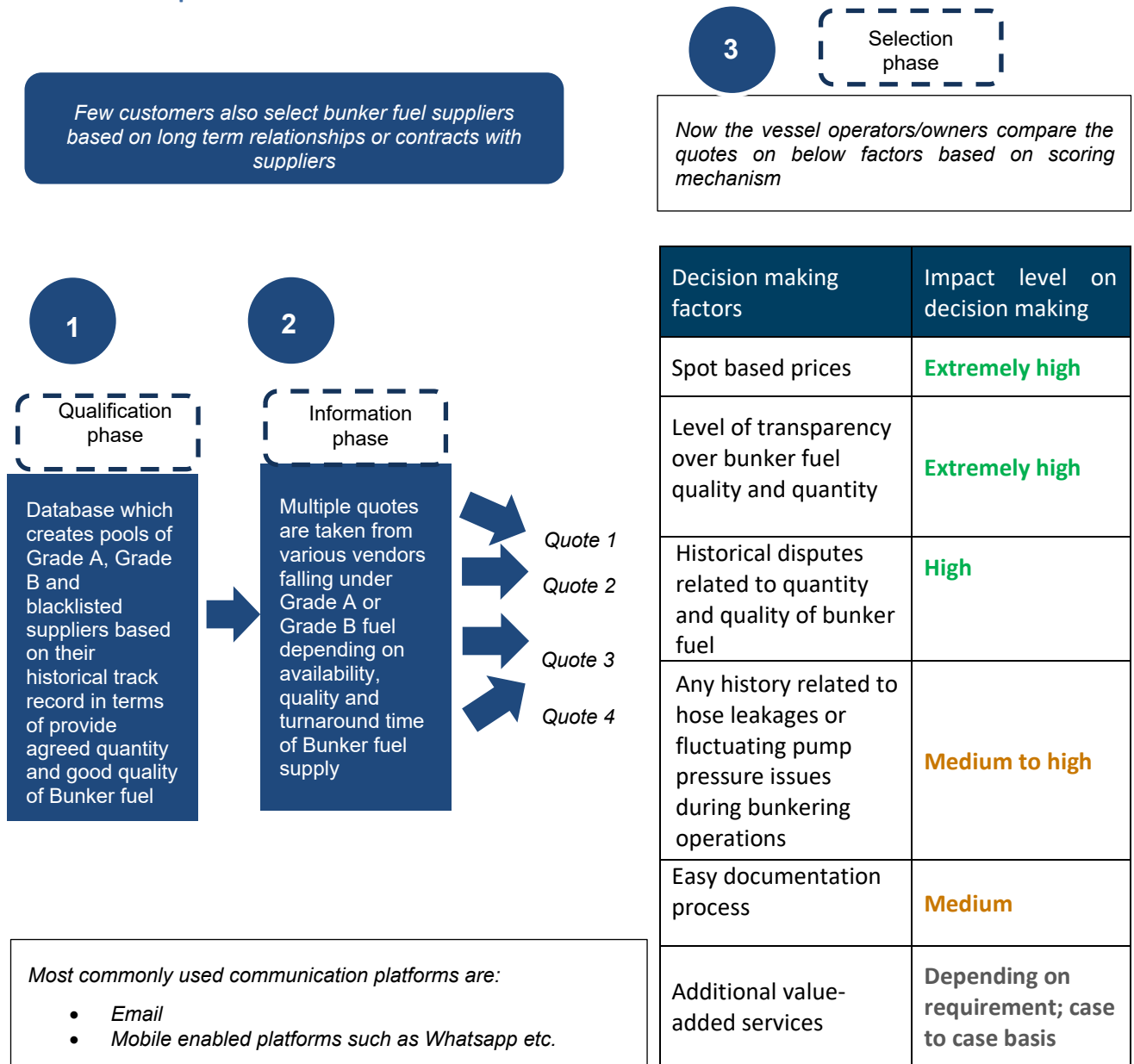


Recommended solutions to address the customer challenges

- Utilization of digital innovative solutions such as Internet of things (IoT), blockchain, marine intelligence tools, smart bunkering solutions such as bunkering optimization tools, price risk management tools, cloud-based collaboration and communication tools and automated data integration systems to reduce operational costs and process inefficiencies and enhance customer value through effective communication, real time decision making and reducing manual errors
- Adoption of suitable innovative electrical and mechanical solutions such as advanced flow meters to bring transparency in quantity and quality of the fuel supplied to the vessel operators
- Providing additional services such as on-board technical support, agency services, ship repair and maintenance, crew services and training over fuel, engine specifications and sustainable practices
- Setting up an example with strict adherence to sustainable environmental practices and international level health, safety and security standards during bunkering operations
- Utilization of financial management tools for effective financial/ working capital management, thus providing better flexibility on credit terms of suppliers
- Digitization of documentation process, partnerships with third party bunker fuel sampling test providers can further reduce the turnaround time during bunkering operations
- Long term and alternate contracts with refiners, bulk suppliers at other bunkering locations helps in managing inventory levels and unexpected supply chain disruptions
- Partnerships with globally recognized technology providers with the objective of bringing more transparency in bunkering operations

Figure 3.2: Overview of Vendor selection process

Vendor selection process



Source: D&B Analysis

3.2 Customer Intelligence (Primary research)

Customers

Customers of bunkering fuel providers are ships and vessels of all sizes

Customer segmentation

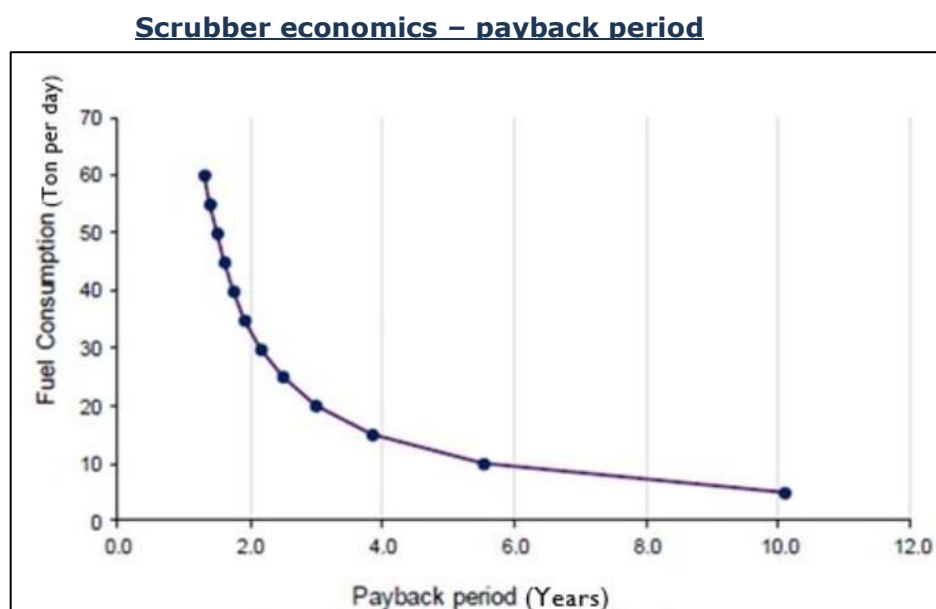
- 1) By Vessel sizes: Small ships, Charters, Ferries, VLCC
- 2) At Fujairah port, approximately 45-50% of demand originates from charterers with remaining share originating directly from ship owners and bunker fuel traders.

Fuel product segments for customers

- **HSFO:** Many of the larger customer ships are fitted with scrubber and therefore go for HSFO oil grade as it is still profitable for them considering the factors like price differential between HSFO and LSFO, payback period of the installed scrubber. Scrubbers reduce the ship's emissions through treatment of exhaust gases which allows the ships to continue using HSFO complying with IMO 2020 regulations.

Scrubbers require a capital investment of USD 2-8 million per ship. ROI of the investment in scrubber is calculated from savings by buying low cost HSFO over high cost LSFO/MGO. As per the analysis done by Wood Mackenzie and Bloomberg, most larger vessels with daily fuel consumption of more than 30 MT per day, the payback period of a scrubber will be 2 years or less. ^{24 25} [Link](#) [Link](#)

Figure 3.3: Economics of EGCS for HSFO



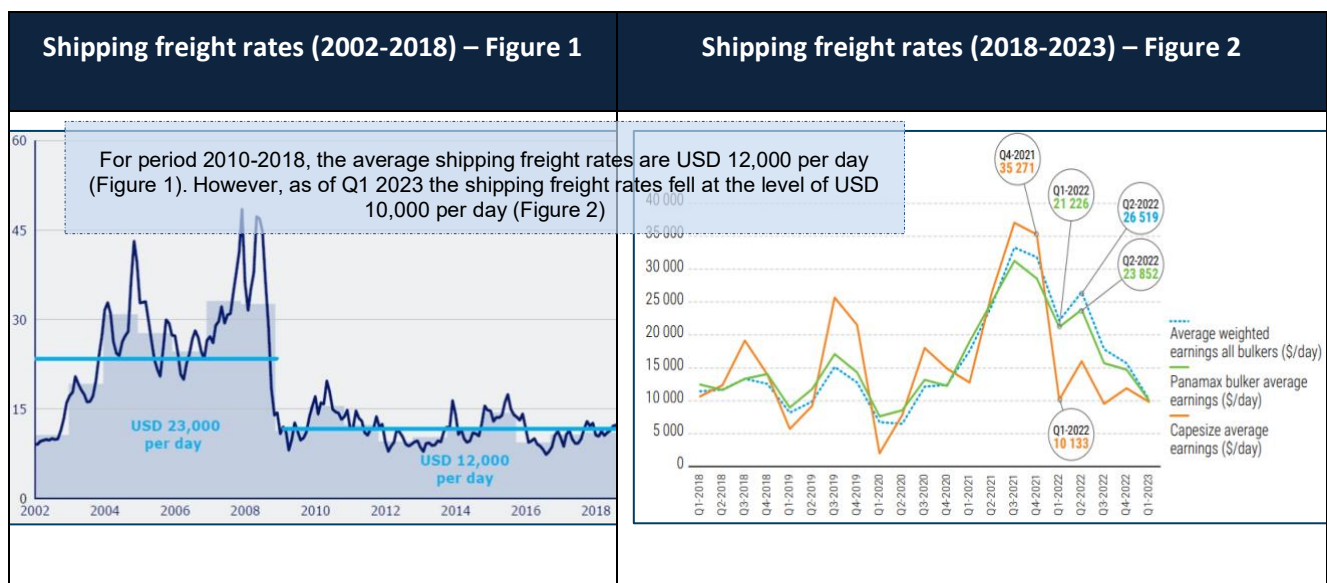
Source: Natixis, Wood Mackenzie, Bloomberg

However, because of the constrained global yearly installation capacity of approximately 3,000 scrubbers worldwide and the rising installation expenses associated with them, opting for scrubbers is no longer financially viable for aging fleets.

After the implementation of IMO 2020 regulations, there has been a significant surge in demand for LSFO, resulting in a price gap of approximately USD 150 to USD 250 per MT between HSFO and LSFO. Consequently, operators of large vessels equipped with scrubbers face an additional annual operating expense of 20-25% if they opt for LSFO over the more economical HSFO. Moreover, they are unable to fully transfer the increased operating costs to end consumers due to the substantial oversupply of ships in the global market, which restricts their bargaining power.

Over the past 15 years, shipping freight rates have shown volatility, and as of Q1 2023, they have only recovered to approximately 50-60% of the level seen in 2008. This situation presents vessel operators with challenging fuel economics. ²⁶ [Link](#)

Figure 3.4: Overview of shipping freight charges



Source: Clarksons, Danish Ship Finance, UNCTAD

Due to the factors such as high price differential between HSFO and LSFO, limited global installation capacity of scrubbers linked with increasing installation cost and low negotiating power over shipping freight rates with the ship owners, it is expected that the large vessel operators/owners will continue using HSFO as their fuel of choice. Large vessel owners will be able to recover the high installation cost of scrubbers and achieve profitability in a timeframe of 2-4 years.

- LSFO:** Majority of customer ships have now moved towards LSFO following the IMO2020 regulation. Approximately 50% of the HSFO sales shifted towards LSFO in the last quarter of 2019, globally. Following the global trend, Fujairah experienced the rapid surge in LSFO demand post the IMO 2020 regulations were implemented. Before 2020, Fujairah port was at a price disadvantage compared with other bunker ports, as bulk of HSFO was made in the older, less complex refineries of Russia and Europe and HSFO prices were less at the ports closer to these production locations. However, the more complex refineries of the ME region and the Asia-pacific, with a high yield of middle distillates will be in a better position to produce blending components for 0.5% Sulphur fuel. Fujairah's proximity to these producers will improve its position on LSFO price among the world's bunker hubs. [Link](#)
 The price differential between HSFO and LSFO becomes the most important factor while choosing a bunker fuel by the vessel operators and owners. ²⁷

To demonstrate the significance of the price gap, consider that fitting a scrubber to a VLCC costs USD6 million, with a lifespan of 15 years. The table below presents four scenarios reflecting varying price differentials between LSFO and HSFO²⁸ [Link](#)

Scenario analysis for large container ships with >80 MT per day of bunker fuel consumption

When the price differential is 20% (Scenario 1) and the price of HSFO is relatively low, the scrubber payback period is just 3.3 years. If the price of HSFO drops, but the differential remains the same (20%), the payback is pushed out to 4.2 years. If the price differential grows to 50%, the payback reduces to 1.3 years (Scenario 3); and if there is no price difference between HSFO and LSFO, then there is no price advantage and therefore zero payback (Scenario 4).

Scenario analysis for large container ships with >50 MT per day of bunker fuel consumption

When the price differential is 20% (Scenario 1) and the price of HSFO is relatively low, the scrubber payback period is just 3.7 years. If the price of HSFO drops, but the differential remains the same (20%), the payback is pushed out to 4.7 years. If the price differential grows to 50%, the payback reduces to 1.5 years (Scenario 3); and if there is no price difference between HSFO and LSFO, then there is no price advantage and therefore zero payback (Scenario 4).

Scenario analysis for large container ships with >65 MT per day of bunker fuel consumption

When the price differential is 20% (Scenario 1) and the price of HSFO is relatively low, the scrubber payback period is just 2.8 years. If the price of HSFO drops, but the differential remains the same (20%), the payback is pushed out to 3.6 years. If the price differential grows to 50%, the payback reduces to 1.1 years (Scenario 3); and if there is no price difference between HSFO and LSFO, then there is no price advantage and therefore zero payback (Scenario 4).

Table 3.6: Scenario analysis of bunker fuel consumption

Scenario analysis for large container ships with >80 MT per day of bunker fuel consumption

	Scenario 1: Base Case	Scenario 2: Lower conventional fuel cost	Scenario 3: Higher low-Sulphur fuel cost	Scenario 4: Lower low-Sulphur fuel cost
Scrubber Installation Cost	USD6 million	USD6 million	USD6 million	USD6 million
HSFO cost per MT	USD450	USD350	USD 450	USD450
LSFO cost per MT	USD540	USD420	USD 675	USD450
Difference	USD90 (20%)	USD70 (20%)	USD225 (50%)	USD 0 (0%)
Cost saving (lifecycle/interval of 15 years)	USD27.2 million	USD21.1 million	USD68 million	0
Annual saving	USD1.8 million	USD1.4 million	USD4.5 million	0
Payback	3.3 years	4.2 years	1.3 years	0

Note: In these calculations, the daily fuel consumption of the vessel is assumed to be 80 MT, with an annual activity level of 70 percent.

Source: Hempel, D&B Analysis

Table 3.7: Scenario analysis of bunker fuel consumption

Scenario analysis for large container ships with >50 MT per day of bunker fuel consumption

	Scenario 1: Base Case	Scenario 2: Lower conventional fuel cost	Scenario 3: Higher low- Sulphur fuel cost	Scenario 4: Lower low-Sulphur fuel cost
Scrubber Installation Cost	USD6 million	USD6 million	USD6 million	USD6 million
HSFO cost per MT	USD450	USD350	USD 450	USD450
LSFO cost per MT	USD540	USD420	USD 675	USD450
Difference	USD90 (20%)	USD70 (20%)	USD225 (50%)	USD 0 (0%)

Cost saving (lifecycle/interval of 15 years)	USD25 million	USD19 million	USD62 million	0
Annual saving	USD1.6 million	USD1.3 million	USD4.1 million	0
Payback	3.7 years	4.7 years	1.5 years	0

Note: In these calculations, the daily fuel consumption of the vessel is assumed to be 50 MT, with an annual activity level of 70 percent.

Source: Hempel, D&B Analysis

Table 3.8: Scenario analysis of bunker fuel consumption

Scenario analysis for large container ships with >65 MT per day of bunker fuel consumption

	Scenario 1: Base Case	Scenario 2: Lower conventional fuel cost	Scenario 3: Higher low-Sulphur fuel cost	Scenario 4: Lower low-Sulphur fuel cost
Scrubber Installation Cost	USD6 million	USD6 million	USD6 million	USD6 million
HSFO cost per MT	USD450	USD350	USD 450	USD450
LSFO cost per MT	USD540	USD420	USD 675	USD450
Difference	USD90 (20%)	USD70 (20%)	USD225 (50%)	USD 0 (0%)
Cost saving (lifecycle/interval of 15 years)	USD32 million	USD25 million	USD80 million	0
Annual saving	USD2.1 million	USD1.7 million	USD5.3 million	0
Payback	2.8 years	3.6 years	1.1 years	0

Note: In these calculations, the daily fuel consumption of the vessel is assumed to be 65 MT, with an annual activity level of 70 percent.

Source: Hempel, D&B Analysis

- **MGO:** Apart from LSFO, the next alternative source of low Sulphur fuel for shippers is MGO. The advantage with MGO is that most of the shippers can quickly shift to MGO with minimal operational change and no significant fixed investment or service time going into installation of scrubbers. However, the huge price differential of USD 250-350 per MT between LSFO and MGO, switching to marine gasoil would significantly increase fuel costs for shippers.
- **Other fuel oils:** Due to shift from HSFO to LSFO, marine industry would demand more of Low TBN Lubricants (total base number) as these are more compatible with the specifications of acid formation of LSFO. ²⁵ [Link](#) [Link](#)

Needs and expectation metrics:

	Needs	Expectations	Decision parameters for vendor selection
Underlying theme	Basic necessity	Buyers' perception of product, services, brands and support that they hold in their mind	Factors that ship operators evaluate for vendor section
Level of requirements	Must have	Mix of must and good to have	Evaluation parameters across needs and expectation elements

- 1) Timely fueling
- 2) Right grade of fuel
- 3) Cost effective
- 4) Compliant fit
- 5) Volume accuracy
- 6) Unadulterated fuel
- 7) Flexibility in supply
- 8) Technical support
- 9) Communication and customer service
- 10) Supply chain efficiency
- 11) Credit terms
- 12) Auxiliary offerings
- 13) Environmental sustainability
- 14) Insurance and liability
- 15) Risk management
- 16) Up-to-date technology

Primary factors

Basic necessity elements are considered default needs, which in case has to be there as per the customers

Expectation factors: These factors are primary the decision-making parameters

Parameters described under the expectations are highly discussed among customers while making decision

3.3 Customer needs and expectations

While bunkering customers have some basic mandatory needs, which are essential to the functioning of their operations, they also have wide expectations from their vendors. Customer expectations are more than the usual customer needs.

All in all, bunkering customers look for the least hassle and hope for good end-to-end bunkering solution experience.

“Customers want less of headache; your commitment to timely delivery is the most important perspective from the customer point of view. It is also very important from the supplier point of view, as any delay in the scheduled delivery can choke the scheduled of upcoming ones, bringing dissatisfaction among customers” – Industry expert

3.3.1 Customer need

Actual needs of bunker customers revolve around securing the right oil at reasonable prices, taking in consideration the compliance needs. Basis the industry experts, defined list of actual needs are listed below:

Table 3.9: Needs of Bunker end consumers

Broader area	Key needs	Description	Customer challenges	Solution / remedy
Core need	Timely filling of oil for the ship operations	Timely filling of bunker oil is the most important need enabling the smooth operation of ships	Cases of delay in oil delivery in ships has cascading effect on the ship operation	- Long terms relationship-based contracts between bunker fuel supplier and vessel operator - Provision for prioritized berthing and fueling for regular customers - Utilization of Marine fuel intelligence tools which provide live bunker prices, quality and availability from ports across the globe helps in effecting procurement and inventory management during shortage of fuel - Suppliers can partner with the third-party sampling labs locally to reduce the turnaround time of bunker fuel delivery, as most of the customers prefer buying bunker fuel post analyzing the bunker report verified by a recognized third-party lab - Suppliers should have quick dispute resolution mechanism which will save customer's voyage time in case of any disputes related to quantity or quality

Broader area	Key needs	Description	Customer challenges	Solution / remedy
Grade need	Right grade of fuel	Abiding by the need of the ship fuel, and defined regulations, right grade is decided	Other than Sulphur content, new compliant fuel categories not able to provide consistent viscosity and stability	- Efficient cleaning of VLSFO and LSFO is required to achieve reliable and economical ship operation - Separator systems can be used to ensure proper fuel treatment and maximum engine protection. For e.g.: Alfa Laval ALCAP separator system with automatic flow control has proven to be an effective solution. ²⁹ Link - Suppliers can partner with technology providers which can install separation systems used for effective treatment of LSFO and VLSFO
Costing need 30 31 Link , Link	Competitive pricing	Customers seek competitive and transparent pricing for bunkers, considering market rates and any applicable taxes or fees	Customers report lack of transparency in pricing	- Involved discussions with customers to convince them about pricing - Agile fuel pricing: Fixed price agreements between supplier and vessel operators gives the opportunity to the vessel owner to know their fixed fuel cost regardless of the daily fluctuating bunker oil prices - Utilization of bunker price optimization tools which can optimize bunker fuel costs for both suppliers and buyers. Also, it can save administrative costs and save time through simplified and transparent procedures. For e.g.: BCG bunkering optimization tool
Compliance need 32 33 Link , Link	Compliant fit	Bunkering suppliers must adhere to local and international regulations governing the bunkering industry, including environmental standards and safety protocols	Meeting and adhering to constantly evolving international and local regulations, especially those related to emissions and sulfur content, can be challenging	- The UAE is working closely with IMO's member states to help prepare a proposal dealing with Maritime Autonomous Surface Ships (MASS) regulations under the supervision of the Maritime Safety Committee and Legal Committee - Suppliers should train their employees over MASS regulations
Transparency 34 35 36 37 Link , Link , Link , Link	Volume accuracy	Transparency in universal quantity measurements so that delivered fuel is as per the contract	Cases of theft and overstated consumption due to manual measurements <i>"The vast majority of quantity measurement is done by rolling a tape down into a cargo tank of a</i>	- Application of Mass Flow Meter (MFMs) that measures the quantity, mass, and density of the fuel - Inspection of fuel oil during the barge, showcasing the customer that no such practices has been undertaken, which will build trust in them - The port authority can mandate the usage of mass flow meters while the suppliers can take that initiative at their end to cultivate more trust with the vessel operators

Broader area	Key needs	Description	Customer challenges	Solution / remedy
			bunker vessel and measuring the height of the fuel line before and after delivery. It's manual, it's archaic, and it's highly subject to manipulation and human error." - Tyler Baron, CEO of physical bunker supplier Minerva Bunkering Cappuccino effect: Introduction of air into the supply line during and/or after bunkering operations to exaggerate high volumetric flow meter readings As per the data from Blue Insight consultancy, short bunker supplies at Fujairah have costed buyers, owners and charterers an estimated USD 100 million in year 2021.	
	Unadulterated fuel	Ensuring the quality of bunkers is a significant challenge. Poor-quality fuel can lead to engine issues, increased maintenance costs, and	Cases of poor-quality bunker oil impacts ships performance by clogging and damaging ships engine	- Purchasing oil from trusted vendor in full transparency of oil quality - Suppliers should bring more transparency during real time sampling of bunker fuel from barge manifold during bunkering operations at anchorage position - Suppliers can partner with third party fuel verification labs which will help in generating quick bunker reports. This will help them to reduce bunker delivery time and win

Broader area	Key needs	Description	Customer challenges	Solution / remedy
		compliance concerns		customer's trust over consistent fuel quality and clean track record

Source: S&P Global, Intertek, Argus Media

3.3.2 Customer expectations and associated challenges

Customer expectations exceed that of their actual needs of mere getting the bunker fuel. In fact, expectations of customers play a key role in decision making of selecting a vendor for bunker oil.

Below is a list of key customer expectation in bunkering oil space, along with the challenges they face in day-to-day business.

Table 3.10: Customer expectations

Broader area	Expectations	Description	Customer challenges	Solution / remedy
Product 35 36 37 Link , Link , Link	Transparent fuel sampling mechanism and consistent good fuel quality	Customer companies expect the same quality of fuel grade as provided to them in the sample while decision making	There have been cases of difference of actual delivered oil of inferior quality than the sample provided initially Ensuring the quality of bunkers is a significant challenge. Poor-quality fuel can lead to engine issues, increased maintenance costs, and compliance concerns	Regular checks on the actual supplied oil across the value chain Provide transparency and training regarding composition of the bunker fuel taken on board Detailed information of the stored and supplied fuel batches Obligation for parties in the supply chain to keep a record of the origin and composition of the blend components and delivered fuel, which will enforce accountability in the event of an incident Setting up of a customer portal where inferior fuel batches can be reported anonymously Representative sampling during the filling up of bunker barges from quality control perspective Commonly used sampling devices during fuel oil bunker transfer are: • Manual and online drip sampler • Flow proportional automatic sampler • Time proportional automatic sampler
	Risk management	Fluctuations in oil prices can have a significant impact on operating costs	Predicting and managing these price changes is a constant challenge	Long term contracts between fuel supplier and vessel owners allows the supplier to provide Bunker fuel at relatively competitive prices

Broader area	Expectations	Description	Customer challenges	Solution / remedy
Supply chain 38 Link	Flexibility in supply	for shipping companies Bunkering suppliers should be able to accommodate varying quantities and delivery schedules to meet the needs of shipping companies	Disruptions in the supply chain, such as geopolitical tensions, natural disasters, or infrastructure issues, can lead to delays and impact the timely delivery of bunkers	Daily real time knowledge of arising geopolitical tensions Alternate arrangements with the other local refiners or storage terminal operators in case of any force majeure risk
	Efficient delivery process	Shipping companies value bunkering suppliers with efficient logistics and a well-organized supply chain to ensure timely deliveries	Communication challenges between brokers, suppliers, shipowners and port authorities Lack of standardized documentation processes and manual errors Lack of real time and accurate information about fuel availability, quality and prices	Real time collaboration and information sharing through digital tools improves decision making and reducing response time Implementation of automated data integration systems reduces manual errors Shifting entirely into real-time cloud-based collaboration tools which enables real time data sharing, centralized access to information for all stakeholders in the supply chain and efficient document sharing Digitizing the documentation processes can significantly enhance communication workflows. Replacing paper-based documents with electronic formats reduces the chances of human errors, allows for easier sharing, and enables efficient tracking of important information. Advanced technologies, such as blockchain, can also be utilized to ensure the authenticity and immutability of crucial documents, reducing disputes and legal complications
Support 32 Link	Technical Support	Availability of technical support and expertise is crucial, especially for vessels that may require specific types of fuel or have unique	Lack of customer knowledge about unique engine requirements, and other technical specifications	Implementation of data integration tools and blockchain technology helps the suppliers to profile their customers providing accurate technical services in terms of fuel grade, composition and engine specifications Provide additional technical advice over fuel usage reduction options etc. on top of the core

Broader area	Expectations	Description	Customer challenges	Solution / remedy
Support		operational considerations		bunkering technical support. For E.g.: Good hull coating minimizes speed loss and reduces fuel consumption
	Communication and customer Service	Effective communication and excellent customer service are essential. Shipping companies appreciate suppliers who can provide accurate and timely information and respond promptly to inquiries or issues	Lack of integrated platforms for real time and accurate data sharing	Define clear communication channels among stakeholders, ensuring swift and accurate information flow Regular meetings and well define reporting and accountability structure between suppliers and ship owners Promote data-sharing initiatives among stakeholders to achieve greater transparency and efficiency. Integrating data from various sources enables informed decision-making and improved supply chain visibility Provide training programs to educate stakeholders about efficient communication practices, new technologies, and industry regulations
	Credit terms	Flexible and reasonable credit terms are often a consideration for shipping companies, as they may prefer payment arrangements that align with their financial planning	Managing credit terms, payment schedules, and financial transactions can be complex, especially when dealing with large quantities of bunkers and fluctuating market prices	Suppliers can implement financial management tools of Bunker supply management software for better financial planning
Financing				
40				
Link				
Services	Value added services (Auxiliary services)	Many of the bunkering suppliers offer a range of related services, such as lubricants and additives, risk management, consultancy, and more to be preferred as a one-stop solutions provider	Customers may face challenges with sole bunkering oil provider, as for other accessory services related to bunkering, they need to spend time and money from other places in port	Wider range of value-added services besides will ensure increased customer satisfaction and repeat business Suppliers can organize various technical trainings to ship crew such as changing over procedure from HFO to LSFO. This will help build more trust with ship owner/operator
	Insurance and liability	Bunkering suppliers must have appropriate insurance	Managing the paperwork and documentation required for	Digitized documentation-based solution can help overcome the
Services				
41				

Broader area	Expectations	Description	Customer challenges	Solution / remedy
Link Environmental 42 43 Link , Link		coverage and clear liability policies to address any potential incidents or accidents during bunkering operations	bunkering operations, including compliance reports, can be time-consuming and may require additional resources	issue related to time wastage due to manual compliance reports
	Environmental sustainability	With the increasing focus on environmental sustainability, shipping companies may prefer bunkering suppliers that provide environmentally friendly fuel options and adhere to green initiatives	Meeting environmental sustainability goals and managing the environmental impact of bunkering operations can be challenging	Suppliers should create bunkering checklist and training programs for their employees to avoid any leakage and pressure fluctuation issues during bunkering operations

Source: GulfNews, safety4sea, Mondaq, Hellenic shipping news, Argus media, LinkedIn

4 Competition Landscape

4.1 Overview

Bunkering fuel competitive landscape structure in Fujairah

Landscape: As of 2023, Fujairah's bunker fuel market exhibits a semi-fragmented landscape with

- 13 licensed bunker fuel suppliers
- 16 storage terminal operators
- 2 refining operations

The top 6 players, namely:

- Akron Trade and Transport
- Oil Marketing and Trading International
- Vitol Bunkers
- Montfort Trading
- Geo Marine LLC
- Al Bakri Bunkering (International Supply)

Collectively control ~70% of the Fujairah's Bunker fuel supply market

Other significant contributors include:

- Medieastern Energy (Al Shareq Al Matwaset Oil Services LLC Fujairah Branch)
- Pearl Marine Logistics
- Al Arabia Bunkering
- Peninsula Fuel Supply
- Minerva Bunkering
- GPS Bunkers
- PetroChina holding

Holds market share of more than 30% of the Fujairah's Bunker fuel supply market

4.2 Competitive landscape evolution

In 2017, there were 16 bunker supply companies officially licensed to operate in the Fujairah market. However, variations in activity levels among these companies were observed, with some being notably more active than others. Additionally, a few of them might not have been active at all during that period.

- **Historical scenario:** In 2017, Al Bakri Bunkering (operating as International Supply), Oil Marketing & Trading International LLC, GP Global, and Chemoil Middle East DMCC (operating under the Fujairah government-owned entity Gulf Petroleum Supplies) were reported as the leading suppliers in terms of volume. Aegean (Fujairah) Bunkering S.A. initially held a prominent position as one of the largest volume suppliers until mid-2017. However, subsequently, they scaled down their presence in the market, reducing the number of barges from seven to just one.
- **Current scenario:** Few participants from the 2017 landscape have risen to prominence as major players in 2023. These include Akron Trade and Transport, Al Arabia Bunkering, Al Bakri Bunkering (International Supply), Oil Marketing and Trading International, Vitol Bunkers, Montfort Trading FZE and GEO Marine LLC.

4.2.1 Mature vs. Emerging players – Fujairah

Figure 4.1: Competitive landscape in Fujairah

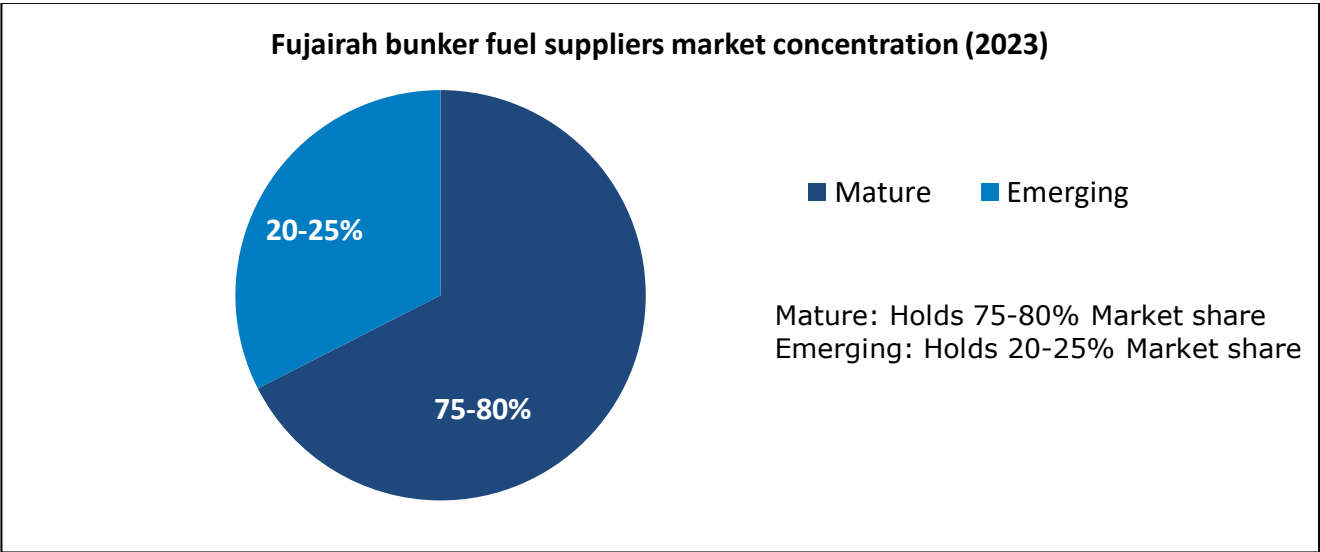


Source: Competitor website, Primary research, Hempel, D&B Analysis

4.2.2 Market share dynamics

Mature players such as Akron Trade and Transport, Al Bakri Bunkering (International Supply), Oil Marketing and Trading International, Vitol Bunkers, Montfort Trading FZE, Al Arabia Bunkering and GEO Marine LLC collectively control 75-80% of the Fujairah’s bunker fuel market. However, the emerging suppliers such as Mideastern Energy, Minerva, Peninsula and Pearl Marine Services etc. control 20-25 % of the market.

Figure 4.2: Fujairah bunker supplier concentration levels



Source: D&B Analysis

4.2.3 Market share and international presence – Fujairah bunker fuel suppliers

Table 4.1: Market share of Bunker suppliers in Fujairah

Market share and international presence - Fujairah suppliers				
Suppliers	Headquarters	Fujairah Market share (2023)	Geographical outreach	Global Rank
Akron Trade and Transport LLC	Fujairah, UAE	17-18%	Gulf, East & West African regions	-
Oil Marketing & Trading International LLC	Fujairah, UAE	15-16%	UAE, Greece, Singapore	-
Vitol Bunkers (s) Pte Ltd - Fujairah branch	Rotterdam, Netherlands	9-10%	Turkey, China, Singapore, Australia, the UAE, North West Europe (ARA), the Caribbean and the US Gulf Coast.	4
Montfort Trading FZE	Geneva, Switzerland	9-10%	UAE, South Africa, Mozambique, East Africa, Singapore	-
G.E.O Marine LLC	Fujairah, UAE	9-10%	18 Ports across the globe	-
Al Bakri Bunkering (International Supply)	Jeddah, Saudi Arabia	8-9%	GCC countries	-
Al Arabia Bunkering	Sharjah, UAE	7-8%	Khorfakkan, Sharjah and Kalba	-
Mideastern Energy (Al Shareq Al Matwaset Oil Services LLC Fujairah Branch)	Dubai, UAE	5-7%	UAE	-
GPS Bunkers	Fujairah, UAE	5-6%	UAE Ports	-
Pearl Marine Logistics UAE FZE	Fujairah, UAE	4-5%	UAE	-
Minerva (Fujairah) bunkering S.A Fujairah branch	Geneva, Switzerland	4-5%	50+ countries	13

Petrochina International (Middle East) Company Limited - Fujairah branch	Dubai, UAE	2-3%	UAE (Fujairah branch)	2
Peninsula Fuel Supply LLC - Fujairah branch	London. The UK	2-3%	UAE	-

Source: Primary research, D&B Analysis

4.2.4 Fujairah's bunker fuel supplier presence in other ports of UAE

With a clear majority, nearly three-fifths of these players are focused only on the Fujairah port for bunkering within the UAE, leaving behind about two-fifths who also have a presence in other ports.

Table 4.4: Details of leading bunker suppliers in Fujairah

Fujairah key bunker fuel suppliers (presence in other ports of UAE)		
Suppliers	Presence in UAE ports	Ports served in UAE
Akron Trade and Transport LLC	1	Fujairah
Al Arabia Bunkering	3	Fujairah, Kalba, Sharjah and Khorfakkan
Medieastern Energy (Al Shareq Al Matwaset Oil Services LLC Fujairah Branch)	1	Fujairah
GPS Bunkers	2	Fujairah and Khorfakkan
G.E.O Marine Llc	1	Fujairah
Al Bakri Bunkering (International Supply)	1	Fujairah
Minerva (Fujairah) bunkering S.A Fujairah branch	1	Fujairah
Montfort Trading FZE	1	Fujairah
Oil Marketing & Trading International LLC	4	Ajman, Dubai, Fujairah, Jebel Ali, Khorfakkan, Mina Saqr, Ras Al Khaimah, Sharjah
Pearl Marine Logistics UAE FZE	1	Fujairah
Petrochina International (Middle East) Company Limited - Fujairah branch	2	Fujairah and Khorfakkan
Peninsula Fuel Supply LLC - Fujairah branch	1	Fujairah
Vitol Bunkers (s) Pte Ltd - Fujairah branch	3	Fujairah, Jebel Ali, Khorfakkan

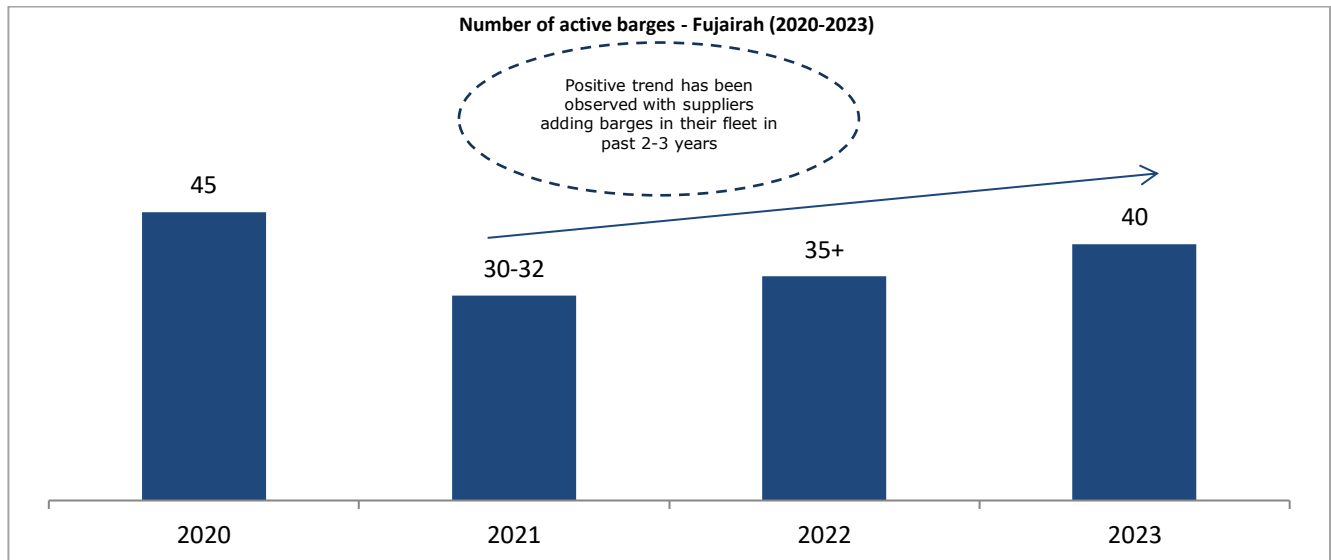
Source: Competitor website, Primary research

4.2.5 Fujairah barging capacity

With the stabilization of bunker fuel demand at Fujairah in 2021, the port experienced a shortage of barges for refueling vessels. Some stakeholders had decreased their barge presence at the port in 2020 due to low premiums and OPEC production cuts in the area. In 2021, the number of active barges was approximately 35-40, compared to 45 in 2020. Additionally, in 2021, bunker suppliers began imposing an extra premium of USD 2 per MT for shorter delivery times of two to three days instead of five days. [Link](#)

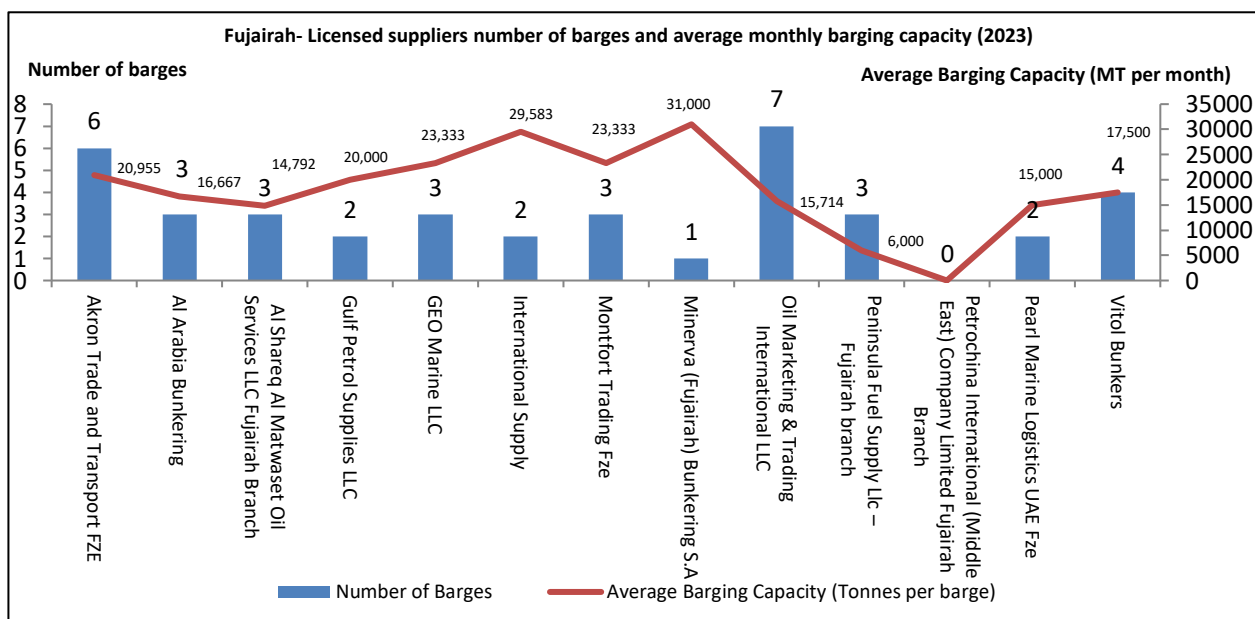
In 2022, few suppliers such as Warm Seas Development and Victoriam were banned due to UAE maritime violations due to which at least three barges were reduced from the port. [Link](#)

Figure 4.5: Barges availability at Fujairah



Source: Primary research, D&B Analysis

Figure 4.6: Bunker supplier wise barge capacities



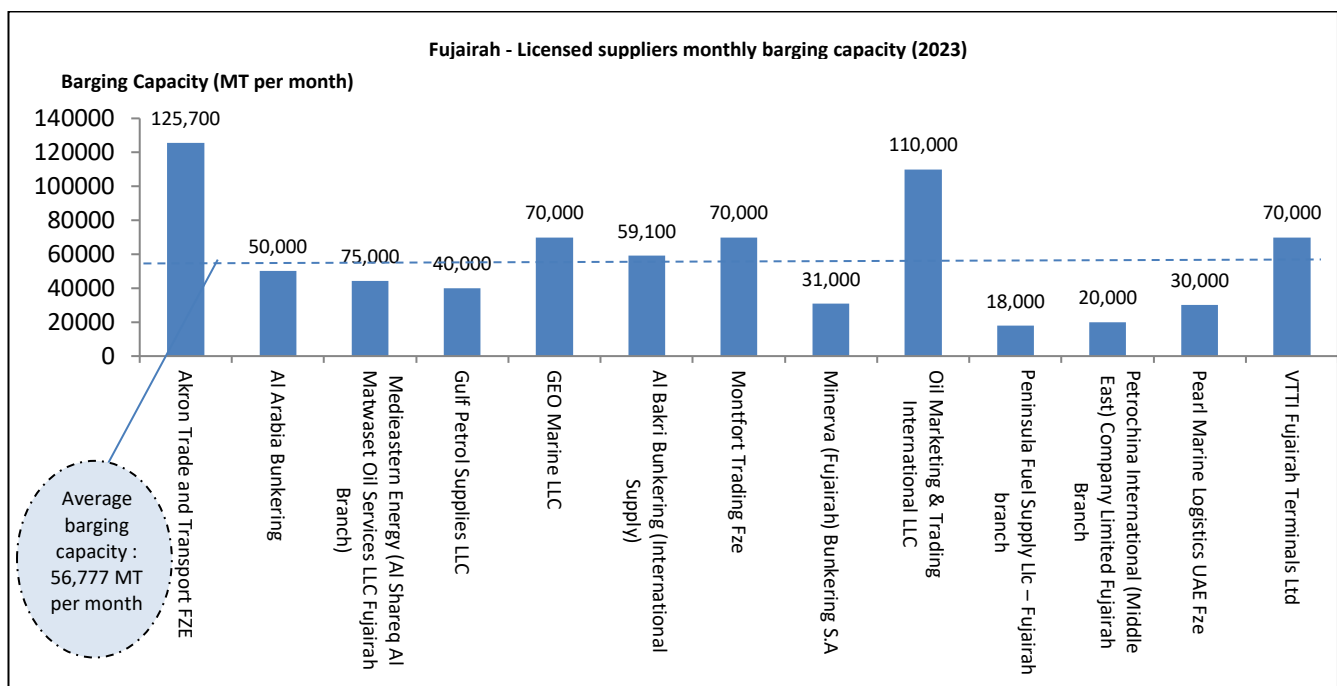
Source: Competitor website, Primary research, D&B Analysis

Note: Petrochina does not own barges, however supplies bunker fuel at Fujairah port

Owing to existing geopolitical tensions and recent assaults on commercial barges in the Red Sea, a vital trade route connecting Europe and North America to Asia via the Suez Canal, exporters and shippers are actively seeking alternative maritime routes. Consequently, ports such as Fujairah, Durban, Port Louis, the Canary Islands, and Salalah are experiencing a surge in demand as ships are redirected to the region.

Shortages of active barges are evident: Currently, Fujairah boasts around 40 active barges and has 13 licensed fuel suppliers in the region. Despite this, they have been unable to meet the demand, resulting in a **waiting period of 5 to 7 days**. Recognizing the escalating demand, key players such as Peninsula, Minerva, and VTTI have expanded their fleets by adding barges. **The current total barging capacity is approximately 700-800 TMT per month, with an average capacity per supplier hovering around 56 TMT per month.**

Figure 4.7: Bunker supplier-wise barging capacity per month



Source: Competitor website, Primary research, D&B Analysis

To address the growing and steady demand, the Fujairah port is in the process of adding new jetties dedicated to servicing liquid bulk vessels, marking the first expansion since 2017. Currently equipped with nine primary liquid bulk jetties, along with a VLCC jetty, the port started incorporating two more jetties in 2021. Authorities expected a surge in the number of oil tankers, expecting 4,800 vessels to call at the port in 2023 compared to the 4,720 recorded in 2022. ⁴⁴ [Link](#)

Table 4.7: List of Licensed Bunker and Lubricant supplier in Fujairah

Licensed bunker suppliers providing compliant fuels Link			
Akron Trade and Transport LLC		Petrochina International (Middle East) Company Limited	
GPS Bunkers		Mideastern Energy (Al Shareq Al Matawaset Oil Services LLC)	

Licensed bunker suppliers providing compliant fuels Link			
Al Bakri Bunkering (International Supply)		Minerva (Fujairah) Bunkering S. A	
Oil Marketing & Trading International LLC		Al Arabia Bunkering	
Vitol Bunkers (s) Pte Ltd - Fujairah branch		Montfort Trading FZE	
Pearl Marine Logistics UAE FZE		G.E.O Marine Llc	
Peninsula Fuel Supply LLC - Fujairah branch			

Source: Fujairah port authority

Licensed lubricant suppliers- Fujairah Link	
Chevron Al Khalij	 
International Tankers Enterprises Co.	 
Gulf Oil Middle East Ltd.	
Petro Gulf Manufacturing	

Source: Fujairah port authority

4.2.6 Product portfolio by supplier – Fujairah

VLSFO (<0.5%) and HSFO (<3.5%) appear to be the most common grade supplied by the Bunker fuel suppliers at Fujairah. LSMGO (<0.1%) and MGO (<0.5%) are the most prevalent grades among the Marine Gas oil product.

Below are the key bunker fuel suppliers operating at the Fujairah port along with their grade wise supply capabilities.

Table 4.8: Bunker supplier wise product portfolio

<u>Bunker fuel - Licensed suppliers of Fujairah port</u>	MGO (<0.1%)	MGO (<0.5%)	LSMGO (<0.1%)	HSFO (<0.5%)	VLSFO (<0.5%)
Akron Trade and Transport FZE					
Al Arabia Bunkering					
Mideastern Energy (Al Shareq Al Matwaset Oil Services LLC Fujairah Branch)					
Gulf Petrol Supplies LLC					
GEO Marine LLC					
Al Bakri Bunkering (International Supply)					
Montfort Trading FZE					
Minerva (Fujairah) Bunkering S.A					
Oil Marketing & Trading International LLC					
Peninsula Fuel Supply LLC – Fujairah branch					
Petrochina International (Middle East) Company Limited Fujairah Branch					
Pearl Marine Logistics UAE FZE					
VTTI Fujairah Terminals Ltd					

Source: Competitor website, Primary research

4.2.7 Storage terminal operators – Fujairah

- Significant volumes of Russian crude oil and refined products have elevated tank storage levels in Fujairah, UAE, resulting in record-high storage fees at this transit and blending hub. Challenges related to Western insurance and ship access have compelled certain traders to blend Russian crude with other resources at storage facilities like Fujairah before proceeding with re-export
- Throughout the years, Fujairah has grown into a prominent hub for storage, blending, and ship-to-ship transfers, due to its strategic positioning near the entrance of the Strait of Hormuz. The strong demand for storage is positively impacting the revenues of operators like Vopak, Vitol's VTTI, Emirates National Oil Company, and Abu Dhabi National Oil Company⁴⁵ [Link](#)
- Demand for storage space is also attracting buyers for these assets. Mercuria Energy group is in the process of evaluating expressions of interest from potential buyers for its storage terminal at Fujairah while GP Global's terminal at Fujairah was sold for about USD 124 million ⁴⁶ [Link](#)
- Below are the storage terminals operators at Fujairah along with their storage capacities.

Table 4.9: Storage Terminal operators at Fujairah

Storage terminals in Fujairah		
S No.	Storage terminal operators	Storage terminal capacity (cubic meters)
1	VOPAK Horizon Fujairah Limited	21,00,000
2	VTTI Fujairah	20,00,000
3	Mount ROW Terminal	10,00,000
4	BPGIC	6,01,600
5	GPS Chemoil	5,50,000
6	Fujairah Oil Terminal	1,17,700
7	Marine Oil Terminal	4,65,000
8	Mercuria Energy Group	4,65,000
9	Ecomar Energy Solutions	4,00,000
10	Emarat	3,83,000
11	GTI Fujairah	3,52,000

Source: Fujairah port authority, competitor website

Note: GPS Chemoil is a JV company which operates a storage terminal at Fujairah port

4.3 Key Competitors' Dynamics and Profiles

- The primary suppliers dominating Fujairah's bunker fuel market share include –
 - Akron Trade and Transport
 - Oil Marketing and Trading International
 - Vitol Bunkers
 - Montfort Trading
 - Geo Marine LLC
 - Al Bakri Bunkering (International Supply)
- These entities play significant roles within the bunkering value chain in Fujairah. They collectively supply between 18-125 TMT of bunker fuel per month to various stakeholders such as tanker fleet owners, container ship owners, bulk carrier owners, general cargo owners, and chemical tanker owners.
- All major suppliers operating in the Fujairah bunker fuel market engage in diverse activities across the value chain, which carries its strategic significance in terms of cost efficiency, customer retention, and real-time decision-making and strong customer relationships.
- Companies like Akron, GPS Bunkers, and Minerva have developed strategies focused on establishing their own dedicated facilities, including storage terminals, shipbuilding facilities, and pipeline networks connecting adjacent ports. These initiatives have enabled them to achieve cost efficiencies and expand their geographical presence in neighbouring ports of Fujairah, such as Khorfakkan and Sharjah based ports. Additionally, they have secured long-term contracts with refineries to ensure a steady and uninterrupted supply of bunker fuel.
- Suppliers like Montfort and Vitol operate their own refineries, allowing them to produce Low Sulphur Fuel Oil and Marine Gas Oil locally. This ownership of refineries aids in reducing their fuel-sourcing expenses and enables them to meet the precise fuel specifications required by their customers. Additionally, many of these companies offer value-added services to retain their customer base. These services encompass ship agency services, oil trading services, ship repair and maintenance, provision of fresh water, arranging supply and

crew boats, delivery services, garbage collection and disposal services, bunker survey and marine survey, as well as providing market knowledge to their customers in a timely manner.

Competitor's profile #1

Akron Trade and Transport	
	HQ: Fujairah, UAE Founded: 1997 Employees: Details not available Market share: ~16-17% in UAE Rev: ~USD 20 million (2022) Capacity: 125,700 MT per month Website link: http://www.akron.ae/index.html
Role in the value chain	Bunker supplier/ Ship management/ Oil trading
Key products and service offerings	Purchase of Marine fuel oil cargoes and physical supply to vessels calling at east coast ports of UAE Manages a fleet of product tankers / IMO Type 2 chemical tankers of cargo capacity ranging from 5000 MT to Suez Max size Types of grades in oil: MGO (<0.5%), MGO (<0.10%), LSMGO (<0.10%), HSFO (<3.5%)
Bunkering operating metrics	Supplying capacity: 125,700 MT per month Number of barges: 6
Target geographies	Gulf, East and the West African regions
Target customers	Oil companies, Ship owners/ Managers/ Charterers, Bunker traders and brokers
USPs / strengths	• Captive fleet of tankers • Long term contracts with leading refineries in the MENA region • Real time information on trends and prices • Owns a shipyard in China (Haidong Shipyard) • Owns a subsidiary named as Emepco FZE which has been building ships since 2002
Positioning	Falls under one of the top 5 suppliers at Fujairah
Value proposition	Market leader in ship management services at Fujairah

Competitor's profile #2

Oil Marketing and Trading International	
	HQ: Dubai, UAE Founded: 2001 Employees: Details not available Market share: ~15-16% In UAE Rev: Details not available Capacity: 1,10,000 MT per month Website link: https://oil-marketing.com/ (Under Maintenance)
Role in the value chain	Bunkering, Oil Trading, Chartering, and Barge Operator
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: MGO (<0.10%), IFO 380 CST Fuel Oil, IFO 180 CST Fuel Oil

	Bunkering operations: Operations are based out at all ports of UAE including Fujairah and UAE
Bunkering operating metrics	Supplying capacity: 1,10,000 MT per month Number of barges: 7
Target customers	Oil Companies, Ship owners/ Managers/ Charterers, Bunker Traders and Brokers
Positioning	Falls under one of the top 5 suppliers at Fujairah
Value proposition	Comprehensive set of services including bunker barging facilities, owned storage terminal, ship agency, ship repair and maintenance network

Competitor's profile #3

Vitol Bunkers / VTTI Fujairah Terminals Ltd (VTTI FTL) FZC	
	HQ: Rotterdam, Netherlands Founded: 2006 Employees: ~600 Market share: ~9-10% In UAE Rev: Details not available Capacity: 70,000 MT per month Website link: https://www.vtti.com/
Role in the value chain	Bunker fuel and lubricant supply services, Refinery operations (82,000 b/d)
Key products and service offerings	Products: Crude, Gasoline, Jet fuel, Kerosene, Naphtha, Gasoil, Diesel, LPG, Fuel oil, VGO Fuel type managed in Fujairah: - VLSFO - LSMGO Bunkering operations: Terminal core activities include - transportation, storage, loading and offloading of oil and petroleum products for its wide customer base. Storage terminals are managed by VTTI and bunker operations are managed by Vitol bunkers, its subsidiary
Bunkering operating metrics	Supplying capacity: 70,000 MT per month Number of barges: 4 Total supply of bunkers per year: ~1,800 Refining capacity: 82,000 b/d 2 captive storage terminals are connected to the Fujairah refinery Limited (FRL) refinery Available storage capacity at Fujairah: 2 million cubic meters # of Jetties: 11 # of Tanks at Fujairah: 66 Annual fuel trading volume: 8 million barrels Total Storage terminals owned globally: 17 (15 countries) Global storage terminal capacity: >10 million cubic meters
Target geographies	Vitol's global bunker supply network now covers Turkey, China, Singapore, Australia, the UAE, North west Europe (ARA), the Caribbean and the US Gulf Coast.
Target customers	Diverse group of ocean-going and coastal ship operators and marine fuel traders, brokers and other end-users of marine fuel and lubricants.

	• Shell • British Petroleum • Maersk
Supplier names	FRL refinery
USPs / strengths	• Provides additional services apart from bunker fuel supply such as Agency services, Bunker and Marine surveys • Vitol manages its own fleet of vessels through its Latvian-based technical management company, LSC, as well as through a Singapore-flagged fleet offering barging services
Positioning	Registered compliant fuel supplier at Fujairah
Value proposition	• VTTI is co-owned by Vitol group, IFM and ADNOC • VTTI has integrated operations setup at Fujairah port, providing access to refinery, storage terminals, fuel supply & delivery along with terminal management services like – Waste water treatment; vapour treatment; soil protection & remediation; compliance module • Vitol bunkers have delivered first bio fuels at Fujairah port • First operator to deploy electric-hybrid tankers Singapore harbour
Strategic focus	• VTTI wants to triple the company's pre-tax earnings by 2028, with half of the total coming from "transitional fuels" and new energies which include LNG • Vitol continues to expand its Biofuel delivery in UAE and abroad • Vitol continued to expand its presence in UAE, including Jebel Ali Port

Competitor's profile #4

Montfort Trading FZE	
	HQ: Geneva, Switzerland Founded: 2021 Employees: ~800+ employees globally Market share: ~9-10% In UAE Rev: Details not available Capacity: 70,000 MT per month Website link: https://mont-fort.com/
Role in the value chain	Global commodity trading and asset management
Key products and service offerings	Fuel products managed at Fujairah: • VLSFO • HSFO • LSMGO Operations: Manages a diverse portfolio of energy products and services including: Crude oil, Distillates, Gasoline, Fuel Oil, Bunkering, LPG, Naphtha, Grains & Oilseeds, metals & Dry Bulk.
Bunkering operating metrics	• Type of Bunkering operations: Trading, Storage and Logistics • Total # of Refuelling barges: 3 • Total supply capacity of barges: 70,000 MT per month • Annual Fuel volume sold at Fujairah: 30 million barrels of LSFO • Capacity of crude processing facility at Fujairah: 65,000 b/d • Marine fuels volume delivered worldwide in 2022: 955,000 MT
Target geographies	UAE, South Africa, Mozambique, East Africa, Singapore etc.

Target customers	Diverse group of ocean-going and coastal ship operators and marine fuel traders, brokers and other end-users of marine fuel and lubricants
Supplier names	FRL refinery
USPs / strengths	- Montfort has integrated bunkering operations systems setup at Fujairah port - Montfort is also investing in renewables like wind & solar energies
Positioning	Offers integrated services that include: Storage, shipping, retail, upstream, refining
Value proposition	Owns a 65,000 b/d crude oil processing unit which has the capacity to produce up to 5 MT per year of low sulphur marine fuel oil, which accounts for 50% of Fujairah market
Strategic focus	- Montfort has joined Energy LEAP, a strategic alliance committed to spearheading initiatives for operational efficiency in the energy industry - Acquired Uniper's UAE based crude oil processing and marine fuel trading business (Uniper Energy DMCC), market leader in the production and supply of IMO 2020 compliant low-sulphur fuel oils to the Fujairah market. The new JV company is named Fort energy refining Middle east DMCC

Competitor's profile #5

Geo Marine LLC	
	HQ: Istanbul, Turkey Founded: 2015 Employees: Details not available Market share: ~9-10% in UAE Rev: Details not available Capacity: 70,000 MT per month Website link: https://geomarinetrade.com/
Role in the value chain	Bunker supplier (Trading)/ Bunker operations/ Environmental solutions/ Technical consultancy
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: MGO (<0.10%), MGO (<0.5%), LSMGO (<0.10%), VLSFO (<0.5%) Bunkering operations: Operations are based out at all ports in ME region
Bunkering operating metrics	Supplying capacity: 70,000 MT per month Number of barges: 3
Target geographies	Middle East
Target customers	Oil companies, Ship owners/ Managers/ Charterers, Bunker traders and Brokers
USPs / strengths	Provides bunkering services at all major ports in ME region
Positioning	Falls under one of the top 5 suppliers at Fujairah
Value proposition	Comprehensive set of services including bunker barging facilities, bunkering operations, environmental and technical services
Strategic focus	The company provides environmental sustainability solutions in alignment with the global priority of reducing bunker fuel and lubricants consumption, exploring sustainable fuelling options for maritime shipping professionals, owners, charterers and ship operators

Competitor's profile #6

Al Bakri Bunkering (International Supply)	
	HQ: Jeddah, Saudi Arabia Founded: 1973 Employees: <i>Details not available</i> Market share: ~8-9% in UAE Rev: <i>Details not available</i> Capacity: 59,100 MT per month Website link: www.bakrienergy.com
Role in the value chain	Bunker supplier (Trading)/ Bunker operations/ Storage Terminals
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: LSMGO (<0.10%), VLSFO (<0.5%) Bunkering operations: UAE, Saudi Arabia and Singapore
Bunkering operating metrics	Supplying capacity: 59,100 MT per month 2 Barges (as of 2023)
Target geographies	Middle East and Singapore
Target customers	Oil companies, Ship owners/ Managers/ Charterers, Bunker traders and Brokers
USPs / strengths	Provides bunkering services at all major ports in UAE, Saudi Arabia and Singapore
Positioning	Falls under one of the leading suppliers at Fujairah
Value proposition	Comprehensive set of services including bunker barging facilities, bunkering operations and storage terminal operations
Strategic focus	The company expanded its storage terminal capacity in ME region and Singapore to capitalize over the financial benefits gained by increasing demand of compliant and low sulphur bunker fuel

Competitor's profile #7

Al Arabia Bunkering Company LLC	
	HQ: Sharjah, UAE Founded: 2011 Employees: <i>Details not available</i> Market share: ~6-7% In UAE Rev: <i>Details not available</i> Capacity: 50,000 MT per month Website link: https://alarabiabunkeringcompany.com/
Role in the value chain	Fuel supply services, Bunkering, Agency services, Fresh water & service provider of supply & crew Boat arrangements, Delivery services, Garbage collection / Disposal services, Bunker survey & Marine survey
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: MGO (<0.10%), VLSFO and HSFO Bunkering operations: Operations are based out at Fujairah
Bunkering operating metrics	Supplying capacity: 50,000 MT per month Number of barges: 3
Target geographies	Owns licenses to operate in Fujairah, Khorfakkan, Sharjah and Kalba

Target customers	Diverse group of ocean-going and coastal ship operators and marine fuel traders, brokers and other end-users of marine fuel and lubricants.
USPs / strengths	Provides additional services apart from Bunker fuel supply such as Agency services, Bunker and Marine surveys
Positioning	Compliant fuel supplier at Fujairah
Value proposition	- Long established relationship with major oil corporations, bunkers, shipping operators and energy providers - Also provides agency services to its customers apart from bunkering operations
Strategic focus	- Plans to expand and start supplies at all Abu Dhabi ports, as official distributors of ADNOC products, under the brand name Majestic Al Arabia Services LLC. - Customer acquisition and retention services through additional services such as agency services

Competitor's profile #8

GPS Bunkers	
	HQ: Fujairah, UAE Founded: 2015 Employees: <i>Details not available</i> Market share: ~5-6% in UAE Rev: <i>Details not available</i> Capacity: 40,000 MT per month Website link: https://www.gpsbunkers.ae/
Role in the value chain	Bunker supplier (Trading)/ Bunker operations/ Logistics and storage
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: MGO (<0.10%), MGO (<0.5%), LSMGO (<0.10%), VLSFO (<0.5%) Bunkering operations: Operations are based out at all ports of UAE including Fujairah and UAE
Bunkering operating metrics	Supplying capacity: 40,000 MT per month Storage capacity: 4.4 million barrels Number of barges: 2
Target geographies	Gulf, East and the West African regions
Target customers	Oil companies, Ship owners/ Managers/ Charterers, Bunker traders and Brokers
USPs / strengths	Provides bunkering services at all major ports of UAE
Positioning	Falls under one of the top 5 suppliers at Fujairah
Value proposition	Comprehensive set of services including bunker barging facilities, owned storage terminal, ship agency, ship repair and maintenance network
Strategic focus	Short term - Addition of two more bunker barges which will double the company's supplying capacity Long term - Creation of pipeline network connecting the main harbour in Hamriyah, UAE and covering other major ports such as Khorfakkan and Jebel Ali for physical bunker supplies The company has plans to move up the value chain by adding more and more terminals, refining and manufacturing units and trading offices worldwide with close synergies amongst all locations and business areas

Competitor's profile #9

Minerva (Fujairah) Bunkering S.A	
	HQ: Geneva, Switzerland Founded: 2014 Employees: ~700+ employees globally Market share: ~4-5% in UAE Rev: Details not available Capacity: 31,000 MT per month Website link: https://www.minervabunkering.com/
Role in the value chain	- Owns and operates storage facilities and bunkering logistics including bunker fuel supply - Financial, operational and environmental solutions (Lowering cost and risk, dedicated logistics, flexible financing and environmental impact solutions)
Key products and service offerings	Fuel products managed at Fujairah: - VLSFO - LSMGO - MGO Proprietary blending network, Equity production, exclusive refinery off take agreements, crude for refined product swaps, spot cargo trading, management of bunkering logistics
Bunkering operating metrics	- Type of bunkering operations: Trading, storage and logistics - Total # of refuelling barges: 1 - Total supply capacity of barges: 31,000 MT per month - Annual fuel delivery capacity: 25 MMT - Annual fuel sourcing volume: > 43 MMT - Both in sea and port bunkering capabilities - Storage capacity: 465,000 cubic meters
Target geographies	Bunkering operations at 50+ countries; 150+ ports; 52+ physical locations
Target customers	750 + customers globally including container ships, dry bulk carriers, cruise ships, tankers, and ferries
USPs / strengths	Owns two subsidiaries named as: MM Marine subsidiary: Dedicated fleet management company MOT (Minerva Oil Terminal Operations): Terminal operations
Positioning	- Offers integrated services that include: Storage, bunker fuel supply and trading - Owns significant storage capacities at Las Palmas, Tanger Med, Morocco and Balboa, Panama
Value proposition	Integrated services including storage, bunker fuel supply and trading

Strategic focus	<u>Expansion strategy</u> • In 2023, expanded its Red sea operations to serve vessels calling at King Abdullah port, Yanbu and Jeddah ports, KSA • Expanded its operations at Suez Canal and Egyptian ports • 2021: Strategic partnership with Ecoslops. Ecoslops and Mercuria are working jointly on a feasibility study for developing collection and recycling port infrastructure (Marpol Annex I) in the ME and South east Asia regions that would meet the highest standards of service and compliance using state-of-the-art technology <u>Innovation</u> • Partnered with rise-x.io adding end to end workflow to the Advanced Delivery Platform (ADP) <u>Sustainable solutions</u> • Introduced carbon neutral marine fuel offering, which enables customers to mitigate the carbon emissions associated with their bunker fuel purchases through verified carbon offsets
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Competitor's profile #10

Pearl Marine Logistics UAE FZE	
	HQ: UAE Founded: 1996 Employees: Details not available Market share: ~4-5% in UAE Rev: Details not available Capacity: 30,000 MT per month Website link: https://pearlmarine.com.sg/about-us
Role in the value chain	Bunker supplier (Trading)/ Bunker operations/ Transportation/Logistics/ Supply Chain and Storage
Key products and service offerings	Operates as a physical distributor as well as reseller of global marine fuel Types of grades in oil: LSMGO (<0.10%), VLSFO (<0.5%) Bunkering operations: ME region and Singapore
Bunkering operating metrics	Supplying capacity: 30,000 MT per month Number of barges: 2
Target geographies	Middle East and Singapore
Target customers	Oil companies, Ship owners/ Managers/ Charterers, Bunker traders and Brokers
USPs / strengths	Provides bunkering services at all major ports in ME region and Singapore
Positioning	Emerging bunker fuel supplier at Fujairah
Value proposition	Comprehensive set of services including bunker barging facilities, bunkering operations, general ship store supplies, inspections, smaller workshop repairs, air and gas cylinder refills, provisions and logistic services
Strategic focus	The company acquires its bunker fuel customer through its other business segments such as store supplies, workshop repairs, shipping and logistic services

Competitor's profile #11

Peninsula Fuel Supply Llc – Fujairah branch	
	HQ: Gibraltar Founded: 1996 Employees: ~400+ employees globally Market share: ~2-3% in UAE Capacity: 18,000 MT per month Website link: https://www.peninsula360.com/
Role in the value chain	Leading physical bunker and lubricant supplier (reseller of global marine fuel (bunker); LNG Bunkering services (since 2021) Physical supply, reselling fuel, cargo procurement, Yacht services
Key products and service offerings	• HSFO, VLSFO & LSMGO Bunkering, storage, services, risk management
Bunkering operating metrics	• Type of Bunkering Operations: Trading, Storage and Logistics • Total # of Refuelling Barges: 3 • Storage capacity: > 1,000,000 cubic meters
Target geographies	18 offices globally. Present in 18 physical ports
Target customers	750 + customers globally including container ships, dry bulk carriers, cruise ships, tankers, and ferries.
USPs / strengths	• Integrated services including sourcing and blending capabilities; logistics; Chartering; fuel quality management; ship agency, market analysis
Positioning	• Offers integrated services that include: Storage, bunker fuel supply and trading • Owns significant storage capacities at Las Palmas, Tanger Med, Morocco and Balboa, Panama
Value proposition	Integrated supply platform with sourcing and blending capabilities; logistics; Chartering; fuel quality Management; ship agency, market analysis (Physical supply, Reselling fuel, Cargo procurement, Yacht services)
Strategic focus	<u>Innovation</u> • Drones Maritime is an innovative JV between Peninsula and Nexus-Iberia. The company explores the benefits of drone technology for use within the maritime sector. <u>Integration</u> • Integrated supply platform with sourcing and blending capabilities; logistics; chartering; fuel quality management; ship agency, Market analysis <u>Unconventional fuel market</u> • Initiated several partnerships for Ammonia, Biofuel and LNG supply

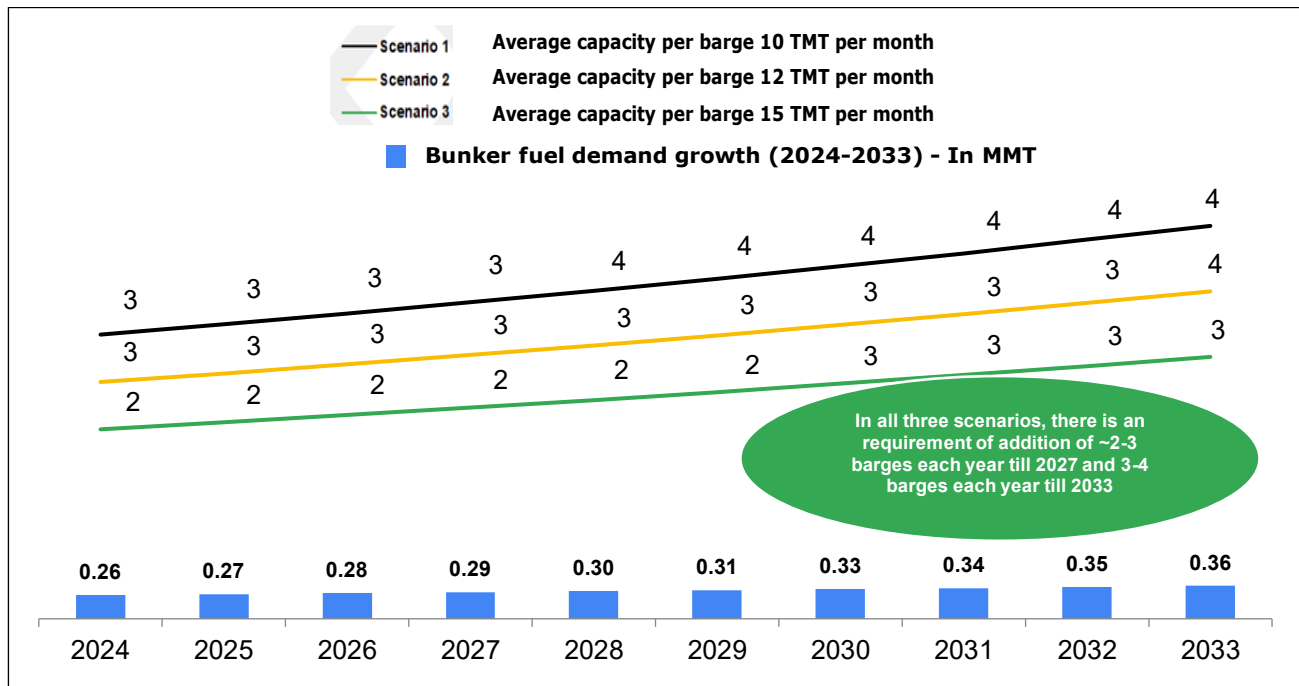
4.4 Survival plays in competition

Is there a scope to increase number of barges by existing players and new market entrants at Fujairah port?

Indeed, considering the projected demand growth of 3.7% y-o-y over the next decade (2024-2033), approximately 260-365 TMT of demand will be added each year at Fujairah port. This growth can be attributed to various factors such as the expansion of sea-borne trade, the increase in the global fleet size, the expansion of tanker fleets due to augmented offshore oil and gas investment, and the anticipated traffic growth stemming from the forthcoming IMEC trade corridor. Additionally, Fujairah's proximity to refineries in Asia and the ME region grants it a distinct advantage in terms of access to readily available LSFO from nearby refineries, which is poised to attract more customers and vessel operators passing the Strait of Hormuz.

Given the anticipated demand growth, there is an opportunity for both existing and new players to expand their presence in the Fujairah market to meet the rising demand. The addition of new barges, ranging from 10-15 TMT per month, will necessitate the inclusion of 2-3 barges annually until 2027, with this number increasing to 3-4 barges each year from 2028 to 2033. This potential market expansion can be capitalized on by existing players seeking to bolster their market share and by new entrants aiming to establish themselves in the Fujairah bunker fuel market.

Figure 4.8: Scenario analysis – Number of barges required to cater the expected demand growth at Fujairah



Source: S&P Platts, Reuters, Fujairah Port Authority, Economic times, UNCTAD, IMARC Group, Source: Competitor website, Primary research, D&B Analysis

How will IOC ME – Axiom Global JV help in achieving some market share from its competitors?

IOC ME – Axiom JV will be relatively more helpful in achieving additional market share from its competitors, through building strengths in below areas:

- Strong financial capabilities & better management of working capital and credit terms with its customers
- Cost optimization through implementation of innovative processes
- Enhanced marketing, sales and R&D capabilities
- Better sourcing of innovative and digital technologies
- Axiom's local market knowledge supplemented with IOC ME capabilities and network

The most common practices which can help the bunker fuel suppliers to increase their competitiveness and market share are:

Figure 4.9: Best practices to increase market share in Fujairah Bunker fuel market from existing players

Focus area		Potential strategy
	Cost optimization/ efficiency	• Captive fleet of tankers and barges • Captive refining, storage and/or blending facility • Long terms contracts with refiners/ blending component providers • Alternate contracts to rent barges in case of demand being unaddressed due to full capacity utilization of owned barges • Partnerships with barge manufacturing companies
	Customer retention & acquisition	• Additional services such as brokerage, ship agency, repair and maintenance, oil trading services, garbage collection and waste disposal services • Long term contracts with customers/ vessel owners for longer period of time • Acquisition of potential new customers expected to be added due to opening of new routes such as IMEC corridor • Long term contracts with offshore oil and gas companies and oil services providers operating in the ME and Africa region to supply Bunker fuel to their tanker fleets
	Innovative technologies	• JVs to implement digital platforms for marine survey, fuel quality management, real time pricing, market knowledge, real time demand prediction tools etc.
	Geographical expansion	• Develop infrastructure to increase its bunker fuel supply capabilities to the adjacent ports such as Khorfakkan, Yanbu, Jeddah, Oman based ports, Egyptian ports
	Other value-added services	• Facilities like providing market knowledge to its customers, providing fresh water, catering services and other facilities to the ship crew
	Effective financial management	• Bunker fuel suppliers should maintain a certain level of working capital through flexible credit terms with its customers • Partnerships with financial management companies/institutions helps the supplier to predict any financial risks and mitigate them in a timely and effective manner

Source: Primary research, D&B Analysis

5 Market Intelligence

5.1 Five force analysis

▪ Bargaining power of suppliers

Force strength (medium-to-high)

Limited suppliers across the supply chain and positive demand outlook of bunker fuels strengthening the suppliers

Limited suppliers and stringent regulations

The marine fuel market is supplied by major oil companies, traders and brokers, independent physical suppliers, national government-owned supply companies and non-major refiners. The major oil companies are the dominant suppliers of marine fuel to the shipping industry. These companies have large, integrated operations in many locations around the world as well as a ready supply of the by-product of their refining process

Limited suppliers

- There are 13 registered complaint bunker fuel suppliers in the Fujairah market and approximately 40 active barges in service, with an cumulative capacity of 700- 800 TMT per month⁴⁷ [Link](#)
- UAE is one of the largest exporters of crude oil, which has a significant share in the tanker fleet sector. With the expected growth in sea-borne trade, the suppliers have therefore high bargaining power in the supply chain of bunker fuels ⁴⁸ [Link](#)

High investment and low margins

- The strategic location of Fujairah, arising geo-political tensions in Suez Canal route and world-class port infrastructure have made it a hub for maritime trade across Europe, Africa, and Asia, which will additionally support the suppliers in the market
- The licensing and permits to the suppliers in UAE are provided by DMCA which is a complex process. Established and reliable suppliers are important to ensure the continuous and secure flow of fuel to vessels. This restricts the entry of the new suppliers in the industry. On top of this, the government controls the access to key ports and regulate the number of suppliers to ensure safety, security, and compliance

Actions by other companies to counter these / Best practices / Recommendations

- Making long term contracts and commitments, strategic alliances with key suppliers for ensuring consistent supply
- The companies are investing in building storage capacities to have more control on inventory. Fujairah port under its expansion plan which is set to complete in 2023 has expanded storage, refining capacities and port infrastructure
- The companies are investing in digital solutions for better inventory management and demand estimation
- Exploring more regional and global suppliers or partners

Impact on IOC ME & Axiom global JV – Positive

- **Supplier power as per the industry standards:** Since the JV will be a supplier to bunkering oil it will have medium-to-high power in the business similar to the industry

Challenges

- Limited negotiation power: The venture may have limited flexibility in negotiating favourable terms, leading to less advantageous payment terms, delivery schedules, and contract conditions
- Low margin business: There is pressure across the industry to increase operational efficiencies and expedite turnaround time during bunkering operations

- **Stability in supply chains:** A supplier with a significant market share and captive refining/ storage capacity offers stability and reliability in the supply chain. This can be beneficial for large and established players and a challenge for the new players
- **Quality assurance:** Since the licensing is highly complex and the industry is regulated, there would be quality assurance standards and service levels that need to be met by the supplier

■ **Bargaining power of buyers**

Force strength (medium)

The bargaining power will be medium as the demand will remain steady for the bunker fuels and buyers can explore global or domestic suppliers

Positive demand outlook

The buyers in the bunkering industry in the UAE, include both local and international entities engaged in various facets of maritime activities and is influenced by factors such as global shipping trends, regulatory changes and the overall economic health of the region

Steady demand

- With expected increase in offshore oil and gas spending, the demand for bunker fuel is expected to hold steady in 2023 and trade activities are likely to increase. ⁴⁹ [Link](#)
- The bunker fuel market in the ME and Africa is expected to grow at a CAGR of more than 12% during the forecast period 2024-2029. The region is also a global leader in crude oil production. With the expected increase in production, the demand for bunker fuel is likely to increase. ⁴⁸ [Link](#)

Standard product and low switching cost

- The bargaining power is medium as there are relatively large number of suppliers for the bunkers across the region and worldwide. Singapore, Rotterdam bunkering industry are giving tough competitions to the bunkering industry in the ME region by providing competitive prices and better services. ⁵⁰ [Link](#)
- At the same time, there has been an intense competition domestically ranging from KSA to Qatar to Bahrain and Egypt entering the market. This opens various options for the buyer's as switching cost is low and the bunker fuel products are standard.

Actions by other companies to counter these / Best practices / Recommendations

- **Continuously enhancing bunkering services:** Technologies such as cutting-edge software, real-time communication tools, cloud-based collaboration tools and electronic Bunker delivery note (e-BDN) systems have been integrated into operations to provide enhanced bunkering services. ⁵¹ [Link](#)
- **Diverse fuel options:** A wide range of fuel grades, including LSFO and MGO, means shipping companies can easily find the most suitable and cost-effective options for their vessels at the same place.

Impact on IOC ME & Axiom Global JV – Positive

- ✓ **Positive demand outlook:** The demand outlook of the industry looks promising: Being a major exporter of crude oil, UAE holds a substantial presence in the tanker fleet shipping sector. Given the anticipated expansion in global sea-borne trade, the country is poised to maintain its prevailing dominance.
- ✓ **Increased oil production:** ME nations, including UAE and KSA, are strategizing to boost crude oil production. The anticipated rise in production is expected to present substantial opportunities in the shipping market and thus for the venture.

Challenges

- ✗ The company will have pressure to keep the prices competitive in the market worldwide as the products are standard and switching cost to the buyer is low.
- ✗ The company must be flexible for the changes and transformation required to shift to the cleaner fuels over time. It should have strategic plans to do so to remain competitive in the market.

■ Threat of rivalry

Force strength (high)

Apart from the competition given by the international players like Singapore, the industry is also facing the competition from the domestic players in the region

Increased competition - from neighbouring ports

In UAE, there are approximately 13 ports and terminals equipped with the necessary infrastructure to provide marine fuel services to vessels in the area. Jebel Ali Port, boasting a container handling capacity of 19.3 million TEU, holds the title of the largest port in both the UAE and the broader ME region. ⁴⁸ [Link](#)

High regional competition

- The UAE's Port of Fujairah, the world's third-biggest bunkering hub, is facing increased competition to supply marine fuels in the region, with new entrants ranging from KSA to Qatar to Bahrain and Egypt. The rising competition among domestic players is chipping away Fujairah's status as bunker hub.
- The emergence of alternative refuelling options at strategic locations around the ME region and a recent growth in bunkering activity outside port limits off the coast of Fujairah could pose further downward pressure on Fujairah's market. ⁵² [Link](#)

High global competition

- Fujairah is now more expensive than Singapore. As of 2022, Fujairah's 0.5% sulphur marine fuel prices have surged to a \$3.88/MT premium compared to Singapore, marking a reversal from the \$3/MT discount observed throughout 2021, as per S&P Global data. ⁴⁹ [Link](#)
- Fujairah's downstream market may face further pressure with the completion of Kuwait's Al-Zour refinery's third and final crude distillation unit. This refinery's potential future LSFO exports could further depress sentiments in Fujairah's downstream market. ⁵³ [Link](#)

Actions by other companies to counter these / Best practices / Recommendations

- The hub is focussing on development of several infrastructures, investment in technology as most of the bunkering operation in Fujairah is still done manually. In contrast, top bunker hub Singapore has stepped up efforts in digitalising bunkering operations, which helps to pave the way for new fuels. ⁴⁹ [Link](#)
- Fujairah has been planning to gear up for alternative fuel markets such as liquefied natural gas (LNG), Biofuels and Methanol. Potential investments involve repurposing tankage to accommodate future fuels. ¹² [Link](#)

Impact on IOC ME & Axiom Global JV – Positive

- ✓ **The rise of Dubai's bunkering industry offering business share for all companies despite competition:** The bunker market is still very positive for UAE due to its strategic position on the Persian Gulf which provides easy access to major trading routes, making UAE a convenient and accessible bunkering location. Despite competition, there will be enough business for the JV to scale
- ✓ **UAE commitment to innovation and expansion:** The UAE's FTA and DMCA continually strive to improve the infrastructure, facilities, and services offered to shipping companies. This has helped oil traders and suppliers, to establish bunkering operations in the region. Even in wake of high competition, the ecosystem good provision of facilities

Challenges

- × Increased competition may create pressure on the venture to keep the prices competitive
- × The potential geopolitical instability in the region can create market turbulence since bunker market is dependent on carriers crossing inbound and outbound the Strait of Hormuz

■ Threat of new entrants

Force strength (low)

The industry requires high investment in infrastructure, technology, is highly regulated and controlled by IMO, DMCA and EU

The threat of new entrant is low

Highly regulated and controlled industry

- The DMCA is responsible for licensing vessels and ships that supply fuel along its territorial waters.
- It has its own bunkering framework and has strengthened its commitment to the local shipping community as it guarantees growth and development.
- There are various permits categorized into several groups:
 - Bunker supplier facility permit
 - Marine vessel processing permit
 - Bunker supplier for marine vessel permit
 - Payload lightering permit
 - Fuel lightering permit
- The bunkering license in Dubai gets issued on an annual basis and is anchored on the requirement that the vessels licensed will meet all the security guidelines and resolutions set out by the DMCA. ^{54 55 56} [Link1](#) [Link2](#) [Link3](#)

High investments

To deliver seamless bunkering services, the ports across UAE are continuously investing in infrastructure and technology.

- The Fujairah port expansions which were completed in 2023, included a new dry bulk export terminal with initial 18 MMT capacity, Etihad Rail connecting Fujairah with Abu Dhabi and Dubai, ADNOC's 42-million-barrel underground crude storage and Fujairah oil terminal's VLCC connection. ⁴⁹ [Link](#)
- CDPQ announced the investment of USD2.5 billion in the Jebel Ali Port in 2022. ⁵⁷ [Link](#)

Actions by other companies to counter these / Best practices / Recommendations

- Key players adopt numerous strategies such as product launch, agreement, partnership, business expansion, and acquisition to stay competitive in the bunker fuel market
- Apart from conventional marine fuel supply facility, the Fujairah port authority plans to introduce a ship-to-ship LNG transfer facility, which is expected to support IMO regulation towards low sulphur content fuels. ⁴⁸ [Link](#)

Impact on IOC ME & Axiom Global JV – Positive

- ✓ **Diversified product portfolio:** The demand for VLSFO is likely to witness significant growth due to environmental concerns and the strict regulations related to emissions from the maritime industry. The JV can capitalize by catering to different customer preferences and compliance requirements by offering broader range of bunkering services and products in the UAE. *It has an edge over any other possible new smaller entrants in the market, as they will be less likely to have capabilities on product portfolio as that of IOC ME ME-Axiom JV*
- ✓ **Partnership:** The strategic partnerships of both the companies with other shipping companies, port authorities, digital technology providers and other stakeholders can strengthen market presence, benefitting the venture.
- ✓ **Increased Investment:** Both **IOC ME** and Axiom Global JV are established brand in the sector, the JV of both the brand will attract business from stakeholders, while any other new entrants will less likely to have these benefits

Challenges

- × The JV being a new entrant in the Fujairah bunkering fuel industry, it will have to face all industry challenges like getting licenses, high capital investment, etc. like any other new entrants

■ Threat of new substitutes / sustainable alternatives

Force strength (high)

The urgency for a maritime industry decarbonization to meet carbon emission reduction targets set out by the U.N.'s IMO is making industry to switch to alternative fuels such as LNG, Biofuels and Methanol

LNG is leading alternative bunker options

High investment in LNG-fueled fleet by ship owners

- The ship industry is making massive investments in new builds and energy supply infrastructure that will impact GHG emissions for the next 25-30 years
- In January 2023, LNG bunkering was available at 185 ports worldwide with a further 50 facilities planned by 2025 ⁵⁸ [Link](#)
- LNG and LSFO is becoming a mainstream fuel choice in shipping segments. The industry has shifted focus to emissions reductions efforts, alternative fuels and more efficient system ⁵⁰ [Link](#)

LNG regulatory compliance

- Aligned with the new regulations set by the IMO (IMO 2020), Dubai has actively promoted the adoption of LSFO within its port. It is focussing on development of LNG bunkering infrastructure, with ambitions to establish itself as a key regional hub for LNG marine fuel by 2030 ⁵¹ [Link](#)
- By 2050, Methanol and LNG are expected to take the top market share of alternative bunker fuels at Fujairah, followed by Biofuels and ammonia, showed at the FUJCON forum ⁵⁰ [Link](#)
- Under the implementation of the IMO's Carbon Intensity Index (CII), LNG-fuelled vessels will be able to continue operating as normal under the CII system until after 2030, while fossil LNG blended with bio-LNG and e-LNG will further extend compliance to 2050 and beyond ⁵⁸ [Link](#)

Actions by other companies to counter these / Best practices / Recommendations

- The companies are establishing the LNG infrastructure at key bunkering locations. These projects are attracting massive investments as they are promising long term benefits to the company ⁵⁸ [Link](#)
- The companies are also looking forward to expand Bio-LNG production. According to the report by Sea-LNG, pure bio-LNG could cover up to 3% of the total energy demand for shipping fuels in 2030 and 13% in 2050 ⁵⁸ [Link](#)

Impact on IOC ME & Axiom Global JV – Positive

- ✓ **Diversify portfolio:** With IOC ME already having capabilities in LSFO, JV stands at a favourable position here. Further, potential investments at the JV should also involve repurposing tankage to accommodate these future fuels; this will not only create long term benefits for both the company but also will help in remain compliant with the stringent regulations formed.

Challenges

- × The bunkering industry is undergoing a significant changes, therefore it has become important to have clear 5–10-year strategy, detailing actions and transformation under alternative fuels also (from LNG to BioLNG to Synthetic LNG) ⁵⁸ [Link](#)

6 Bunkering Operations Review

6.1 Storage & logistics

6.1.1 Key Facts - Fujairah Oil Industry Zone

Location

- **Location:** New Energy corridor that stretches from the east of the Suez Canal to Asia, including deep water port and an oil industry special economic free zone
- **Area:** ~10 sq km. (Inland); and 0.66 sq km. (Sea) [Link](#)

Overview

- **Establishment of Fujairah Oil Industry Zone (FOIZ):** 2011
- **Global rank in terms of bunker fuel sales:** 3rd largest bunkering port (As of 2023)
- **Unique Selling Proposition:**
 - Strategic location on the Gulf of Oman, just outside the Strait of Hormuz
 - Economies of scale with multitude of bunkering business opportunities
 - Wider space of independent liquid bulk storage terminals

Evolution of Fujairah port [Link](#)

- **1978:** Initial construction of the port started as part of the economic development of the UAE
- **1983:** Full operations commencement of Fujairah port
- **1994:** Oil storage in Fujairah started with a storage capacity of 550,000 cubic meters
- **2006:** Oil Terminal 1, commissioned in January 2006, there are three berths with a draft of 15 meters.
- **2010:** Second oil terminal commenced operation in June 2010
- **2011:** Fujairah oil industry zone was established in a special economic zone
- **2017:** The Port of Fujairah and Abu Dhabi Port signed a 35-year concession agreement granting Abu Dhabi Ports' exclusive rights to provide services for container, general cargo, RoRo and cruise ships
- Fujairah sales was down pertaining to Qatar diplomatic prices, as it imposed a ban on vessels flying flags of Qatar or vessels heading to Qatar or arriving from Qatari ports.
- **2019:** Commissioned a new service harbor
- **2020:** Announced expansion plans worth AED 750 (USD 204.2 million) over the next two years to expand and upgrade its oil handling infrastructure
- **2022:** Total number of vessels arriving at Fujairah offshore anchorage area was close to 12,500 vessels and the number of varied service companies increased to more than 430

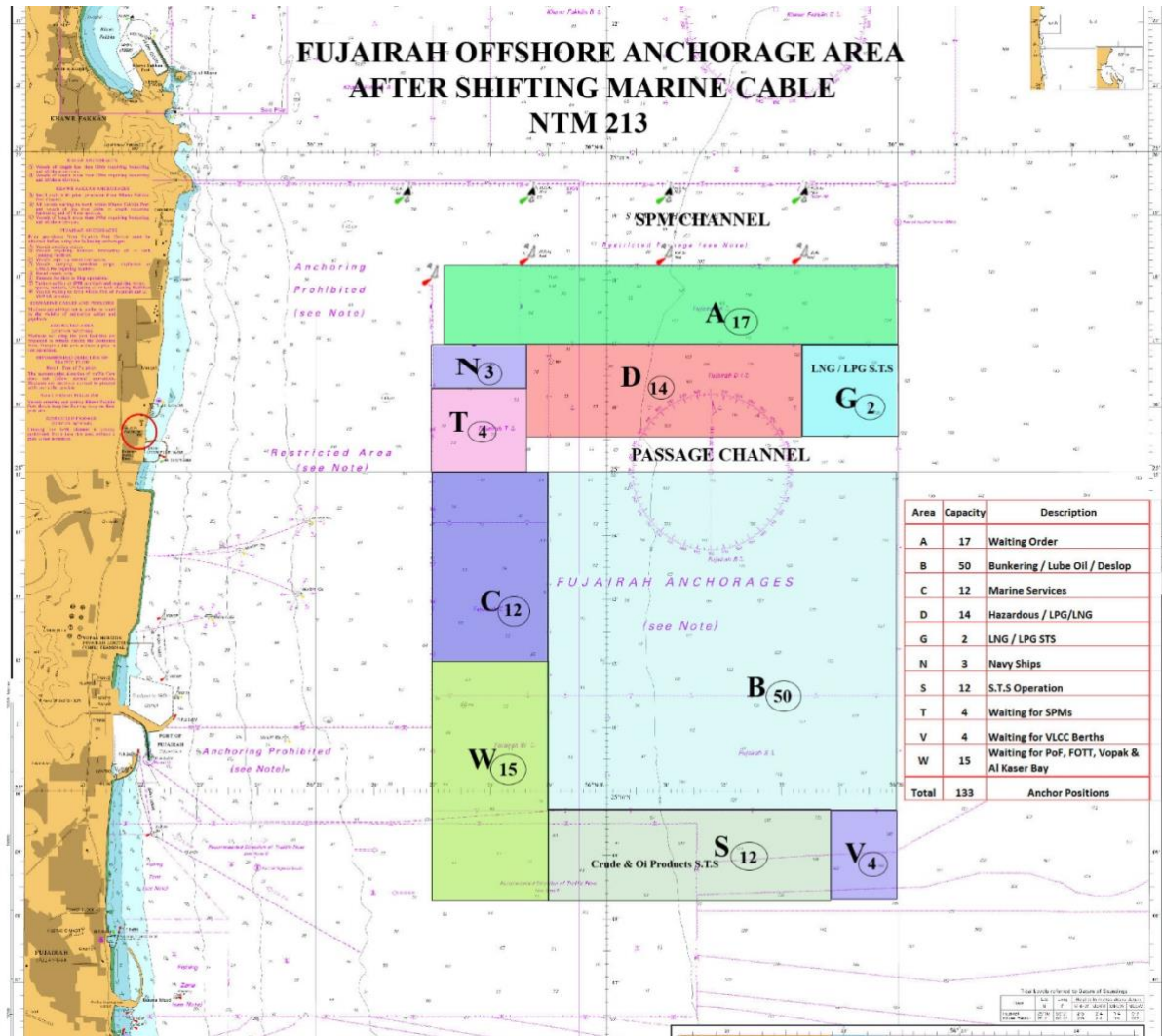
Infrastructure and capacity

1. **Crude oil pipeline:** 380 Kms Habshan-Fujairah pipelines (Abu Dhabi to Fujairah)
2. **Number of refineries:** 4 (2 refineries are currently supplying LSFO and VLSFO)
3. **Storage Tanks:** 366
4. **Number of barges for LSFO and HSFO:** ~40
5. **Oil Berths:** 9 Main Berths / 14 Wing Berths
6. **Bunker Barge Berths (Low Sulphur oil tankers):** 4 Berths
7. **Capacity of Anchor positions at one time:** 133
8. **Cumulative refining capacity:** 28,000 tonnes per day (0.5 million b/d)

9. **Crude Oil and product storage capacity:** 17 million cubic meters (as of 2023)
10. **Cargo Handling capacity:** 100 million Tonnes of cargo / per year
11. **Expected terminal capacity by 2030:** 16 million ton (720,000 TEUs); and 1.3 MMT of general cargo

Anchorage, Port area and bunker barge berths

The Fujairah Port's anchorage area is as shown in the figure below

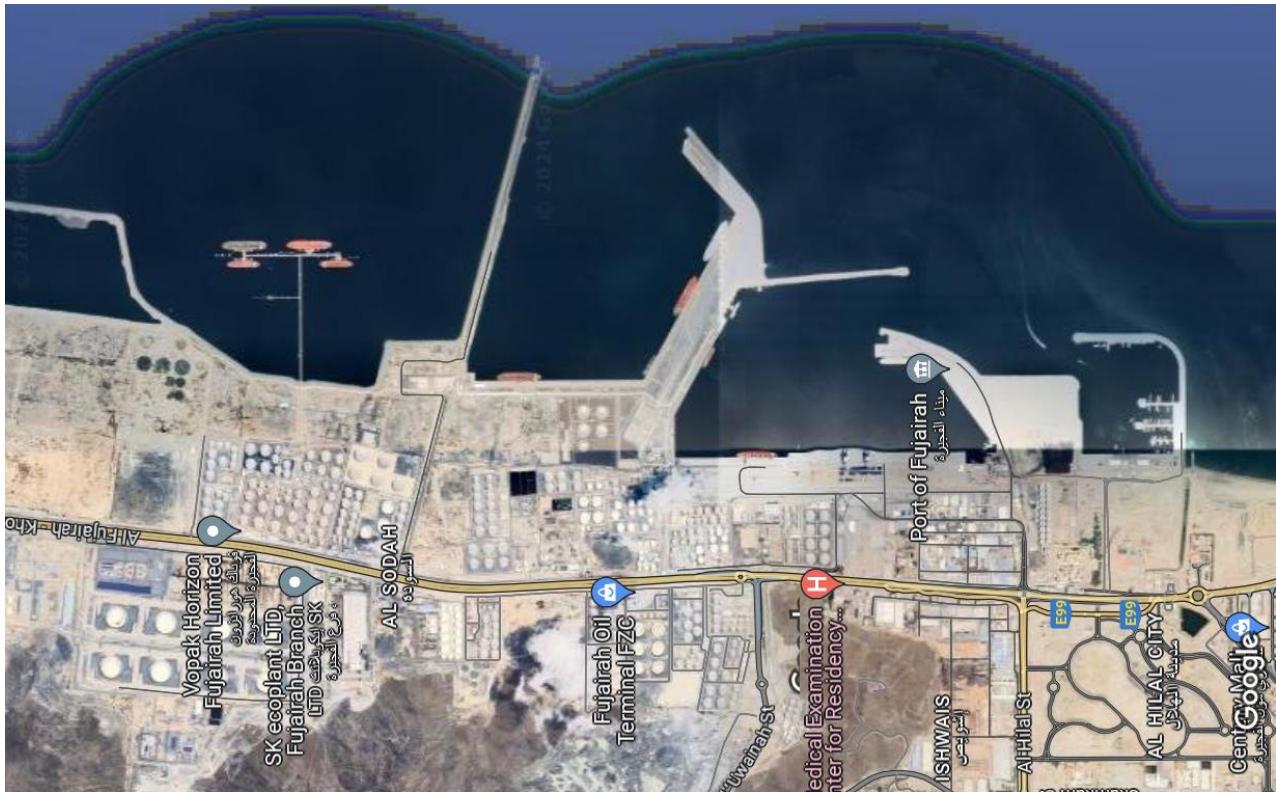


The above figure indicates the specific allotted anchorage areas for various types of vessels handling different products. Bunkering vessels anchorage is within the area B – which provides a total of 50 anchorage positions of the total 133.

Berths 3A, 3B, 3 East and 3 West are situated at Oil Terminal 1, with a draft of 15.00 meters. These berths are capable of accommodating vessels between 10,000 – 20,000 DWT. Each berth has a black line and a white line connecting to Matrix Manifold 1, which connects all user's pipelines. The four berths are dedicated for the MGO and VLSFO loading and discharge operations.

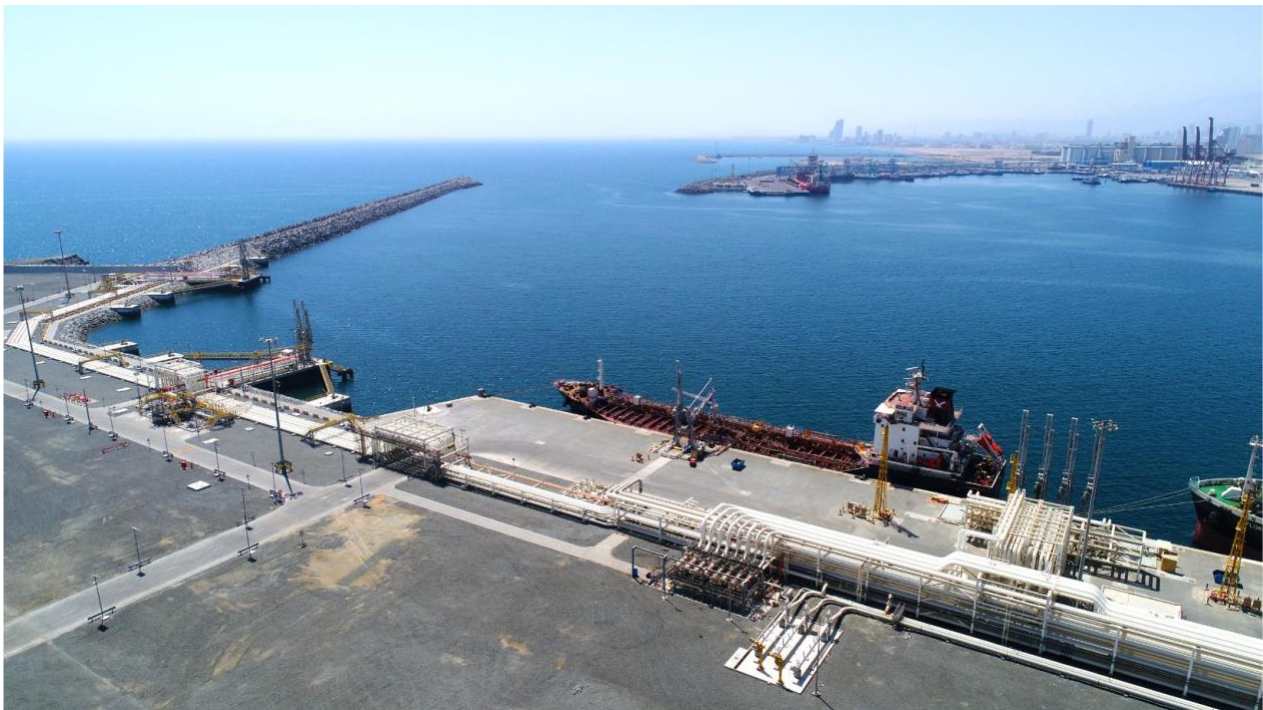
The overview of the Fujairah port area and the bunker barge berths are shown in figure below.

Fujairah port overview



Source: official website of Fujairah Port

Bunker Barges berths area



Source: official website of Fujairah Port

Fujairah port has two Matrix Manifold systems which provide full connectivity to any of the FOTT berths. The storage terminals help to connect a minimum number of black oil and white oil lines to one of the Manifolds which in provides access to all the 9 main 9 and 14 wing 14 berths via FOTT's dock lines.

The port has two Oil Terminals and the Matrix Manifold system enables the product flows between two terminals via its manifolds and interconnectivity lines.

Bunkering statistics

1. **Vessel calls in 2022:** ~ 12,500
2. **Terminal storage companies:** 15
3. **Operating tank storage companies:** 10
4. **Varied service companies:** 430+
5. **Licensed bunker fuel suppliers/traders:** 13
6. **Registered lube oil suppliers:** 4
7. **Operating companies/ partners:** 18
8. **Customers:** 15 (VTTI, GTI, Marine Oil Terminal Corporation, GPS Chemoil, Vopak, Ecomar Energy, Emarat, Fujairah Oil Terminal FZC, BPGIC, Horizon Terminals, Mount Row, ADNOC, ENOC, Montfort (Formerly Uniper), Vitol Energy)

6.1.2 Fujairah Bunker & Lubricants supply statistics

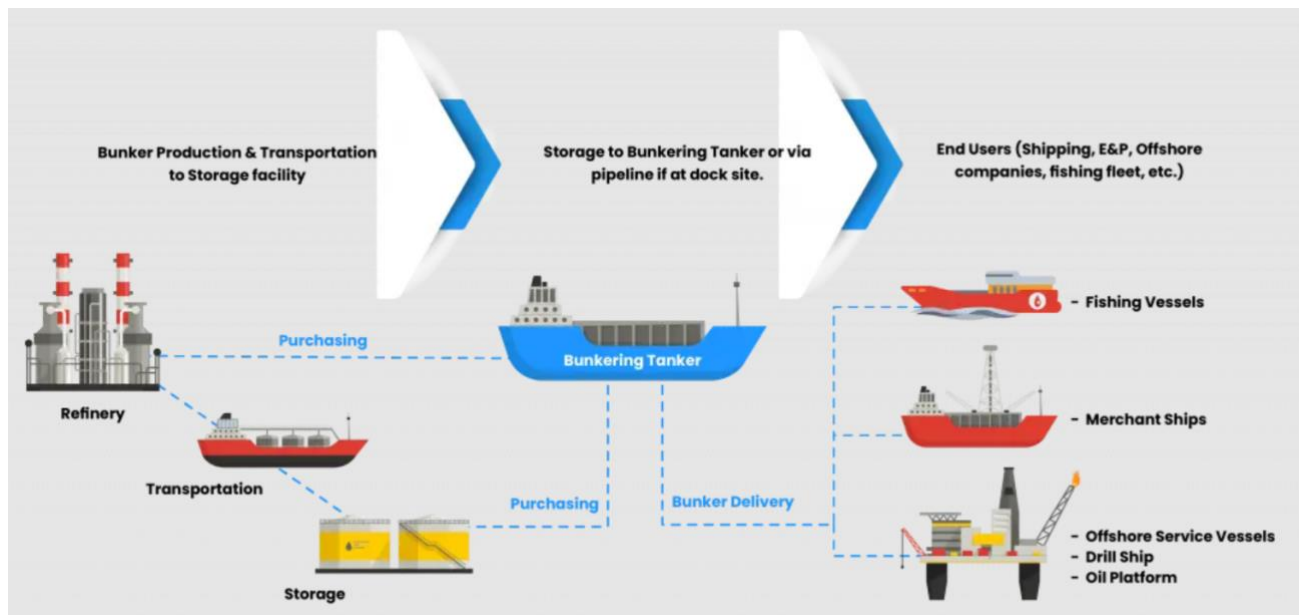
Table 6.1: Product wise bunker supplies at Fujairah (2023)

(MTs)								
Month	180cst LSFO	380cst LSFO	380cst HSFO	MGO	LSMGO	Total Bunker Sales	Lubricants	Total
23-Jan	300	453,703	151,055	796	27,169	633,023	4,039	637,062
23-Feb	545	389,544	159,908	713	21,661	572,371	4,549	576,920
23-Mar	250	451,405	81,058	538	26,549	559,800	3,475	563,275
23-Apr	480	441,912	126,943	257	26,738	596,330	4,270	600,600
23-May	170	468,635	126,586	2,021	22,976	620,388	4,723	625,111
23-Jun	573	449,264	127,573	1,088	26,876	605,374	4,732	610,106
23-Jul	1,272	459,665	174,227	1,387	23,015	659,566	4,151	663,717
23-Aug	-	450,420	196,343	459	30,281	677,503	4,587	682,090
23-Sep	-	425,942	172,945	1,688	28,064	628,639	3,944	632,583
23-Oct	792	449,652	164,107	1,699	37,903	654,153	4,499	658,652
23-Nov	513	407,503	171,438	2,430	28,614	610,498	4,305	614,803
23-Dec	2,409	457,756	167,321	685	30,467	658,638	4,414	663,052
Total '23	7304	5,305,401	1,81,9504	13,761	330,313	7,476,283	51,688	7,527,971
Jan-24	341	436,604	207,124	122	30,441	674,632	4,372	679,004

6.1.3 Various options for bunkering storage and logistics

The storage and logistics facilities or infrastructure are an integral and a very important component and part of the process value chain in trading business of bunkers and Oil & Gas industry products. The overview of standard process of procuring, storage and supply of bunkers is shown in figure below.

Figure 6.1: Bunkering - Overview value chain



The dynamics of deciding the storage and logistics facilities include key aspects such as **location, products, capacity, type and scale of storage.**

There are multiple options of ensuring that these key requirements are met. These can be arranged through **short term services / deployed / developed.**

- **Short duration contracts / on-demand services:** In case of a bunkering traders operating on a small scale and not capable of having capital outlay allocations towards these facilities, owing to relatively low volumes of business, the storage and logistics facilities may be arranged on the basis of on-demand approach / model, wherein the equipment, facilities, services are arranged against a specific order or for a specified very short duration.
- **Deployment on lease basis:** For traders who are relatively established, engaged in considerable higher level of volumes of transactions, and catering to multiple clients and through repeat orders, but still cannot plan for huge investments towards captive in-house facilities, these facilities can be secured by having the optimal no. of barges and in land logistics support on mid to long term lease. Through this option, the company can have certain storage and logistics arrangement within their control for ready disposal, which can aid in servicing the clients with commitments and certainty.
- **Captive / in-house development of facilities for long term business plan:** For players, who are engaged in activities across multiple nodes of value chain, engaged in refining, operating on a much larger / global scale, and certain of contributing towards a considerable share of trade / business in a given geography, they shall propose having secured storage and logistics facilities within their complete / higher control. This shall help them in controlling the quality, time and cost aspects and also plan and manage quantity variations and volatilities across multiple storage points.

6.1.4 Guiding parameters for the proposed JV

- The details of key aspects pertaining to proposed JV are as follows:
 - **Location:** Both the companies – AGOT and IOC ME, **are focusing on Fujairah Port**, as the sole location for initiating the proposed JV business.
 - **Products:** Although AGOT has experience and IOC ME also has access to a variety of products, presently the prime product would be marine bunkers with bunker trading as focus area, through the Fujairah

Port. As a secondary focus area, sale of lubricants also shall be looked at, as AGOT is already engaged in similar activities with IOC ME, and at other ports in South Asia.

- **Capacity:** AGOT is presently engaged in bunker trading focusing on South and South East Asian (SEA) regions, with IOC and other key players. In line with its present volumes handled, the proposed **JV plans to target a modest, reasonable trading volume of about 60,000-80,000 MTs per month** to start with, as it shall study the Fujairah port live competitive landscape during initial phase of JV, and grow to a peak of 150,000 MTs per month.
- **Storage:** Reviewing the present and proposed future refining capacities in Fujairah and overall in the UAE, considering the availability of bunkers in local markets and more focussed from refineries in close proximities, considering AGOT's present experience in sourcing and handling the fuels locally in UAE region – from varied suppliers, **it is envisaged not to have a dedicated storage facility** for initial couple of years, and to have an allocation, to avail these facilities, in case of opportunities for blended products or hedging or any exigencies. The bunker offtake shall be managed through barges (Ship to Ship). This shall offer flexibility to supply bunkers to ships in anchorage area also, rather than booking and waiting for the berths availability in case of Pipe to Ship transfer.

Storage and barge capacities proposed

- Based on the considerations as proposed for the JV, it may not have any dedicated storage facilities at tank terminal, for the initial phase of 2 years, and only availability of tankers need to be ascertained for local transport of bunker up to the barges.
- JV can look for 3-4 barges each of 10,000 MT capacity to support the sale and transfer of bunker to the end use vessels.

6.2 Product sourcing strategy

Sourcing of bunker fuel

Bunker being one of the critical cost components for marine transportation, the end users are always keen to keep a close watch on the costs, resulting in bunkering business to be attributed with stiff competition within a bunkering hub as well as competing against the other bunkering hubs. This makes it important to have clarity in sourcing / procurement strategy, to be monitored in view of product demand – supply dynamics.

- The location, product mix and domestic production and supply capability, its competitiveness with global markets are critical factors deciding the source of procuring the products for bunkering.
 - For this JV's initial phase, the location is specified to be the Fujairah Port
 - Post IMO 2020, the product wise demand mix has shifted more towards VLSFO, and reduced share of HSFO and marginal share of MGO. In line with this, the similar supply trends have been observed in Fujairah Port's past trends.

Looking at the forecast trends which are likely to remain same or more buoyant towards VLSFO, (as HSFO based vessels would reduce further), the proposed shall have to look at the below said mix:

 - VLSFO : 60-65%
 - HSFO : 30-35%
 - MGO : 5-10%
- In terms of supply source mix, it is noted that UAE has sufficient production output volumes, logistics and storage capacities, for all the three broad products – VLSFO, HSFO and MGO. Hence largely the sourcing can be from domestic sources with support from import sources to address the gaps and also to leverage on competencies of refiners from other Asian geographies. The proposed sourcing plan can be as indicated below.

Table 6.2: Product wise sourcing plan for proposed JV

Supply Source	VLSFO	HSFO	MGO
Domestic	70-75%	85-90%	95-100%
Imported	25-30%	10-15%	0-5%

- AGOT already has sufficient understanding about the local suppliers and an operational relationship with many suppliers, due to its presence in this region through the group company – SPR Global. Hence the proposed JV can also leverage on this on-field presence, local market know-how, existing relationships and aim to have a faster and seamless sourcing / procurement aspect secured for the new location of proposed JV.

The Fujairah port has a storage capacity of about 17 million cubic meters (Mn cbm), (approx. 15 Mn MT), covering all products – all types of oils, bunker products, lubricants. The capacity has increased in recent 1-2 years from 11 Mn cbm to 17 Mn cbm. There are about 15 licensed entities holding storage terminal in the Fujairah Port area, dealing with all products and about 13 entities for bunkering products.

- It is noted that the Middle East is relatively slower in adopting the LNG bunkering as compared to Singapore and European region. Fujairah Port too indicates similar patterns. It is yet to have dedicated large capacity LNG terminal and storage capacities to be developed. Recent developments indicate that the plan of a LNG terminal project which was initially proposed to be developed at Fujairah Port, has been changed and its now proposed to be established at Al-Ruwais. This may further defer/ delay the LNG port facility and capacities at Fujairah, which will impact the regional LNG bunkering business.
- Although, LNG is not the key focus products for this JV, presently, it is expected that globally, in a span of 5-7 years, LNG bunker demand would considerably grow across regions.

Sourcing of Lubricants

As compared to the bunkers, lubricants consumption in volume terms, is relatively lower, but there are different types of lubricants used in marine industry, addressing needs of different types of vessels, ships and liners. The lubricant products are packed in smaller quantities as it involves considerable retail markets (B2C) too, along with the B2B, and hence the inland and overseas logistics is relatively easier, organized and can be managed through much higher no. of service providers.

Considering the product characteristics, packaging sizes and options, it is noted that Lubricants can be sourced from imported or domestic sources, both of which are equally attractive options. The dynamics of pricing, availability, quality also play an important role for lubricants. Although consumed in lesser quantities, lubricants play a very crucial role in functioning of the critical equipment on vessels (engines and allied support utility systems), which ensure the continuity and timeliness of a vessel's transportation and thus the overall business. Negligence towards use of lubricants, can impact the overall operations and schedules of a vessel's navigation plan from source to destination.

6.3 Pricing

The bunker pricing, apart from the basic product related costs, depends upon quite a few parameters, which play a dynamic role in deciding the effective suitability from the end users point of view, till the bunker reaches the vessel. Below are the key primary attributes and the sub components of each of these, which play a role in pricing of bunkers.

Key parameters affecting the overall pricing of bunkers

- Price**
- Quality**
- Operating cost at Ports**
- Port Congestion levels**
 - Anchoring and docking availability
 - Supply waiting time
 - Provision of services, port congestion

- Port tariffs
- Prices of complementary services for fuel supply at berth (pilotage, mooring, etc.)
- Prices of complementary services for fuel supply at anchorage

■ Geographical

- Geographical advantage
- Accessibility to port (from sea side)
- Hinterland proximity

- Port access waiting time

■ Port procedures and regulations

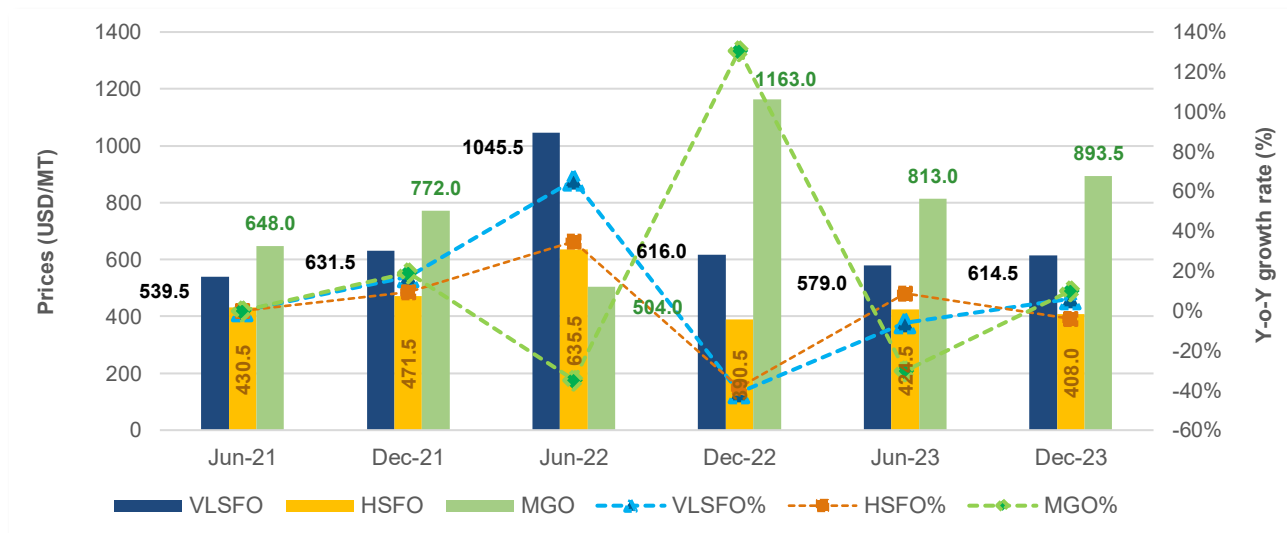
- Ship inspection thoroughness
- Simplicity of crew changes
- Presence of restrictive environmental regulations
- Customs strictness
- Clear and precise information about services
- Port security
- Organizational tradition and capacity
- Industrial disputes

6.3.1 Pricing approach for long term contracts for bunker fuel grades.

Apart from the need to manage the dynamics among the above-mentioned parameters, for long terms contracts, usually it is preferred by the vessel owners / operators to have a fixed fuel price for a specific amount (off take quantities) for a defined tenure / duration, with a main aim to mitigate the fluctuating spot prices and uncertain fuel consumption between the ports. But as the actual market prices are volatile and governed by global external dynamic factors, so, to facilitate and account for these variations, usually a dynamic programming model to is proposed to optimize the total bunkering cost covering both scenarios - when fuel consumption and / or fuel prices are uncertain.

Historical prices and y-o-y growth rates of bunkers

Figure 6.2: Bunker price trends



Bunker fuel is a derivative of crude and therefore there has close correlation to **crude oil price trends, refining costs and its volatilities**. Bunker prices also depend on the availability of the product of different grades in the market.

Also, any major geopolitical upheavals / downturn events regionally or globally impact the prices of crude and bunkers, as the production and import / export of these products are closely linked to key macroeconomic indicators globally. In the year 2022, the crude and bunker prices were volatile due to the scenario of gradual return to normalcy post COVID-19 and the Ukraine-Russia war and remained volatile throughout 2022.

7 Regulatory environment and JV Partner Review

7.1 Regulatory environment for bunkering

7.1.1 Policy and regulatory overview

The UAE's bunkering industry holds a significant market share in global maritime bunkering fuel industry. But for investors and JVs, navigating this sector, demands a thorough understanding of its complex regulatory landscape.

The maritime sector in UAE including bunkering business is regulated by **ministries, government appointed bodies and authorities** which work together to ensure the implementation of IMO regulations such as MARPOL, SOLAS, ISGOTT etc. Agencies like Federal Transport Authority (FTA) ensure compliance & regulations related to overseas navigation as per the standards and best practices.

The ground level implementation of maritime and bunker fuel regulation at Fujairah port is done by the local authorities such as Fujairah Environment Agency, Fujairah oil industry zone and Marine department of Port of Fujairah.



International Maritime Organization (IMO)

- Frames international maritime laws and regulations for 175 member countries including UAE
- The IMO is a specialized agency of the United Nations and its primary purpose is to develop and maintain a comprehensive regulatory framework for shipping including maritime safety, environmental concerns, legal matters, technical cooperation, maritime security and shipping sector efficiency related matters



The Supreme Council for Financial and Economic Affairs in Abu Dhabi

Sets policies at Abu Dhabi level

- Regulatory oversight of Abu Dhabi's financial, investment, economic, petroleum and natural resources affairs
- This entity holds policy-making responsibility for the petroleum industry in Abu Dhabi and exercises management control over the Abu Dhabi government owned oil company, ADNOC. The council plays a crucial role in shaping the overarching policies that govern the sector ⁶⁰ [Link](#)



Ministry of Energy and Infrastructure

Promotes and implements IMO rules and regulations at UAE ports and overall maritime sector

- Implementation of IMO & MARPOL regulations at UAE ports
- Launch initiatives related to UAE maritime sector growth & decarbonization
- This ministry is involved in planning the growth of maritime sector in UAE and promotes the attractiveness of UAE ports. Recently in 2023, the ministry was elected to the executive council of the IMO in category B for third time in a row. It promotes adoption of modern technologies and digital solutions, consolidates and directs the port authorities to comply with international standards of maritime safety, fuel quality and specifications and protects the marine environment within the UAE ports ⁶¹ [Link](#)



Ministry of Climate Change and Environment (MOCCAE)

Manages all aspects related to international and UAE level climate change affairs

- Enforces regulations related to air and water quality including the emissions from bunker fuel
- This ministry enforces environmental regulations for air and water quality, impacting fuel specifications and emission standards for bunker fuels. Its role becomes crucial in ensuring sustainable bunkering practices ⁶² [Link](#)



Federal Transport Authority (FTA)

Ensuring compliance related to overseas navigation regulations with the international standards at UAE level

- Framing rules related to navigation in the state ports for good traffic and provision of the proper marine facilities
- Maintain regulatory oversight related to maritime safety and security standards
- The FTA oversees maritime safety and security standards across the UAE, including aspects like bunkering operations, vessel inspections, and personnel certification. Its regulations set the national framework for safe and secure bunkering practices ⁶³ [Link](#)



Federal Customs Authority (FCA)

Emirati governmental authority which draws custom policies in cooperation with customs administrations

- The FCA regulates customs procedures for goods entering and leaving the UAE, including bunker fuels. Oversees bunkering licenses and complying with customs clearances falling under their purview ⁶⁴ [Link](#)



Fujairah Environment Agency (FEA)

Concerned entity for environmental affairs management in the Emirate of Fujairah

- Monitoring and enforcement of environmental regulations specific to Fujairah's bunkering activities
- This agency plays a critical role in monitoring and enforcing environmental regulations specific to Fujairah's bunkering activities. The FEA conducts inspections, issues permit, and levies penalties for non-compliance, ensuring environmental protection within the port ⁶⁵ [Link](#)



Fujairah Oil Industry Zone (FOIZ)

Regulates the companies operating in Fujairah Oil Industry Zone

- This free zone dedicated to the oil & gas industry facilitates bunkering activities and provides regulatory oversight within its jurisdiction. Companies operating within the FOIZ must comply with its specific regulations in addition to national frameworks ⁶⁶ [Link](#)
- Regulation of all aspects of the hydrocarbon industry with absolute care to health, safety, quality and environment
- Ensuring compliance for the companies operating within the FOIZ in alignment with national and international frameworks

Regulates the bunker fuel business at Fujairah port

- Issuance of Bunker fuel supplier license
- Ensuring adherence to bunker fuel quality specifications as per MARPOL Annex VI
- Selection of independent bunker surveyors and testing companies
- Ensuring safety and security standards of vessels calling at Fujairah port, Oil terminals, SPMs and Anchorage area, consistent with SOLAS¹, MARPOL regulations and ISGOTT² and OCIMF³ guidelines
- Guidelines and certifications related to communication channels from vessels calls till port clearances
- Guidelines related to supplier portal guidelines
- Issuance of marine certificates and health and safety standards
- Manages the port infrastructure and operations in Fujairah, including bunkering facilities. They issue bunkering licenses, regulate storage and handling of bunker fuels, and ensure adherence to safety protocols within the port premises ⁶⁷ [Link](#)

¹SOLAS: International convention for the safety of life at sea, 1974

²ISGOTT: International Safety Guide for oil tankers and terminals

³OCIMF: Oil Companies International Marine Forum

7.1.2 Key acts and laws

Regulatory framework and compliance

The UAE's regulatory framework in the bunkering sector is geared towards maintaining high standards of operation, focusing on safety, environmental protection, and quality assurance.

The recent drive towards establishing unified bunkering standards, led by the DMCA (Dubai Maritime City Authority), underscores the commitment to enhancing the quality of bunkers and enforcing stringent regulations that minimizes environmental footprints and enhances safety in the industry. ⁶⁹ [Link](#)

Here are some environmental, quality, and safety regulations that are important to know and understand:

Environmental regulations

- **MARPOL Annex VI:** This international regulation restricts emissions from ships, impacting bunker fuel specifications. The UAE implements these regulations stringently, with the FEA conducting regular inspections and enforcing strict penalties for non-compliance. ⁶⁹ [Link](#)
- **IMO 2020 Sulphur Cap:** This regulation, effective January 2020, limits Sulphur content in marine fuels to 0.5% globally. The UAE proactively adopted this, making it a frontrunner in environmental compliance. Since then, statistics show a significant decline in Sulphur emissions from ships bunkering in Fujairah. ⁷⁰ [Link](#)

Safety regulations

- **IMO safety standards:** The UAE adheres to strict IMO safety regulations, covering aspects like bunker quality, equipment, and personnel training. The FEA conducts regular audits and certifications to ensure compliance. ⁷⁰ [Link](#)
- **Bunker Delivery Agreements (BDAs):** These legally binding agreements detail specifications, delivery procedures, and liability clauses for bunker fuels. Ensuring clear and comprehensive BDAs is crucial for mitigating risks and resolving disputes. ⁷¹ [Link](#)

Quality regulations

- **ISO standards:** The UAE implements various ISO standards for bunker fuel quality, testing, and analysis. For example, ISO 8217 specifies fuel properties to ensure compatibility with ship engines. Stringent adherence to these standards guarantees reliable fuel performance and minimizes engine damage. ⁷² [Link](#)
- **Fuel testing and inspection:** Accredited laboratories are critical in verifying bunker fuel quality. Fujairah boasts several independent laboratories offering pre-and post-delivery testing services, protecting both suppliers and buyers. ⁷³ [Link](#)

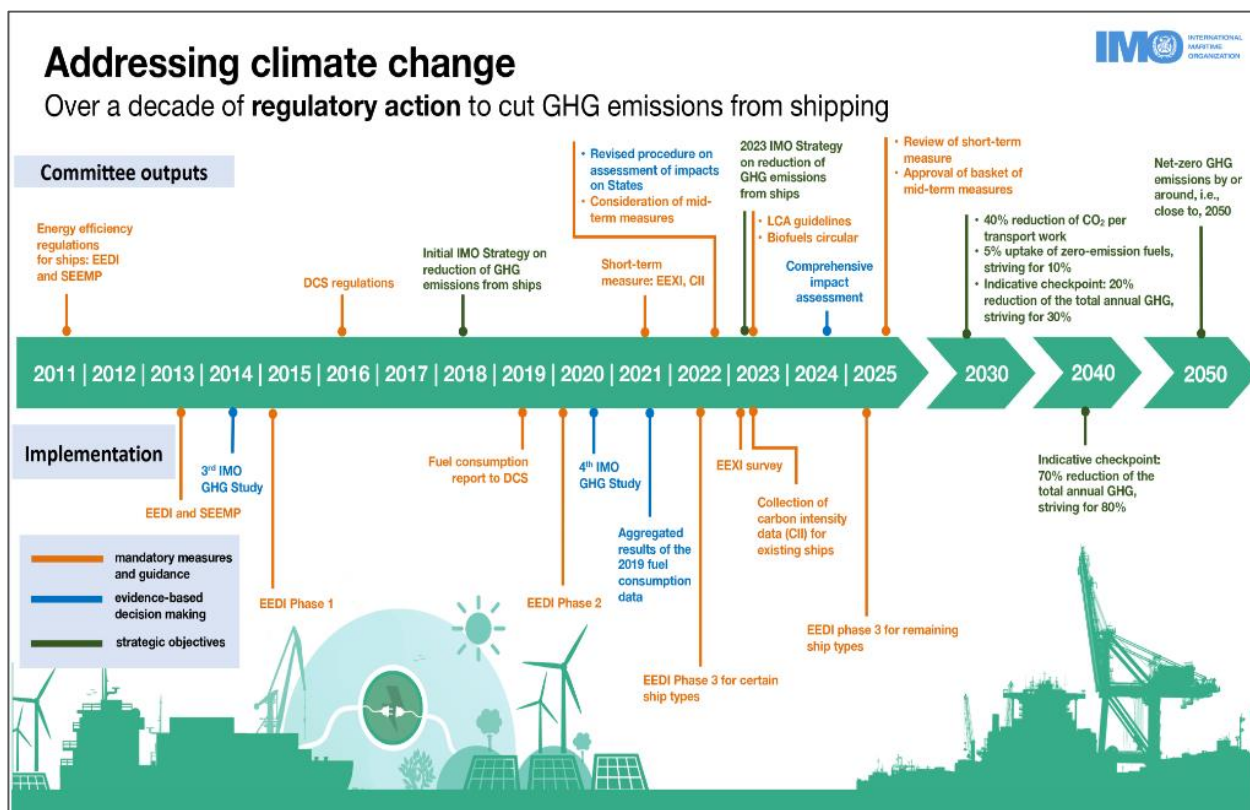
Additional regulations

- **Waste management:** The UAE has robust regulations for handling and disposing of used oils and oily waste generated during bunkering operations. Companies must adhere to strict guidelines to prevent environmental pollution. ⁷⁴ [Link](#)
- **Maritime security:** The UAE implements comprehensive maritime security measures to combat piracy and terrorism. Bunkering operations fall under these regulations, requiring specific procedures and protocols to ensure the safety of vessels and personnel. Maritime security and piracy follow International Ship and Port Facility Security (ISPS) code. Divided in two sections, the ISPS code contains detailed security related requirements for governments, port authorities and shipping companies in UAE. ⁷⁵ [Link](#)

7.1.3 Regulatory challenges

- Regulatory hurdles are a significant concern in the UAE bunkering industry. Fujairah's 2020 implementation of the IMO 2020 Sulphur Cap led to a significant 50% decrease in high-Sulphur fuel sales compared to pre-2020 levels [Link](#) ⁴⁹. However, on the other hand, IMO 2020 Sulphur cap regulations favors Fujairah market due to its close vicinity to Asian and ME based complex refineries which are capable of producing and supplying relatively cheaper LSFO in comparison to Europe based refineries.
 - The 2023 IMO GHG Strategy specifically outlines a goal to decrease the carbon intensity of international shipping, aiming to reduce CO2 emissions per transport work by at least 40% by 2030, calculated as an average across the global shipping industry. This elevated ambition focuses on the adoption of zero or near-zero greenhouse gas (GHG) emission technologies, fuels, and/or energy sources, with the aim that they constitute a minimum of 5%, with a target of 10%, of the energy utilized by international shipping by 2030.
- Stringent air and water quality regulations by MOCCA (UAE Ministry of Climate Change and Environment) add complexities to fuel specifications and emission standards. While crucial for environmental sustainability, these regulations often require substantial investment in technology and processes to ensure compliance
- The implementation of the International Air Pollution Prevention Certificate (IAPP) as mandated by the UAE Federal Maritime Administration is a case in point, requiring ships to comply with strict Sulphur content limits in bunker fuels ^{76 77} [Link](#) [Link](#)
- Delays in obtaining environmental permits from the FEA cause operational bottlenecks for some suppliers

Figure 7.1: IMO regulatory plan of action to cut GHG emissions from shipping



Source: IMO

Safety and security assessment

The UAE's bunkering industry prioritizes stringent safety and security measures, given the critical nature of oil and fuel operations. This focus is evident in various aspects, from operational protocols to emergency response strategies.

Operational safety standards

- **Compliance with international safety standards:** The UAE adheres to international safety standards such as the International Safety Management (ISM) Code. According to the IMO, compliance with these standards significantly reduces maritime accidents and pollution. ⁷⁸[Link](#)
- **Investment in safety training and equipment:** As reported by the Emirates News Agency, substantial investments have been made in safety training programs and safety equipment for personnel involved in bunkering operations, aligning with global best practices. ⁷⁹[Link](#)

Security measures against external threats

In response to regional geopolitical tensions, the UAE has invested in advanced maritime security systems. As per the UAE's Ministry of Energy and Infrastructure, these include surveillance technologies and coordinated patrols, substantially reducing security incidents in ports like Fujairah.

Also, UAE has joined the International Alliance for the safety and maritime navigation, a US led coalition to secure sea lanes and maritime trade in the region. The coalition plan includes protecting the broader Persian Gulf region and surveillance of Strait of Hormuz. ⁸⁰ [Link](#)

Safety in fuel handling and quality control

The Emirates Authority for Standardization and Metrology (ESMA) mandates regular fuel quality inspections. These inspections have led to a marked improvement in fuel quality, as indicated in ESMA's annual reports.

Fujairah utilizes multiple third-party bunker fuel inspection service agencies to ensure bunker fuel quality and quantity.

Also, the suppliers are introducing digital solutions such as Advanced Delivery Platform (ADP) which provides better visibility into all details of their bunkering operations including quality of fuel delivered, operational timelines and quantity received as per mass flow meter profiles.⁸¹ [Link](#)

Cyber security measures

With the digitization of port operations, the UAE has increased its investment in cyber security. According to the Telecommunications and Digital Government Regulatory Authority, cyber security budgets in maritime operations have grown, enhancing the protection of digital assets and operational data.

Emergency response and disaster management

The UAE has established rapid response mechanisms for oil spills. The National Emergency Crisis and Disasters Management Authority (NCEMA) reports a significant reduction in response time to maritime incidents, enhancing environmental safety and operational continuity.⁸² [Link](#)

Environmental safety considerations

The UAE's compliance with MARPOL Annex VI, aimed at reducing Sulphur oxide emissions, is a testament to its commitment to environmental safety. The Federal Transport Authority - Land & Maritime (FTA) mandates strict adherence to these regulations in all UAE ports.

Country risk and investment climate analysis

The UAE presents a favorable investment climate for the bunkering industry, characterized by political stability, economic diversity, a strong legal framework, commitment to sustainability, and a strategic geographical location.

However, staying attuned to regional geopolitical dynamics and global economic shifts is vital for navigating the investment landscape effectively. Let's evaluate the important factors.

7.1.4 Regulatory Risk Assessment

Political stability and governance

Risk: Medium to low

World Bank's UAE Political Stability Index score: 0.7 (as of 2022) (-2.5 weak; 2.5 strong)⁸³ [Link](#)

The UAE is known for its political stability, a key factor in attracting foreign investments. The World Bank's governance indicators consistently rank the UAE in the medium to high range in terms of political stability and absence of violence.

As per Economist intelligence forecast, UAE will be politically stable through the 2024-28 period.⁸⁴ [Link](#)

Additionally, the UAE government actively supports the maritime and bunkering sector, with initiatives like the **Dubai Maritime Vision 2030** to enhance the industry. Such government backing reduces the political risk for investors.

The Ministry of Energy and Infrastructure has launched a strategic partnership with Marihub, a leading e-commerce provider, to develop value-added services and digital innovative solutions to boost the competitiveness of the UAE's maritime sector. [Link](#). Also, the ministry launched the "**Blue Pass**" project in collaboration with Marihub, which will create quality packages and facilities aiming to increase the foreign direct investment in UAE's maritime sector.⁸⁶ [Link](#)

The UAE's Ministry of Energy and infrastructure actively participates in the IMO council and falls under the Category B which includes – '10 states with the largest interest in international sea borne trade'.⁸⁷ [Link](#)

Economic factors and market dynamics

Risk: Low

The UAE's efforts in diversifying its economy beyond oil, as outlined in the UAE Vision 2021, reduce dependence on a single sector and add to the economic stability, positively impacting the bunkering industry.

Also, as a critical trading hub, the UAE's strategic location at the crossroads of major international trade routes bolsters the bunkering sector. This is reflected in Fujairah port's growth, one of the world's top bunkering hubs, as reported by the Fujairah Oil Industry Zone (FOIZ).

As per UAE's GDP growth forecast by IMF and SwissRe institute, the UAE's economy is projected to grow at a healthy growth rate of **approximately 4% in 2024 and 2025** which is **well ahead in comparison to GDP growth forecast for KSA, GCC countries, US and EU countries**.^{88 89} [Link](#) [Link](#)

The Ministry of Energy and Infrastructure has initiated the UAE Maritime cluster with the aim of advancing the global presence of the UAE's maritime sector and developing alternative electronic components to complement the oil sector. Its primary goal is to establish a consortium that unites various organizations and individuals within the UAE maritime industry, harnessing their collective capabilities to bolster the country's global standing.⁹⁰ [Link](#)

Legal and regulatory framework

Risk: Low

The UAE has a well-developed legal framework governing maritime and bunkering operations. The Federal Maritime Law and local emirate-level regulations provide clarity and consistency, reducing legal risks for investors. Compliance with international maritime laws, such as MARPOL Annex VI, and initiatives to develop local regulations in line with global standards, foster a regulatory environment conducive for investment.

The regulatory authorities of UAE such as Ministry of Energy and Infrastructure and UAE Federal Maritime Administration have made the rules stricter for HSFO bunkering operations in accordance to regulation 4 and 14 of Annex VI of Marpol. Implemented in 2022, the ships planning to receive bunker with Sulphur content of more than 0.5% need to provide the port authority with the copy of International Air Pollution certificate (IAPP Certificate), however under the previous system, suppliers were only required to take a confirmation from the receiving vessel that the HSFO will be used in a MARPOL compliant way.⁵⁴ [Link](#) [Link](#)

Environmental compliance and sustainability

Risk: Low

The UAE's commitment to sustainability, as part of its Energy Strategy 2050, impacts the bunkering industry by promoting cleaner fuels and environmentally friendly practices, aligning with global decarbonization trends.

The increasing investment in renewable energy sources in the UAE also positions the country as a leader in sustainable energy, offering opportunities in green bunkering solutions.

Fujairah's proximity to ME and Asian refineries and cheap availability of middle distillates gives it an added advantage of low Sulphur fuel oil production at the port at competitive price. Also, in 2023, Vitol completed its first Biofuel delivery in Fujairah aligning with the vision set by United Nations Climate Change Conference COP28 in Dubai. Players like Dubai's Tristar Group has placed an order with Turkey's Akdeniz Shipyard to build an electric hybrid bunker barge for the UAE market.^{91 92} [Link](#) [Link](#)

Geopolitical risks and global economic trends

Medium to Low

While the region has its geopolitical complexities, the UAE maintains a relatively stable position. However, investors need to stay informed about regional developments that could impact maritime trade routes and the bunkering market.

Recent attacks against commercial vessels in the Red Sea and disruption of key trade route linking Europe and North America with Asia through the Suez Canal, will favor the increase in bunkering demand at Fujairah port, as the exporters and shippers look for alternative ocean transportation route. ⁹³ [Link](#)

Global economic trends, including oil price fluctuations and international trade policies influence the UAE's bunkering sector. Understanding these trends is essential for making informed investment decisions.

7.2 Historical JVs

The UAE's bunkering landscape, particularly Fujairah's, has witnessed numerous JVs woven through its history. These collaborations, forged between international oil giants and local expertise, have fueled the sector's rise to prominence. Key reasons for some of the historical JVs within the bunkering fuel market include.

Strategic alignment	Aligning the goals and objectives of both local and international partners ensuring mutual benefits and long-term sustainability of the venture
Entry into new market	Utilizing local market knowledge capabilities and know-how
Technology and innovation	Embracing technological advancements and innovative practices has been a key driver in the success of these ventures, particularly in adapting to environmental regulations and operational efficiency

7.2.1 Examples of key JVs and the impact on the entities

JV Partners	Year of JV	Objective	Impact on JV	Benefits/ Synergies
ENOC and Vopak (Fujairah)	1996	Partnership started with the commencement of Jebel Ali Fujairah Terminal, a leviathan of storage capacity that attracted new clients and cemented Fujairah's position as a global bunkering hub	The storage terminal has gone through an expansion phase as well, since its inception at Fujairah port	ENOC uses the tanks run by the Vopak Horizon venture for fuel trading and uses tank space it holds on its own to store and blend fuel for domestic use
BP and Sinopec (Started with China and later the JV entered Fujairah) ⁹⁷ Link	2015	<u>BP and Sinopec</u> Leveraging each other's existing bunkering locations and activities to deliver marine fuel to their customers <u>BP</u> Enter into restricted Chinese market and utilize Sinopec's local capabilities	The JV has been successful and the two companies are extending their cooperation in other areas such as fuel sales, oil and gas trading, and upstream activities	- BP utilizes this JV to enter Chinese market and utilize Sinopec's bunker fuel production capabilities and presence at bunkering locations - Sinopec will utilize BP's expanded network to increase its presence globally

JV Partners	Year of JV	Objective	Impact on JV	Benefits/ Synergies
DP World Jebel Ali Port and Maersk Supply Service ⁹⁴ Link (Jebel Ali Port, UAE)	2018	Collaboration on various aspects of service delivery and work towards a common goal of decarbonizing logistics	• This partnership comes post a strategic partnership with the Maersk Mc-Kinney Moller center for zero carbon shipping • This second partnership portrays their confidence in each other to achieve their common goal of de-carbonization	Prioritized berthing for Maersk vessels, support for Maersk's customers and implementation of new processes to improve quayside productivity, all leading to faster turnaround times at Jebel Ali port and reduced bunker fuel consumption
GPS and Chemoil ⁹⁸ Link (Fujairah, UAE)	2021	JV between Gulf Petrol Supplies and HG Storage International	The JV runs a storage terminal at Fujairah which has gone through an expansion phase since its inception	JV focused on utilizing the partner's core business strengths and geographical presence Chemoil, listed Bunker fuel supplier at Singapore enters ME/UAE market through this JV, as GPS has a strong presence in ME
Forecast Technology Ltd (FTL) and Blockchain Labs for Open Collaboration (BLOC) ⁹⁵ Link	2019	Joint development of an integrated solution for end-to-end tracking and tracing of the marine fuel supply chain	The jointly developed product started getting utilized within a year of launch and got its first customer as Marfin management S.A.M	Utilization of each other's technological capabilities i.e. FTL: Synthetic DNA tracing solutions BLOC: Blockchain technology Co-offering a product named as BunkerTrace . Traceability is achieved through introducing FTL's synthetic DNA into the fuel and recording data in a secure, private blockchain to create an immutable digital record of the fuel, its quality parameters, physical movements and alterations

JV Partners	Year of JV	Objective	Impact on JV	Benefits/ Synergies
Trafigura in Fuel JV with Frontline, Golden Ocean Link	2019	Trafigura secured long term contract with its customers i.e. Frontline and Golden Ocean which enabled it supply bunker fuel at competitive prices	This JV is still active and Trafigura enjoys economies of scale with its JV partners or customers	The JV enabled increased base volumes (economies of scale) and greater access to both infrastructure and credit which enabled Trafigura to provide competitive bunkering supply services to its venture partners

Benefits: Collaborations with international oil companies under concession agreements have enabled the infusion of technology, expertise, and capital, thereby bolstering the UAE's position in the global oil market.

Challenges: However, these ventures have also faced challenges, such as navigating complex regulatory environments, managing cultural and operational differences between partners, and adapting to rapidly changing market conditions.

7.2.2 Synergy evaluation – Proposed JV between Axiom Global and IOC ME

Axiom Global is encountering numerous business challenges, including compliance with stringent environmental regulations, maintaining competitive pricing, enhancing customer service effectiveness and transparent communication mechanisms, adapting to the emergence of alternative refueling options, ensuring consistent availability of high-quality, unadulterated fuel, managing supply chain disruptions, and reducing operational expenses

Current and future challenges from supplier's perspective in UAE bunkering market	Mitigation strategy	Gained synergies through Axiom Global and IOC ME JV
Strict environmental regulations related to bunkering operations and increasing LSFO demand due to IMO 2020 sulphur cap regulations	– Maintaining availability of LSFO through relationship based long term contracts with ME and Asia based refiners – Implementation of best practices related to environment friendly bunkering operations	<u>IOC ME</u> IOC ME can utilize its' LSFO production capability to a better extent as Fujairah is one of the major suppliers of LSFO and VLSFO as a bunker fuel in ME region <u>Axiom Global</u> – Axiom can utilize IOC's LSFO Bunker fuel and middle distillates production and supplying capabilities – Also, it can utilize the IOC's local network of ME and Asian based refiners ensuring LSFO availability through flexible and alternate long-term supply contracts

Current and future challenges from supplier's perspective in UAE bunkering market	Mitigation strategy	Gained synergies through Axiom Global and IOC ME JV
Emergence of bunkering options at adjacent ports in UAE such as Khorfakkan and ports in neighboring countries like KSA, Qatar, Oman and Egypt	– Maintaining availability of LSFO through relationship based long term contracts with ME and Asia based refiners	<u>IOC ME</u> IOC ME will be able to increase its presence and secure a certain percentage of fixed market share in Fujairah and Khorfakkan port through this JV. Also, it can explore other JV partners in the ME region and target a certain market share at other emerging bunkering locations in the ME region <u>Axiom Global</u> – JV with IOC ME can help Axiom increasing its customer base in other neighboring countries utilizing IOC's base of customers
Price wars due to increasing number of suppliers	– Competitive pricing through long term contracts with refiners – Better demand prediction through digital tools – Maintaining long term relationship with customers and vessel operators	<u>IOC ME</u> IOC ME will be able to save marketing and sales expenses through securing fixed contracts for supplying bunker fuel at Fujairah port. It can utilize its savings for increasing its presence at other adjacent ports <u>Axiom Global</u> – Axiom will be able to provide relatively competitive prices to its customers through low cost procurement of bunker fuel from its' JV partner IOCL – Establishing long-term relationships for bunker fuel procurement will bolster Axiom's confidence in its supply capabilities, particularly during periods when customers expect rapid turnaround of bunkering operations – JV will help in predicting price fluctuations well in advance

Current and future challenges from supplier's perspective in UAE bunkering market	Mitigation strategy	Gained synergies through Axiom Global and IOC ME JV
Adapting to emerging alternative refueling options such as LNG, Methanol, Ammonia and LPG	– Build capabilities related to LNG and Biofuel and alternative fuel bunkering	<u>IOC ME</u> IOC ME will get a good marketing and promotional platform for its alternative bunker fuel products through this JV. It will be easier for the organization to test its alternative bunker fuel products with the long-term customers it will be able to capture through this JV <u>Axiom Global</u> – Axiom can utilize IOC's knowledge related to alternative refueling options – It can also utilize IOC production capabilities as the company has already ventured into the production of alternative fuels, biofuels and other innovative products
Availability of compliant and high quality and unadulterated fuel	– Ensure availability of high quality and compliant Bunker fuel	<u>IOC ME</u> – IOC ME will be able to increase its brand image by supplying high quality and compliant Bunker fuel. Also, it can implement a better feedback mechanism with the JV customers over bunker fuel quality which will help the firm to further enhance its product quality through its R&D capabilities and consistent user-based inputs – IOC ME will get benefitted from Axiom Global's expertise in shipping, trading, bunkering, and risk management which help them to capture a significant market share in ME market <u>Axiom Global</u>

Current and future challenges from supplier's perspective in UAE bunkering market	Mitigation strategy	Gained synergies through Axiom Global and IOC ME JV
		– Procurement of bunker fuel through fixed contracts with IOC will help in providing consistent and customized high quality fuel as per the customer's requirements – Partnership will ensure zero variability between the quality of fuel during sampling phase and actual supply during bunkering operations
Supply chain disruptions due to geopolitical tensions and other factors such as non-availability of bunker oil due to high demand	– Build alternate arrangements/long term contracts with multiple refiners in ME and Asia region	<u>Axiom Global</u> – JV with IOC ME will help Axiom in building alternate arrangements with other refiners based in ME and Asia region which can be utilized during supply chain disruptions
Enhanced customer service and transparent communication mechanism	– Clear communication channels ensuring accurate and transparent flow of information	<u>IOC ME</u> JV will ensure consistent and regular feedback over bunker fuel quality and any additional factors <u>Axiom Global</u> – JV will ensure clear communication channels and accurate information flow about origin and composition of the blend components and actual delivered fuel – It will enhance decision making in terms of implementation of innovative and digital technologies related to customer service – It will be easier to establish a standardized and transparent communication process with a JV partner
Reduction of operational expenses and managing working capital	– Reduction in operational expenses through optimum utilization of barges	<u>IOC ME</u> Partnership with Axiom will help IOC ME to save its marketing expenses and increase its market

Current and future challenges from supplier's perspective in UAE bunkering market	Mitigation strategy	Gained synergies through Axiom Global and IOC ME JV
	– Flexibility in paying the fuel procurement cost to the refiner during the times of payment delays from its customers i.e. vessel operators and owners	share by offering relatively cheaper bunker fuel to the JV customers <u>Axiom Global</u> – Partnership with a Bunker fuel producer will help Axiom in managing the working capital as it will have relatively more flexibility in terms of paying the fuel procurement cost to IOC – Regular and fixed contract based supply from IOC will help Axiom in planning the optimum utilization of its barges
Challenges associated to be up-to-date in terms of innovative technologies related to environment sustainability, ship engine efficiency & speed loss and alternative fuels	– Collaborative R&D over innovative technologies with its partners	<u>IOC ME</u> – IOC ME will be able to better utilize its R&D capabilities as it will get consistent, reliable and regular inputs from JV customers – Also, it will be easier for the firm to conduct pilot tests for its R&D based products and technologies <u>Axiom Global</u> – Axiom can complement its local market knowledge with R&D capabilities of IOC. IOC has done huge investment in R&D and runs Asia's best R&D facility in the areas of lubricants and bunker fuel production processes Link

Source: Source: Competitor website, Primary research, D&B Analysis

7.2.3 Strategic insights for potential investors in the UAE bunkering industry

The UAE's bunkering industry, particularly in Fujairah and other major ports, represents a dynamic and strategically significant sector in the global maritime landscape.

For potential investors and JV partners, the insights gleaned from this comprehensive analysis offer a roadmap to navigate the complexities and harness the opportunities within this industry.

1. Leverage political and economic stability: Investors should capitalize on the UAE's political stability and progressive economic policies. As per economic vision 2030, Abu Dhabi aims to achieve a 64% contribution to GDP by the non-oil sectors such as Maritime, tourism and other sectors. [Link](#)

The stability and growth prospects are reflected in the UAE's GDP, which, according to the World Bank, has been on a steady rise, indicative of a robust economy.

2. Navigate regulatory frameworks: Understanding and adapting to the UAE's regulatory landscape is crucial. The implementation of international standards like MARPOL Annex VI by the Federal maritime authority underscores the country's alignment with global environmental and safety norms. Investors should prioritize compliance and leverage regulatory knowledge for strategic advantage.

3. Capitalize on the strategic location and infrastructure: The strategic location of Fujairah and other UAE ports, serving key global shipping routes, presents significant opportunities for bunkering operations.

According to the Port of Fujairah, the volume of bunkering services has seen consistent growth, positioning it as a key player in the global market. Investors can tap into this potential by developing services and solutions that cater to the diverse needs of international shipping companies.

Recent (2023) attacks at Red Sea are expected to increase demand at Fujairah port while its proximity to Middle East and Asian refineries favors the production of LSFO locally which provides an advantage to the suppliers to supply LSFO at competitive rates in comparison to other European and Russian ports.

4. Embrace sustainability and innovation: With the global shift towards sustainability, the UAE's focus on green initiatives and renewable energy investments offers avenues for innovative bunkering solutions. The Energy Strategy 2050 paves the way for sustainable practices in the industry. Investors should explore opportunities in alternative fuels and eco-friendly technologies to align with this trend.

5. Mitigating geopolitical and market risks: While the UAE is politically stable, the broader regional geopolitical climate can influence the bunkering sector. Investors must stay informed about regional developments and global economic trends, such as oil price fluctuations, to mitigate risks. Utilizing market intelligence and risk assessment tools will be key in making informed investment decisions.

6. Fostering collaborative ventures: The success of past JVs in the UAE highlights the value of collaboration. Combining local expertise with global insights can lead to successful ventures. Potential investors should seek partnerships that align with their strategic objectives and complement the UAE's market dynamics. The local players should look for partnerships which can enable implementation of innovative and digital solutions encouraging transparent processes related to bunker fuel quantity and quality. They should also look for international partners which can implement smart bunkering solutions and can further streamline and digitize the bunkering process.

The UAE's bunkering industry is not just navigating change; it is driving it. It presents opportunities for potential investors.

By marrying strategic foresight with actionable intelligence, investments can contribute to and thrive in the UAE's growing bunkering ecosystem.

7.3 Proposed JV Partner

7.3.1 Financial and operational evaluation

Axiom Global Oil & Gas Trading DMCC (AGOT), is a free zone establishment limited liability company, incorporated on 16th April 2019, with an aim to engage in the businesses of trading of oil, bunkering and providing logistic and consultancy services. It is registered under the registration body of Dubai Multi Commodities Centre (DMCC), with its office situated at Unit No. 304, Indigo Icon Tower, Cluster F, Plot No. JLT-PH1-F3A, Jumeirah Lakes Towers, Dubai, United Arab Emirates (UAE). Its other incorporation, trade license details are as presented in Table below.

Table 7.1: Key corporate details of AGOT

Parameter	Details
Legal structure	Free Zone Establishment with Limited Liability
Registered	April 16, 2019
Start date	April 16, 2019
Control date	April 16, 2019
Trade license number	DMCC-673948
Registration number	DMCC-164822
License expiry date	April 15, 2024
Registration Body	Dubai Multi Commodities Centre, Dubai
Chamber of Commerce no	100504246800003
SIC	5171-0000; 5172-9900; 5172-9901; 5172-9902; 5172-0203; 5172-9905; 4213-9908 Trading of oil and bunkering and service provider of logistic and consultancy.
TAX Registration No	100504246800003
LEI Number	984500WC1BEC063A1460

AGOT primarily deals in the following product categories and services:

- Crude Oil
- Refined Petroleum Products
- Marine Fuel Oil
- Gasoil
- Naphtha
- Bunker Trading & Broking
- Ship Chartering
- Base Oil & Chemicals
- Consulting on Fuel Oil Blending

7.3.2 Review of assets and facilities

In terms of assets and facilities of AGOT, as a company engaged largely into commodities trading business, it is noted that majority of these are allocated towards current assets and liabilities, in form of working capital components, i.e. trade receivables, advances to suppliers, trade payables and advances from customers, and with minor components allocated towards fixed assets and long term liabilities. The salient features of the key assets and liabilities as of December 2022 (latest audited Annual Report) is summarised below.

- **Fixed Assets:** Has two broad components – assets for general business, office setup and its, operations; and investments towards business properties.
 - Assets worth USD 134,114, towards furniture & fixtures, computers and IT assets, motor vehicles and other office equipment
 - Investment in properties worth USD 142,954. These involve three charter long term bunkers taken on lease. As per discussions with AGOT team, this no. has increased to 5 no. of barges. each with a capacity of 11,000 Metric Tonnes (MT)
- **Current Assets:** Out of the total current assets aggregating to USD 22.92 Mn, the key components comprise of the following:
 - Trade receivables: USD 11.76 Mn
 - Advances to suppliers: 4.54 Mn
 - Bank Balance: USD 3.36 Mn and
 - Due from related parties: USD 2.22 Mn
- **Current Liabilities:** Out of the total current liabilities of USD 6.40 Mn, the following are key components
 - Trade payables: USD 3.94 Mn
 - Term loans: 1.49 Mn
 - Advances from customers: 0.88 Mn
- **Non-Current Liabilities:** The long term liabilities are USD 89,313, which include bank loans and provisions for staff gratuity
- **Equity (Networth):** AGOT's network in 2022 was USD 16.71 Mn, which has increased considerably, as compared to USD 3.72 Mn as in 2021. It can be attributed to the y-o-y growth of 138% in revenues coupled with increase in the net profit margin growing to 3.41% in 2022 from 0.84% in 2021

The key observations on the assets & liabilities are as below:

- AGOT's past trend indicates that it's the past two years have been quite different from the initial two years, in terms of allocation of funds towards assets, and also the operating, working capital cycles.
- While the initial two years indicate asset allocation towards company setup, incorporation related activities; there are additions towards property investments for operations in 2022.
- It has successfully grown every year in top line as well as bottom line, contributing in a good manner towards net worth.
- Till year 2021 the working capital cycles were a bit balanced / relaxed with trade receivables being sufficiently balanced by trade payables, whereas in year 2022, the receivables are considerably higher at USD 11.76 Mn (from USD 9.60 Mn in 2021) against payables much lower at USD 3.94 Mn (from 9.83 Mn in 2021). The infusion of term loans (of USD 1.49 Mn in 2022), are of unsecured in nature attracting high interest rates.
- All these aspects indicate, that due to considerable increase in business, there has been need of higher working capital (higher utilization of limits), as compared to that in 2021.

The summary of the key assets and liabilities of AGOT are presented in table below.

Table 7.2: AGOT's Assets – Liabilities overview scenario

Particulars	2020	2021 [#]	2022	2023*
Assets				
Fixed Assets	123,217	127,519	134,114	523,908
Investments	72,975		142,954	863,297
Other Assets	213,564			
Total Non-current Assets	409,756	127,519	277,068	1,387,205
Trade receivables	4,182,360	9,598,526	11,761,524	11,324,398
Advances to suppliers	1,881,400	2,723,668	4,538,764	9,294,882
Due from other parties	6,624	632,442	2,218,096	
Inventory / Goods in transit				250,542
Bank Balances	42,169	2,001,608	3,359,443	1,911,245
Other assets	137,455	557,479	1,041,745	
Total current assets	6,250,008	15,513,723	22,919,572	22,781,067
Total Assets	6,659,764	15,641,242	23,196,642	24,168,272
Liabilities				
Non-current liabilities	69,790	179,977	89,313	1,377,571
Trade payables	1,087,372	9,833,276	3,935,144	4,731,989
Advances from customers	2,301,864	754,953	877,354	
Term loans	-	1,098,715	1,491,347	
Other current liabilities	525,033	55,428	97,290	
Total current liabilities	3,914,269	11,742,372	6,401,135	4,731,989
Total Liabilities	3,984,059	11,922,349	6,490,448	6,109,560
Equity (Net Worth)				
Equity	13,615	13,615	13,615	13,615
Shareholder's	790,341	445,678	1,264,950	1,091,806
Retained	1,871,749	3,259,600	15,427,627	16,953,290
Total	2,675,705	3,718,893	16,706,192	18,058,711
Total Liabilities & Equity	6,659,764	15,641,242	23,196,642	24,168,272

* provisional results for Jan-Jun 2023; # unaudited results of FY 2021

7.3.3 Analysis of financial performance and key ratios

This section discusses the key highlights / salient features of AGOT's financial performance.

- AGOT was incorporated in April 2019, and hence gradually commenced its business in 2019. Thus, effectively, its first full year of operations was 2020.
- It is noted that AGOT achieved a top line (operating turnover) of USD 3.6 Mn in 2019, USD 74.6 Mn in 2020 which further increased to a level of USD 393.6 Mn. This demonstrates steep positive leaps y-o-y AGOT has been able to achieve in a short span of 4 years, which also is due to relatively higher increase in product prices in 2022.
- Along with scaling up the business growth, AGOT has been able to improve its profit margins too in 2022, as compared to the previous years.
- Also, it is pertinent to note that the year 2 and 3 (2020 and 2021) have been the COVID-19 years, which impacted global industrial operations and logistics, and likewise has impacted the Indian region and AGOT's operations. Amid these volatile challenging years too, AGOT has been able to maintain growth through and beyond these years, which is a positive sign of having strong understanding and good insights of their core focus business areas and the geography.

- Coupled with this, 2020 also had brought in the new regime (IMO 2020 regulations) with change in Bunker fuel compliance requirements for sulphur content permissible limits being reduced. This actually proved to be a positive turn for AGOT as Indian refineries are well poised to provide the low sulphur fuels
- The relative stretched profit margins in 2021, despite the considerable growth in top line, is evident of the same; but for the year 2022, AGOT achieved gross profit margin of 4.35% and PAT margin of 3.41%.
- Majority of operating revenues are from trading (in the range of 96% to 99%), with a very marginal income from consultancy services, forming the balance share of revenues.

The key financial performance parameters and key ratios are presented in table below.

Table 7.3: Review of AGOT's past performance

Particulars	2019	2020	2021	2022	H1-2023*
(All figures in USD)					
P&L ratios					
Revenue	3,594,484	74,630,781	165,282,756	393,568,449	135,661,001
Gross Profit	736,301	2,390,306	3,587,062	17,116,958	3,379,461
Gross Profit Margin (%)	20.48	3.2	2.17	4.35	2.49
EBITDA	328,703	1,410,554	2,086,628	15,213,317	2,199,731
Operating Profit Margin (%)	9.14	1.89	1.26	3.87	1.62
EBIT	346,652	1,584,840	2,182,916	15,358,336	2,199,731
EBIT Profit Margin (%)	9.64	2.12	1.32	3.90	1.62
PAT	346,652	1,525,097	1,394,475	13,418,027	1,518,792
Net Profit Margin (%)	9.64	2.04	0.84	3.41	1.12
Other key ratios					
Liquidity ratios					
Quick ratio	2.74	1.59	1.32	3.57	
Current ratio	2.79	1.6	1.32	3.58	
Solvency ratios					
Long term debt to equity	-	0.02	0.04	0	
Total debt to equity	-	0.17	0.34	0.09	
Net debt to equity	-0.02	0.15	-0.28	-0.11	
Efficiency ratios					
Interest coverage ratio	0	26.53	2.77	7.92	
Collection period (days)	103	14	15	10	
Payment period (days)	11	3	12	7	
Operating cash flow margin	-15.17	-0.38	1.41	0.85	

* provisional results for Jan-Jun 2023; # unaudited results of FY 2021

- Presently AGOT manages working capital through internal accruals and commercial debts, which attract quite high interest rates.
- It can be noted that the liquidity ratios have improved significantly in 2022, owing to more than doubled business revenues, with limited and stabilized fixed operating costs, leading to better operating realizations, supported by the lower collection periods. But at the same time, the payment periods also has almost halved from 12 days to 7 days.
- Thus, although there is a 33% improvement in collections period, there is almost a 40% reduction in payments period. Further detailed ratios relating to liquidity ratios are indicated in **Appendix B**.
- For the first half of FY 2023, (Jan-Jun 2023), AGOT has achieved a topline of USD 135.66 Mn, with a EBITDA margin of 1.62% and PAT margin of 1.12%. At these lower revenues and operating scale too, its trade

receivables stand at USD 11.32 Mn, which indicates that the collections of past receivables are stretched. From the details in annexures of provisional statement for this period, it is noted that about USD 3.0 Mn of receivables are more than 4 months old, i.e. prior to 1st March 2023.

7.3.4 AGOT's SWOT Analysis

Strengths

- AGOT is promoted by a highly experienced professional of Oil & Gas industry, having worked extensively in India, Asia region and has in-depth understanding of global markets, with experience spanning across different product segments and across the value chain.
- It is supported by suitably experienced and capable team leaders, traders and analysts, who bring have experience of more than 10-12 years, in relevant trading business of bunkering and Oil & Gas products.
- Having managed business in the focus region of east coast of UAE, in the Fujairah and Khorfakkan ports, AGOT has demonstrated through its high growth in a short span, of good understanding about these markets and has ground level / hands-on expertise, covering the connects with clients as well as the bunker suppliers, and co-ordination with other stakeholders in the value chain.
- AGOT's senior management has good understanding of IOCL having worked with IOCL over two decades heading various business units and operations and India and Sri Lanka. It also has existing business relationship with IOCL through procurement and trade of bunkers in the South Asia region.
- Through all these activities, AGOT has developed a business platform, with capability to grow further in future, adapting to the changing demands of bunker fuel consumption patterns globally.

Weaknesses

- Due to its fast paced growth in last two years, AGOT experiences intermittent shortfalls in working capital, with limited banking limits and support. This in a way, restricts to some extent, its future plans and growth potential.
- AGOT is relatively a young firm of 4 years, and hence although with experienced professionals, would be required to pass through few more years to develop its brand and elevate to the levels of leading players in this region.

Opportunities

- The Middle East region is observing a very dynamic economic and business environment with foreseen growth in the O&G sector, post the recent past volatilities.
- Along with this, massive national level development programmes by few leading countries in ME is expected to lead to considerable increase in domestic and regional transportation requirements, which is indicates an expected upward trend for shipping industry. This in turn is expected to lead to renewed increasing demand for bunkers.
- The changing domestic business scenario is expected to turn the local and regional bunker / lubricants market more attractive. Thus, shall provide more opportunities to cater to in the local / regional markets.
- In view of the JV partners' experience and knowhow in the South Asian region, this would help in tapping the growing opportunities in these regions also, as a secondary focus area, which may bring in a modest contribution to its revenues

Threats

- Few other countries – Qatar, Kingdom of Saudi Arabia (KSA), Egypt & Oman are also envisaging enhancing / developing bunkering ports, although would require time to develop and catch up with Fujairah.
- The Fujairah port presently has few well established players in the bunkering business operating on a larger scale, as compared to present AGOT's levels, and hence it would have to compete directly with these players , if it has to grow its business.

To mitigate these threats, it is essential for AGOT firstly to have presence at the established bunkering hub – Fujairah so as to strengthen / stabilize its position and increase its bunkering volumes, which would require more funds. This time span should also be used to study the regional competing bunker ports more closely in terms of product availability and consumption trends and strategize for the suitable regional opportunities.

Further, as the local / regional market demand is expected to grow in near future, which would imply increase volumes, AGOT can aim to have a share of this pie of business, along with partial contribution coming in as a share from existing overall supplies from Fujairah.

Summarizing the SWOT aspects, it can be noted that a business partner capable of supporting AGOT in scaling up the business through financial / management support, and who enjoys a larger brand awareness / outreach would help AGOT to address these gaps and allow the JV Partner to leverage on AGOT's operational capabilities and business acumen.

8 Business Models for proposed JV

8.1 Business Model

Looking at the scenario wherein IOC ME intends to explore suitable opportunities and modes of entry into the bunkering business in Middle East, it is noted that there are multiple options such as:

- Diversification, solely through its captive capabilities
- Acquisition of an operating business / firm
- Forming a joint venture with an operating and experienced firm.

The critical parameters to be taken note of, in assessing business growth plans are as listed below:

- Present experience of the Company / Group in the proposed line of business
- Know-how of the company about the focus geography, in relation to the proposed products and business activities.
- Required skill sets - status of in-house manpower availability or time and cost required to develop team of skilled manpower.
- Investment levels required
- Gestation time for implementing the plan and commencing operations
- Knowledge and liaison capabilities for confirming compliance to regulatory and local laws
- Level of freehand in decision making during project and operations stage.
- Management control over the proposed business plan
- Exit route options and ease for the same.

Considering these parameters and the three options for business growth / diversification, a qualitative comparative assessment highlighting the pros and cons of each of these key options is shown below.

Table 8.1: Comparative assessment of different business models

Assessment parameters	Captive development	Joint Venture	Acquisition
Present experience of the Company / Group in the proposed line of business	✓	✓✓✓	✓✓✓
Company's know-how about the focus geography, in relation to the proposed products and business activities.	✓	✓✓✓	✓✓✓
Required skill sets - status of in-house manpower availability or time and cost required to develop team of skilled manpower.	✓	✓✓	✓✓✓
Preparedness of the Company to develop these teams	✓	✓✓✓	✓✓✓
Relative scale of investments required	✓✓	✓✓✓	✓
Gestation time for implementing the plan and commencing operations	✓	✓✓✓	✓✓
Knowledge and liaison capabilities for confirming compliance to regulatory and local laws	✓	✓✓✓	✓✓✓
Level of freehand in planning, decision making during project and operations stage.	✓✓✓	✓✓	✓✓✓
Management control over the proposed business plan and realizations / returns from it	✓✓✓	✓	✓✓✓
Exit route options and ease for the same.	✓	✓✓✓	✓

The above table depicts the relative positioning of the different routes / modes for business growth.

■ **Captive development route**

- As can be seen, the captive development business model, to be successful, requires for all the key activities to be arranged / managed on its own – covering the relevant in-house expertise to be developed, training provided to a team, to develop a thorough understanding about the focus markets and products, allocate for the full investments required. For all these tasks and efforts, considerable time and costs have to be allocated, which demands further involvement of more senior members. All these add up to the project costs and contingencies.
- Apart from all these efforts, the exit route also shall be relatively more challenging, as having devised the strategy, having invested higher amount of funds, and implemented the project in its own management and functioning style
- This route brings in one advantage i.e. of the management control and freedom to experiment with own O&M models and optimize as per internal plans.

■ **Joint Venture or Acquisition routes**

- As compared to captive development, the Joint Venture or acquisition route helps in many aspects, in saving time and costs, which otherwise would have to be allocated towards all these developments.
- Also these options can assist in having a developed market base, with partial clientele being suggested or brought in by JV partner, based on their existing experience and business relations.
- It shall help in having relevant and suitably trained team members.
- In both these options, the synergies of key management practices, corporate code of conduct, key policies related to finance, manpower and operations have to be in place, for a successful operation, post a JV or acquisition, as different entities shall come together to have shared authorities and responsibilities.
- In case of acquisition, it is to be noted that, in case of trading businesses, it would be relatively difficult to successfully identify and acquire a good running division or firm, as it is less capital intensive and no one would be willing to part away with a successful ongoing trading business.
- The JV route shall allow the prospective partners to focus on their own strengths and focus areas, and seek complementing support from the other. In this way, the partner bringing in its expertise and experience, maintains its presence in business which acts as a win-win situation.
- Between these two alternate routes, the JV route provides more flexibility, in terms of funds to be allocated, and also as a relatively easier exit route, in case the JV would not have progressed and achieved results as envisaged.

8.1.1 Preferred route for the proposed bunkering business in Fujairah

- Based on review of all options, as above, it is preferable for IOC ME to opt for the JV route, considering its presence in UAE, its strengths of parent group, presence in other allied product segments, but limited knowhow of the proposed business in this geography. The synergies of both JV partners align in terms of business and operational philosophies.
- The JV route can provide the above said advantages in terms of time and cost savings and also allow to venture into this segment with a relatively ready platform for kick-starting the operations much faster.
- Apart from options of sourcing bunkering products from local suppliers, the JV can also source from these form IOCL whenever there is a favourable pricing scenario.
- On the other hand, the JV, leveraging its geographical advantage, can support sourcing and supply of bunker products from UAE local suppliers for IOML and Lanka IOC on need basis, thus scaling its demand requirement for better product price advantage.

Looking at the various aspects of the product / industry, markets, preferred route and proposed JV Partner, an overview rationale for IOC ME, towards the proposed project is presented in the figure below. .

Axiom Global:

In view of AGOT's past performance, its senior leaders' experience in UAE, presence in Fujairah region, in-depth understanding of local markets, its past association and existing business relationship with IOC / IOC ME, the synergies with IOC, indicate it to be a preferred Partner for the proposed JV Company.

Bunkering:

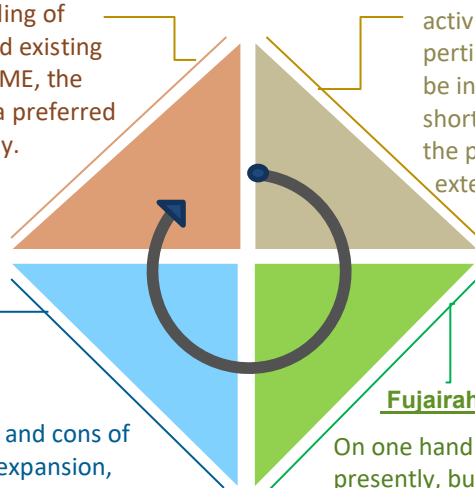
Bunkering is a core and integral part of the shipping industry / value chain and operations. Considering the growing business activities in South Asia, ME, Africa, and the industrial activity dynamics within ME countries, it is pertinent that the demand for bunkers would be increasing in future, irrespective of the short / midterm volatilities and transitions in the product formats. It also blends well as an extension to IOC's other allied products

JV Route:

As explained above, looking into pros and cons of alternative business models (captive expansion, acquisition, JV), looking into the scenario of proposed JV partners - aims and goals, preparedness, complimenting expertise areas and knowhow about products and markets, time required for venturing / establishing presence in new markets, exit options, their synergies, cost of implementation and operations, sharing of authority and responsibilities, risks and rewards, it can be noted that JV option indicates to be a balanced and advantageous route -

Fujairah:

On one hand it is the 3rd largest bunkering hub presently, but considering the volumes, it would be preferable / attractive location for any new entrant into bunkering, as compared to Singapore and Rotterdam (handling much higher volumes). IOC ME already has presence in UAE; well-studied geography, region by AGOT with existing business. UAE, KSA growth plans indicate good local / regional opportunities. Other ports plan for bunkering, so it's a good time, to enter the leading hub in this region, as to have time to develop strong foothold in this region. Its location is strategic, in terms of route options for vessels, as compared to other proposed bunker ports.



- Following this, JV partners shall look at the following steps as to progress to further stages.
 - Continue detailed discussions, at Senior and Middle level of managements, along with completing the in-house due diligence with fine prints (punch lists issued and addressed).
 - Round of discussions on a recheck on legal and regulatory compliance of both the entities, as to be ready to enter into the JV, and to complete any pending documentation, if any, at respective ends. Seek board approvals for the JV and any specific terms / clauses to be addressed, clarifying the no-compete / No Objection conditions, if any.
 - Appointment of consultants, advisors adhering to the local guidelines, as to complete the registration and licensing, permit formalities.

- Agree on the corporate structure at senior level, Board of Directors, Department / Functional heads (HODs), sharing of responsibilities, Standard Operating Procedures for divisional, departmental functioning, approvals, etc.
- Define on overall roles and responsibilities for each of the key positions, key minimum requirements in terms of –duration and format of availability of members required
- Review and consent on the composition of key team members and source – either in-house to be deputed or to be externally hired.
- Indicative quarterly and annual budgets for capex, opex, revenues and profitability; review and control measures, and directives for mitigation measures.

8.1.2 Overview of incorporation process

It is noted that, in UAE, freezones / SEZs are provided with more liberal corporate laws / norms, which makes it easier for foreign entities to setup new companies. The standard benefits through a freezone are as below:

- Zero corporate tax and personal income tax. As per the details on Dubai Multi Commodities Center (DMCC) site, a corporate tax of 9% is applicable for profits more than AED 375000.
- 100% company ownership
- 100% repatriation of funds
- No currency restrictions

The Department of Economic Development (DED) is the nodal authority issuing approvals for setting up new entities in the freezones. The proposed business entity is required to complete the following steps to complete its incorporation, registration and commence operations.

- Define the nature of business activity: Usually, there are three types of licensees
 - Commercial License – for trading activities
 - Professional License – for service providers
 - Industrial License – for manufacturing facilities
- Decide on a Freezone
- Choose a Company name
- Apply for initial approval from DED: Standard document requirements for this step are:
 - Business registration and licensing form
 - A copy of your passport or ID
 - A copy of your residence permit/ visa
 - The company's articles of association
 - Feasibility study of the project
- Open a Corporate Bank Account
- Setup your office location
- Apply for final approval
 - Initial approval receipt and all the previously-submitted documents
 - The lease contract provided by the Real Estate Regulatory Agency (RERA)
 - Duly attested service agent contract (for civil establishments and companies that are 100% owned by non-GCC nationals), the UAE involving a local service agent
 - Approval from other government entities concerned

A given freezone has a set of defined Company rules & regulations, guidelines and Licensing Rules which need to be referred to by the proposed entity while filling in its application form, which will elaborate all the finer details of setting up and operating a business in that freezone.

For DMCC – the documents - *DMCCA Company Regulations – Update Date: 1 January 2022 – Version 2, and Licensing Rules 2020*, provide the insights

It is important to note that a Freezone company can engage into local business (in terms of geography), with another freezone company, but cannot directly trade / transact with a UAE local company registered in the mainland of UAE. To access the actual local markets, a freezone company has to either enter into a commercial agency agreement with a UAE national/company wholly owned by UAE, or establish an onshore company in UAE with majority (51%) shareholding held by a UAE based partner entity. Since recent years, in some cases FDI is allowed, and the minimum shareholding conditions are exempted.

8.2 Corporate Structure

Consultants have had discussions with senior members of both the proposed JV Partner organizations, have reviewed the standard industry practices in this industry, and have assessed the general requirements. Based on this, the key features of the proposed JV are mentioned as below.

8.2.1 Optimal equity structure of JV as per the local regulations for this particular business

- Both the entities proposing to enter into the JV partnership are incorporated and operating in UAE as per the local laws and regulatory compliance. Hence prima facie, there would be no issue in terms of limitation of shareholding on either of the promoter organizations.
- Each of the proposed JV partners is clear on their own prime interests, capabilities, and key focus areas envisaged, and hence, there are no conflict of interest areas.
- IOC ME has more interest in terms of being a partner bringing and support, senior management and in financial support, monitoring the operations at board level, and financing activities, whereas AGOT has interests towards its core strengths of operations – supply chain, trade and customer relationships, marketing & sales, local market knowhow and existing on field presence.
- Hence, it is proposed to have a 50:50 % shareholding to initiate with this JV and equal dividends distribution to the respective shareholders.

8.2.2 Best-fit Organization structure for the JV in UAE market

- The Board of Directors shall comprise an equal number of Directors from both partners, which would be two each, i.e., a total of four Directors. As the two promoting partners would have equal share and shared responsibilities, it would be challenging to have a Chairperson from only one side continuously for a long term. Hence the Chairperson and CFO of the JV Company, can be from one partner and the CEO from another partner. This arrangement would be on rotation basis, for every three years period.
- The JV partners can decide on the sequence of appointing Chairperson. However, considering the UAE location, the nature of business wherein AGOT is presently operating and to kick start the JV business faster and smoother, it may be looked at having the 1st tenure of Chairperson (for initial three years), represented by Axiom. This shall allow IOC ME members to be more familiar with the UAE and bunker trading business from ME perspective and gradually stabilize in the finance functions, and monitoring arrangement.
- As the JV proposes to adopt an asset light business model, i.e. with less capital outlay and expenditure towards development or procurement of long term fixed assets, and more focused towards operations based trading business model, it is better to have a lean structure, with lesser vertical hierarchies, and a limited team addressing the core requirements, as quite a few of the services of storage / logistics would be outsourced.

8.2.3 Manpower requirement

The overview manpower requirements for the proposed JV, indicating the key positions and personnel, is as listed below.

Table 8.2: Manpower requirement for the proposed JV

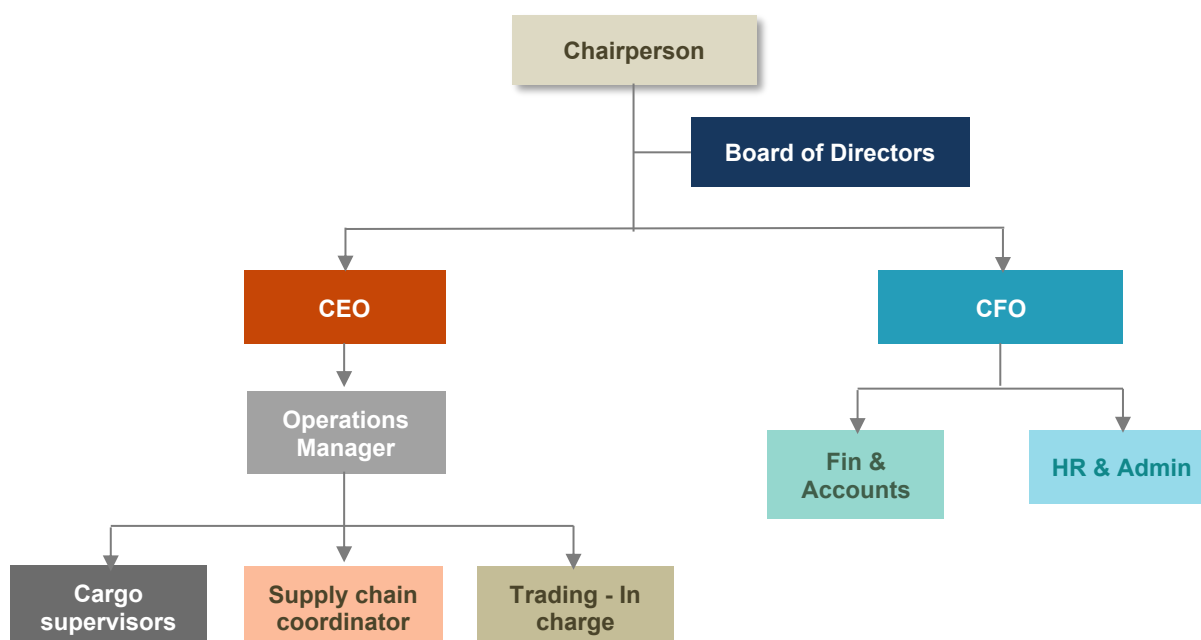
Position	Nos.	Remarks
Chief Executive Officer (CEO)	1	From AGOT
Chief Financial Officer (CFO)	1	From IOC ME
Operations Manager	1	Employee of JV Company and reporting to CEO
Cargo Supervisor	8	Fresh hiring to be done locally
Supply chain coordinator	1	Fresh hiring to be done locally
Finance and Accounts Managers	2	Locally posted and reporting to CFO
Human Resources and Admin	1	To be hired locally
Sales Head	1	Employee of JV Company and reporting to CEO

Apart from these key positions, there would be some mid and junior staff members required for office, administration activities and day-to-day functioning of company.

8.2.4 Organization Structure

Based on the manpower requirements defined above, the proposed organogram indicating key positions is as shown below.

Figure 8.1: Corporate structure for the proposed JV



9 Financial assessment

9.1 Capital Investment Plan

9.1.1 Project Cost

This chapter presents the financial analysis of this Project, covering the capital expenditure plan and the projected operating performance. The summary of capital expenditure is as shown in table below.

Table 9.1: Proposed Capital Expenditure

Sr. No.	Description	Amount (USD Mn)
1	Land and Land Development	-
2	Buildings and Civil works	0.005
3	Plant & Machinery	2.71
4	Miscellaneous Fixed Assets	0.01
5	Contingency	0.27
6	Preliminary & Pre-operative expenses (PPE)	0.76
8	Working Capital Margin Money	8.72
Total Project Cost		12.47

Source: Client / CRIF

It is to be noted that, as this project largely entails trading business, there are no capital costs towards land. Also, it is envisaged not to invest in creation of building, P&M assets, and instead, the required transportation and mobile equipment assets (barges) are proposed to be taken up on lease basis. Hence only the following hard costs are envisaged:

- **Buildings:** Office rent for 3 months – USD 4,500
- **P&M Costs:** Barge lease costs for small initial tenure of 3 months, till the JV company's actual operations kick start, are considered for the P&M costs, amounting to USD 1.36 Mn.

Apart from this, the capital costs include:

- **Miscellaneous Fixed Assets:** USD 10,000 towards IT assets – Hardware and software
- **Preliminary & preoperative expenses (PPE):** Major portion proposed towards securing the initial lot of products, for trade, and other preliminary & preoperative costs of company incorporation, office setup costs, travel and transportation, only the lease actual capital asset formation

Details of Preliminary and Preoperative expenses are as below.

Table 9.2: Assumptions for PPE

Particulars	Unit Cost (USD)		Amount (USD)
Annual License fee	500,000	1	500,000
Company incorporation expenses	15,000	1	15,000
Manpower expenses for 3 months	70,000	3	210,000
Travel and conveyance for 3 months	10,000	3	30,000
Total			755,000

9.1.2 Means of Finance

The project is proposed to be funded through Equity sources, with a proposed equal contribution by both JV Partners.

9.2 Projected Operating Performance

This section discusses about the projected financial performance of the proposed project.

9.2.1 Overview

- The project involves bunker trading as the key business area.
- In view of the market trends, experience profile of the Client - JV promoters, and the dynamics of the proposed location, it is proposed to focus on the following three bunker products – VLSFO, HSFO and MGO
- Client envisages, if suitable, to consider including lubricants in its product portfolio, as a secondary product line. Hence, the base case overall financial analysis is based on consideration of bunker products only.

9.2.2 Capacities and Product mix

Being a trading business, although there is no such specific installed capacity limitations, based on the envisaged business plan, the JV envisages, targeting about 1.8 MMT of sales covering all bunker products. In view of the product demand, consumption patterns and supply capabilities of the focused location, the following product mix is envisaged.

Table 9.3: Product mix and volumes proposed

Product	Product Mix	Sales Volumes (MTPA)
VLSFO	65%	1,170,000
HSFO	30%	540,000
MGO	05%	90,000
Total		1,800,000

Source: Client / AGOT

The proposed business plan envisages to aim a peak of around 60,000 - 150,000 MT of sales per month, amounting to 1.8 MMTPA. This is aimed to be achieved in four years, starting with 0.72 MMT in year 1, 0.90 MMT by year 2 (with a 25% y-o-y increase), and 1.2 MMT by year 3 (with a 33% y-o-y increase) and 1.8 MMTPA by year 4. Further from year 5 onwards, a y-o-y growth of 3% has been considered.

The key assumptions, output, and resulting performance analysis throughout this chapter are presented for the initial few years, and the financial details/statements for broader project tenure are presented in Appendix C.

Table 9.4: Feedstock availability and plant utilization

Year	1	2	3	4	5	6
Sales Volumes						
y-o-y sales growth	0%	25%	33%	3%	3%	3%
Monthly sales Qty.	60,000	75,000	100,000	150,000	154,500	159,135
Months	12	12	12	12	12	12
annual sales volumes	720,000	900,000	1,200,000	1,800,000	1,854,000	1,909,620
Product wise volumes						
VLSFO	432,000	540,000	780,000	1,170,000	1,297,800	1,336,734
HSFO	252,000	315,000	360,000	540,000	370,800	381,924
MGO	36,000	45,000	60,000	90,000	185,400	190,962
Total	720,000	900,000	1,200,000	1,800,000	1,854,000	1,909,620

Source: Client / CRIF

9.2.3 Sales Mix and unit selling prices

Further to the product mix apportionment, based on the geographical business trends and end user's origin, it is noted that the JV would be required to cater to both the markets – domestic and exports. Hence, the proposed sales mix and unit product prices are shown in the table below.

Table 9.5: Sales Mix and Unit Selling prices

Product	Sales (MTPA)	Sales Mix		Unit Selling Prices (USD/MT)	
		Domestic	Exports	Domestic	Exports
VLSFO	1,170,000	70%	30%	600	610
HSFO	540,000	50%	50%	425	430
MGO	90,000	30%	70%	875	925

A modest 2% per annum growth rate has been considered for the selling prices.

9.2.4 Operating Revenues

Based on all the above-mentioned assumptions, covering the business plan of sales volumes envisaged, the product and sales mix, annual growth, product pricing, annual increase in rates, and a 2-day inventory, the expected sales revenues from the bunker trading are arrived at. as shown in table below.

Table 9.6: Operating Revenues

Year	UoM	1	2	3	4	5	6
Sales Volumes							
VLSFO							
Domestic	MTPA	300,720	378,000	504,000	519,120	534,694	550,734
Export	MTPA	128,880	162,000	216,000	222,480	229,154	236,029
Total	MTPA	429,600	540,000	720,000	741,600	763,848	786,763
HSFO							
Domestic	MTPA	125,300	157,325	209,708	216,265	222,753	229,436
Export	MTPA	125,300	157,325	209,708	216,265	222,753	229,436
Total	MTPA	250,600	314,650	419,417	432,530	445,506	458,871
MGO							
Domestic	MTPA	10,740	13,485	17,975	18,537	19,093	19,666
Export	MTPA	25,060	31,465	41,942	43,253	44,551	45,887
Total	MTPA	35,800	44,950	59,917	61,790	63,644	65,553
VLSFO	MTPA	429,600	540,000	720,000	741,600	763,848	786,763
HSFO	MTPA	250,600	314,650	419,417	432,530	445,506	458,871
MGO	MTPA	35,800	44,950	59,917	61,790	63,644	65,553
Total	MTPA	716,000	899,600	1,199,333	1,235,920	1,272,998	1,311,188
Sales Value							
VLSFO							
Domestic	USD Mn	180	236	321	337	354	372
Export	USD Mn	79	103	140	147	154	162
Total	USD Mn	259	339	461	484	509	534
HSFO							
Domestic	USD Mn	53	70	95	99	105	110
Export	USD Mn	54	70	96	101	106	111
Total	USD Mn	107	140	190	200	210	221
MGO							
Domestic	USD Mn	8	10	14	15	15	16
Export	USD Mn	19	25	33	35	37	39
Total	USD Mn	27	35	47	50	52	55
VLSFO	USD Mn	259	339	461	484	509	534
HSFO	USD Mn	107	140	190	200	210	221
MGO	USD Mn	27	35	47	50	52	55

Year	UoM	1	2	3	4	5	6
Total	USD Mn	393	514	698	734	771	810

9.2.5 Expenses

The operating expenses include the landed costs of procuring the products, other costs incurred in managing the entire operating cycle - expenses related to the activities and access charges at the Port area. In addition to this, the fixed costs covering the annual license fees, monthly agency charges, office and administrative costs, travel and transport, manpower, management fees, business promotion/ marketing, subscriptions, etc.

For the products procurement a mix of domestic and imports has been considered as per the mix shown in table below. A gross margin spread of 7% has been considered on the selling price of products, i.e. 93% of costs has been allocated towards the landed cost of procurement of products.

Table 9.7: Procurement source distribution

Product	Source		Spread (%)	
	Domestic	Imports	Domestic	Imports
VLSFO	70%	30%	93%	93%
HSFO	50%	50%	93%	93%
MGO	95%	05%	93%	93%

The assumptions for other costs have been presented in the table below.

Table 9.8: Assumptions for Operating Expenses

Parameters	Assumptions			
Barge Hire charges				
	USD / barge / day	no. of barges	no. of days	USD / year
Lease rent charges	10,000	3	365	10,950,000
Bunker for barges				
	no. of barges	no. of days	USD/MT	USD/year
VLSFO (@ 1 MT/barge/day)	4	365	560	817,600
MGO (@ 1 MT/barge/day)	1	365	765	279,225
Total				1,096,825
Marine Insurance				
		Annual tonnage	USD/MT	USD/year
		1,800,000	0.02	36,000
Wharfage				
		MT/year	USD/MT	USD/year
Outward wharfage costs		1,800,000	0.15	270,000
Other Expenses				
		USD		USD/year
Agency charges per month		3000	12	36,000
Testing costs per month		10000	12	120,000
Annual License fee		500000	1	500,000
Office & Admin per month		10000	12	120,000
BD & Marketing per month		10000	12	120,000
Manpower expenses per month		69,917	12	839,000
Management Expenses per annum		300000	1	300,000

A y-o-y cost increase of 2% has been considered for the product procurement costs, and the operating costs, except the manpower expenses, for which 5% annual increase has been considered. Based on these assumptions, the other operating costs estimates are presented as below.

Table 9.9: Operating Expenses

(All values in USD Mn)

Operating costs	1	2	3	4	5	6
Procurement costs						
Domestic						
VLSFO	168.74	224.16	304.86	320.29	336.50	353.52
HSFO	49.80	66.16	89.98	94.53	99.31	104.34
MGO	27.83	36.25	49.30	75.42	158.48	166.50
Imports						
VLSFO	73.52	97.67	132.83	139.55	146.62	154.04
HSFO	50.39	66.94	91.04	95.64	100.48	105.57
MGO	1.55	2.02	2.74	4.20	8.82	9.26
Total procurement cost	371.83	484.28	669.02	1,023.61	1,135.03	1,192.46
Barge Hire charges	10.95	11.17	11.39	11.62	11.85	12.09
Tank hire charges	0.00	0.00	0.00	0.00	0.00	0.00
bunker for barges	1.10	1.12	1.14	1.16	1.19	1.21
Marine Insurance	0.04	0.04	0.04	0.04	0.04	0.04
Wharfage costs	0.27	0.28	0.28	0.29	0.29	0.30
agency charges	0.04	0.04	0.04	0.04	0.04	0.04
testing costs	0.12	0.12	0.12	0.13	0.13	0.13
license fee	0.50	0.51	0.52	0.53	0.54	0.55
Office & Admin	0.12	0.12	0.12	0.13	0.13	0.13
BD & Marketing	0.12	0.12	0.12	0.13	0.13	0.13
manpower expenses	0.84	0.88	0.92	0.97	1.02	1.07
Mgmt. Expenses	0.30	0.31	0.31	0.32	0.32	0.33
Total Other Expenses	14.39	14.70	15.02	15.35	15.69	16.03
Total Operating Costs	386.2	499.0	684.0	1,039.0	1,150.7	1,208.5

9.2.6 Other key assumptions

Table 9.10: Assumptions for Project Evaluation

Description	Unit	Value
Funding – Equity	%	100%
Cost of Equity	%	15.00%
Interest Rate on LT Loans	%	8.00%
Cost Of Capital	%	15.00%
Interest Rate on Working Capital	%	8.00%
Income Tax Rate	%	9.00%
Depreciation Rates		SLM
Land	%	-
Buildings	%	5.0
P&M	%	15.0
MFA	%	25.0
Inventory Holding Norms		Days
Finished Goods		02
Debtors		30
Creditors		-
Margin Money		25%

9.2.7 Profit & Loss Statement

Based on the above elaborated details of sales, operating expenses and other assumptions of depreciation, interests and taxes, the Profit & Loss statement is as presented below.

Table 9.11: Profit & Loss Statement

		(All values in USD Mn)					
Year		1	2	3	4	5	6
Income							
Operating Revenues	USD Mn	398.8	511.1	706.1	1080.0	1201.2	1262.0
Other Income	USD Mn						
Total Revenues	USD Mn	398.8	511.1	706.1	1080.0	1201.2	1262.0
Expenditure							
Operating costs							
Procurement costs	USD Mn	371.8	484.3	669.0	1023.6	1135.0	1192.5
Other Operating costs	USD Mn	14.4	14.7	15.0	15.3	15.7	16.0
Total Operating Costs	USD Mn	386.2	499.0	684.0	1039.0	1150.7	1208.5
Operating Profit	USD Mn	12.5	12.1	22.0	41.0	50.5	53.5
Profit Margin	%	3.1%	2.4%	3.1%	3.8%	4.2%	4.2%
Depreciation	USD Mn	0.0	0.1	0.1	0.1	0.1	0.1
Interest - LT Loan	USD Mn	0.0	0.0	0.0	0.0	0.0	0.0
Interest – WC	USD Mn	2.1	2.7	3.7	5.7	6.3	6.6
Sub Total		2.1	2.8	3.8	5.8	6.4	6.7
Total Expenses		388.3	501.8	687.9	1044.7	1157.1	1215.2
Profit Before Tax		10.4	9.3	18.2	35.2	44.1	46.8
Tax	USD Mn	0.9	0.8	1.6	3.2	4.0	4.2
Profit After Tax (PAT)	USD Mn	9.5	8.5	16.6	32.1	40.1	42.6
PAT Margin (%)	%	2%	2%	2%	3%	3%	3%
Dividend	USD Mn	1.4	1.3	2.5	4.8	6.0	6.4
Retained Earnings	USD Mn	8.1	7.2	14.1	27.3	34.1	36.2
Margin (%)	%	2%	1%	2%	3%	3%	3%

Based on these estimates of projected performance, the key performance indicators of this project are as below:

- Average EBITDA Margin (%) : 4.43%
- Average PAT Margin (%) : 3.55%
- Net Present Value (NPV) : USD 238.3 Mn.
- Internal Rate of Return (IRR) : 372.9 %
- Debt Service Coverage Ratio (DSCR) : 7.0

The detailed financial statements are indicated in **Appendix C**.

9.2.8 Summary of Ratio analysis

Table 9.12: Ratio Analysis

Parameters	1	2	3	4	5	6	7
Profitability Ratio							
Gross Profit Margin	7%	5%	5%	5%	6%	6%	6%
Operating Profit Margin	3%	2%	3%	4%	4%	4%	5%
EBIT Margin	3%	2%	3%	4%	4%	4%	5%
Net Profit Margin	2%	2%	2%	3%	3%	3%	4%
Y-o-Y growth							
Growth in revenues		28%	38%	53%	11%	5%	17%
Growth in Net profits		-11%	96%	93%	25%	6%	34%
Liquidity ratios							
Current Ratio	1.3	1.5	1.7	1.9	2.2	2.6	2.9
Quick Ratio	1.2	1.4	1.6	1.8	2.1	2.5	2.8
Other key Ratios							
Inventory Turnover Ratio	12.2	12.2	12.2	12.2	12.2	12.2	12.2
Asset Turnover Ratio	11.6	10.1	8.7	8.0	6.7	5.7	5.1
Return on Assets	24%	12%	16%	19%	19%	16%	17%
Return on Net Worth (RONW)	39.3%	55.0%	70.2%	81.9%	87.9%	91.0%	93.4%
Return on equity (ROE)	39.3%	25.9%	33.7%	39.5%	33.0%	26.0%	25.9%
Interest Coverage Ratio	6.0	4.5	5.9	7.2	8.0	8.1	9.2
Debt Service Coverage Ratio	5.5	4.2	5.5	6.7	7.4	7.4	8.4

9.2.9 Sensitivity Analysis

The results of sensitivity analysis carried out for variations in important parameters such as decrease in sales volumes or revenues or product margin spread is presented in the table below. As the other operating costs form a very marginal portion of overall operating costs, variations of 5-10% would not lead to any noticeable changes in key indicators.

Year		Base Case	Varying Revenue	Varying Pricing	Varying Margin
Income			Decline by 5%	Decrease by 5%	Decrease by 3%
EBITDA Margin	%	4.43%	4.36%	4.36%	1.45%
PAT Margin	%	3.55%	3.48%	3.48%	0.84%
NPV	USD Mn	238.3	221.77	222.05	48.67
IRR	%	373%	288.83%	288.83%	34.50%
WC	USD Mn	94.46	89.74	89.74	94.63
CR	ratio	3.71	3.69	3.7	1.64
Inventory ratio	ratio	12.17	12.17	12.2	12.2
Return on Assets (RoA)	%	14%	13%	14%	5%

10 Risk Assessment

Risk	Risk level	Remark and Mitigation
Experience and Capability Risk	Low	Both the JV partners have in-depth experience in the O&G industry, having dealt into marketing, trading of various products including bunkers, apart from equal experience in overall operations in this industry. Hence, they bring in considerable experience in understanding and managing this business.
Pricing Risk	Low to medium	Owing to nature of this industry and products, being part of the commodity markets; the pricing shall always be volatile, involving an intricate dynamics of multi-dimensional / various critical macroeconomic, microeconomic, geopolitical, geographical, industry and market factors. But as the business model involves largely trading activity, without major investments across the value chain components, it largely minimizes the risks. But, to maintain the planned margins, and business continuity, it is critically important to be able to develop and deploy suitable pricing strategies for different types of target customer segments – indicating unique requirements in terms of order volumes, lead time, quality, supply practices & infrastructure required. Thus distinct plans and monitoring, review and mitigation mechanisms shall be required for spot demand and sales, sales through hedging, or long term contracts, access to different locations through tie-ups, etc).
Credit Risk	Medium to High	The present business model being a focused trading business, this risk also is one of the most critical risk factors, as good as governing the existence, growth, success and continuity (sustainability) of this business. Operating within a narrow range of margins, amidst the fierce competitive markets, it is critically essential for any company to continuously control, monitor the working capital management with an aim to maintain it healthy, against the set aimed parameters and levels. This is a much inter-linked parameter to other factors – pricing and bargaining power of a company. Also, to secure working capital to support its trade requirements, and to manage the cost of credit, plays an important role to have a healthier bottom line.
Supply Chain Risk	Medium	In terms of supply chain, on one hand, the Company would always have multiple options of suppliers, to review and choose the best fit of product availability and supplies against each requirement. But, on the other hand, without having captive resources for supply chain, it is always and externally dependable factor, requiring routine monitoring, review to have these systems in place. Any disturbances in these arrangements, can derail the business cycle, for a small duration. Hence to have quality and secured supply chain infrastructure, good due diligence and relationship with suppliers and capability to maintain creditors' level within control is very important.
Operational Risk	Low to medium	In view of the experience of the promoters, senior management and the experienced operations team, the operational risk can be low to medium. With a robust understanding of the trading business and activity, knowing about the internal and external involuntary factors, its own boundaries of activities and responsibilities, the JV company should be able to manage the overall business operations smoothly. Down the line, suitable focus and investments should be allocated towards having access to intelligence systems of Bunker Management Systems, insights towards contractual, product quality dynamics and trends.
Insurance Risk	Low	Post the IMO 2020, the trade terms, norms on acceptability, time, location, quality, quantity of supplies, have been more stringent and any lapses may lead to increases financial / capital management risks. Hence to track the dynamics of products and trends in marine / bunker insurance would be critical to insulate

Risk	Risk level	Remark and Mitigation
		/ secure the Company from any major commercial/ financial damages. A suitable amount of in-house focus, deploying standard good practices, supported by good advisory system can take care of this, and hence the risk is termed as Low.
Demand Risk	Low	The demand trends too, in line with this overall nature of this industry, is relatively volatile, but being into trading, the Company would be always watchful of the micro details and medium level shifts in trends in the geography, products, volumes, and types of customers. Having sound know-how of these practices, based on their extensive experience, it is expected that Company would be able to track and address the changing requirements of clients, for different products, backed by sufficient options of suppliers. Apart from this, any major variations in market demand levels, shall affect / impact equally, the entire set of players in a given geography, and hence its mitigation measures shall be in line with the trends in industry's best players.
Statutory Approvals Risk	Low	The key compliance requirements and parameters are clearly defined and bring in this industry the JV partners/ promoters would be well aware of these. A standard procedures approach would help in addressing the requirements and securing all the permits, approvals for the Company. In terms of operations, relatively higher focus would be required towards the compliance of mobile assets, staff and crew members, as these would be leased, and it is important to check that these satisfy the requirement conditions and adhere to operational protocols.
Force Majeure	Low	This is part of the commercial and insurance risk management, which shall be dealt with equal importance while entering into contracts, and executing the orders.
Geopolitical Risk	Medium	The intermittent occurrence and resurfacing of geopolitical volatility in the Middle East region, affects businesses across industries and sectors and energy sector is more at the core of it. Although the proposed business activity is focused towards trading and not much of capital asset risks involved, but such instabilities, disturbances, volatilities lead to shipping industry eyeing other trade route options despite facing higher logistics costs, with a need to secure its assets, goods and maintain certainty to their business. Such geopolitical tensions, depending upon the inter-relations of various nations within Middle East, may affect business – with duration varying from bring a very small duration, to a medium tenure of few days / months. Such unforeseen risks, may affect the business cycle, top line and cash flow cycles of a Company.

Appendix

A Methodology for demand and supply calculations

Following the below framework, demand and supply calculations/representations were done for the historical and forecasted sections

Historical demand- Fujairah (2013-2023)

Approach for 2013-2020 figures: Grade-wise data was estimated based on the actual average split ratio for the years 2021-2023 apart from LSFO. LSFO sales was estimated based on the yearly %age share of total fuel sales at Singapore.

Approach for 2021-2023 figures: Actual figures were available through S&P Platts reporting.

Step 1	• Total Bunker Fuel sales: Total yearly actual demand for Fujairah port was found through various press releases and secondary research sources which were later on validated through primary interviews
Step 2	• 2021-2023 bunker fuel sales data: 2021-2023 grade wise actual sales data was found through secondary research • 2020 LSFO sales: LSFO actual sales was available through a press release
Step 3	• LSFO sales share of total (%) was researched for Singapore. Singapore was taken as a sample as it holds the biggest market share (~35-40%) of total bunker fuel sales. • Singapore LSFO share of total (%) was applied to Fujairah total figures to estimate LSFO sales for 2013-2020 period
Step 4	• Other fuel sales (380 cst Marine fuel oil (HSFO), MGO and Lubricants) was estimated by subtracting low sulphur fuel sales from total sales at Fujairah port • Other fuel (380 cst Marine fuel oil (HSFO), MGO and Lubricants) sales was estimated by applying average grade wise ratio split (%) for 2021-2023 period, within the other fuel sales category

Historical supply (2013-2023)

Approach for 2013-2023 figures: Grade-wise data was estimated based on the actual production capacities for LSFO. However, for other fuel categories such as HSFO, Marine gasoil and lubricants, production data was available through UN Data.

Step 1	• LSFO production capacities (MT per month) were available through secondary sources for the four refineries operating at Fujairah along with the production start year. Combined production from all the refineries operating at Fujairah was utilized to arrive at yearly LSFO sales, starting 2021
Step 2	• Yearly production of HSFO, LSFO and Lubricants was researched through UNData and other secondary sources

Forecasted demand (2024-2033)

Approach - The below factors along with respective weightages were applied to come up with the demand forecast for 2024-2033 period

Framework		
Quants/Qual	Weightage	Parameters
Quantitative	25%	Global GDP Growth rate
Quantitative	50%	Shipping (Tankers/Cargo/Containers) Market growth
Quantitative	5%	Offshore Oil and Gas Spending (Dominated by MENA, Asia and South America)
Quantitative	20%	Global Maritime/seaborne trade growth

Step 1	Final growth rate was calculated based on weighted average of the above-mentioned factors and was applied to the total 2023 sales at Fujairah
Step 2	Grade-wise splits for the future years (2024-2033) based on the qualitative insights from secondary research and primary/expert interviews
Step 3	Projected grade wise splits were applied to arrive at the forecasted sales figure for various grades falling under bunker fuel category and lubricants

Forecasted supply (2024-2033)

Approach - Future refinery expansions were considered to arrive at the future supply capacity and production of HSFO, LSFO, MGO and lubricants

Step 1	Upcoming refinery expansions in UAE (Post 2023) were tracked based on the secondary sources along with the expanded capacity and expected production start year
Step 2	Upcoming production for various grades was calculated based on ME average refinery utilization, expanded refinery capacity, crude oil to HFO and Gasoil conversion ratios
Step 3	Upcoming LSFO production was calculated based on the assumption that 60% of Heavy fuel oil production in UAE will get converted into LSFO
Step 4	Upcoming production for respective grades were added in the last year production for the year 2024 and 2025

B Key Financial Ratios

B.1 Current assets – break-up of key components

Current Assets Break-up				
	In %			
	2019	2020	2021	2022
Cash on hand	-	-	1.95	0.01
Balances in current accounts with banks	1.67	0.67	12.90	14.66
Due from related parties	-	0.11	4.08	9.68
Trade receivables	78.43	66.92	61.87	51.32
Advances to staff	0.92	1.82	0.75	1.27
DMCC portal deposit	0.07	0.02	-	-
Prepayments	1.97	0.36	0.31	0.17
Unearned revenue	9.46	-	-	-
Salary advance	0.06	-	-	-
Other deposit	0.08	-	0.43	0.26
Margin deposits	-	-	0.15	2.84
Advances to suppliers	7.35	30.10	17.56	19.80
Total Current Assets	100	100	100	100

B.2 Current liabilities – break-up of key components

Current Liabilities Break-up				
	In %			
	2019	2020	2021	2022
Trade payables	18.80	27.78	83.74	61.48
Advances from customers	73.71	58.81	6.43	13.71
Accruals	3.27	1.49	0.03	0.52
Purchase consideration payable	-	1.86	-	-
Payable to employees	-	-	0.19	0.22
Leave salaries	2.75	-	-	-
VAT	0.01	0.09	0.07	0.42
Salary	0.99	-	-	-
Airfare	0.46	-	-	-
Term loans	-	-	9.36	23.30
Vehicle loan	-	0.50	0.18	0.36
Loan from a business associate	-	9.46	-	-
Total current liabilities	100	100	100	100

C Financial Statements

C.1 Profit & Loss Statement

(USD Mn)

Year		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Income																		
Operating Revenues	USD Mn	398.8	511.1	706.1	1080.0	1201.2	1262.0	1471.6	1546.1	1624.3	1706.5	1792.9	1883.6	1978.9	2079.0	2184.2	2294.7	2410.8
Other Income	USD Mn																	
Total Revenues		398.8	511.1	706.1	1080.0	1201.2	1262.0	1471.6	1546.1	1624.3	1706.5	1792.9	1883.6	1978.9	2079.0	2184.2	2294.7	2410.8
Expenditure																		
Annual Operating costs																		
feedstock procurement	USD Mn	371.8	484.3	669.0	1023.6	1135.0	1192.5	1384.4	1454.5	1528.1	1605.4	1686.6	1772.0	1861.6	1955.8	2054.8	2158.8	2268.0
Other Operating costs	USD Mn	14.4	14.7	15.0	15.3	15.7	16.0	16.4	16.7	17.1	17.5	17.9	18.3	18.7	19.1	19.5	20.0	20.4
Total Operating Costs	USD Mn	386.2	499.0	684.0	1039.0	1150.7	1208.5	1400.8	1471.2	1545.2	1622.9	1704.5	1790.2	1880.3	1974.9	2074.3	2178.7	2288.4
Operating Profit	USD Mn	12.5	12.1	22.0	41.0	50.5	53.5	70.7	74.9	79.1	83.6	88.3	93.3	98.6	104.1	109.9	116.0	122.4
	%	3.1%	2.4%	3.1%	3.8%	4.2%	4.2%	4.8%	4.8%	4.9%	4.9%	4.9%	5.0%	5.0%	5.0%	5.0%	5.1%	5.1%
Depreciation	USD Mn	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Interest - LT Loan	USD Mn	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Interest - WC	USD Mn	2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.4	9.9	10.4	10.9	11.5	12.0	12.6
Sub Total		2.1	2.8	3.8	5.8	6.4	6.7	7.8	8.2	8.6	9.1	9.5	10.0	10.5	11.0	11.6	12.1	12.8
Total Expenses		388.3	501.8	687.9	1044.7	1157.1	1215.2	1408.6	1479.4	1553.8	1631.9	1714.0	1800.2	1890.8	1985.9	2085.9	2190.9	2301.2
Profit Before Tax		10.4	9.3	18.2	35.2	44.1	46.8	62.9	66.6	70.5	74.6	78.8	83.3	88.1	93.1	98.3	103.8	109.7
Tax	USD Mn	0.9	0.8	1.6	3.2	4.0	4.2	5.7	6.0	6.3	6.7	7.1	7.5	7.9	8.4	8.8	9.3	9.9
Extraordinary income																		
Profit After Tax (PAT)	USD Mn	9.5	8.5	16.6	32.1	40.1	42.6	57.3	60.6	64.2	67.9	71.7	75.8	80.1	84.7	89.5	94.5	99.8
PAT Margin (%)	%	2%	2%	2%	3%	3%	3%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%
Dividend		1.4	1.3	2.5	4.8	6.0	6.4	8.6	9.1	9.6	10.2	10.8	11.4	12.0	12.7	13.4	14.2	15.0
Retained Earnings		8.1	7.2	14.1	27.3	34.1	36.2	48.7	51.6	54.5	57.7	61.0	64.5	68.1	72.0	76.0	80.3	84.8
Margin (%)	%	2%	1%	2%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	4%	4%

C.2 Balance Sheet

(USD Mn)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Liabilities																	
Shareholders Fund																	
Capital	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Share Application money																	
Reserve and Capital	8.1	15.3	29.4	56.6	90.7	126.9	175.6	227.1	281.6	339.3	400.3	464.8	532.9	604.9	680.9	761.3	846.1
Total	20.6	27.7	41.8	69.1	103.2	139.4	188.0	239.6	294.1	351.8	412.8	477.2	545.4	617.4	693.4	773.7	858.6
Loan Funds																	
Secured Loan	26.2	33.6	46.3	70.8	78.8	82.8	96.5	101.4	106.5	111.9	117.5	123.5	129.7	136.3	143.2	150.4	158.0
Term Loan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
WC Loan	26.2	33.6	46.3	70.8	78.8	82.8	96.5	101.4	106.5	111.9	117.5	123.5	129.7	136.3	143.2	150.4	158.0
Unsecured Loans	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other Loans	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	26.2	33.6	46.3	70.8	78.8	82.8	96.5	101.4	106.5	111.9	117.5	123.5	129.7	136.3	143.2	150.4	158.0
Deferred Tax Liability																	
Total Sources	46.7	61.3	88.2	139.9	181.9	222.1	284.5	340.9	400.6	463.7	530.3	600.7	675.1	753.6	836.6	924.1	1,016.6
Assets																	
Fixed Assets	12.5	11.5	9.8	8.3	7.1	6.0	5.1	4.4	3.7	3.1	2.7	2.3	1.9	1.6	1.4	1.2	1.0
Gross Block	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Less: Cumulative Depreciation	-	0.9	2.7	4.1	5.4	6.5	7.4	8.1	8.8	9.3	9.8	10.2	10.5	10.8	11.1	11.3	11.5
Capital WIP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Investments																	
Current Assets, Loans & Advances	34.9	44.7	61.8	94.5	105.0	110.3	128.6	135.1	142.0	149.2	156.7	164.6	173.0	181.7	190.9	200.5	210.7
other current assets	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inventories	2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.3	9.8	10.3	10.8	11.4	11.9	12.5
Debtors	32.8	42.0	58.0	88.8	98.7	103.7	120.9	127.1	133.5	140.3	147.4	154.8	162.6	170.9	179.5	188.6	198.2
Loans and advances																	
Cash	(0.6)	5.0	16.6	37.1	69.8	105.8	150.8	201.5	254.9	311.4	370.9	433.8	500.2	570.3	644.3	722.4	804.9
Less: Current Liabilities	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Creditors	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Provisions																	
Net Current Assets	34.2	49.8	78.4	131.6	174.9	216.1	279.4	336.6	396.9	460.5	527.6	598.4	673.1	752.0	835.2	922.9	1,015.6
Miscellaneous Expenditure																	
Total Applications	46.7	61.3	88.2	139.9	181.9	222.1	284.5	340.9	400.6	463.7	530.3	600.7	675.1	753.6	836.6	924.1	1,016.6

C.3 Depreciation

(USD Mn)

Year		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
GFA																		
Land		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Civil		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
P&M		12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4	12.4
MFA		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total		12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
Dep																		
Land	0.0%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Civil	5.0%	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
P&M	15.0%	-	0.9	1.7	1.5	1.2	1.1	0.9	0.8	0.6	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2
MFA	25.0%	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total		-	0.9	1.7	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2
Cumu Dep																		
Land		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Civil		-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
P&M		-	0.9	2.7	4.1	5.4	6.4	7.3	8.1	8.7	9.3	9.7	10.1	10.5	10.8	11.0	11.2	11.4
MFA		-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total		-	0.9	2.7	4.1	5.4	6.5	7.4	8.1	8.8	9.3	9.8	10.2	10.5	10.8	11.1	11.3	11.5
NFA																		
Land		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Civil		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
P&M		12.4	11.5	9.8	8.3	7.0	6.0	5.1	4.3	3.7	3.1	2.7	2.3	1.9	1.6	1.4	1.2	1.0
MFA		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total		12.5	11.5	9.8	8.3	7.1	6.0	5.1	4.4	3.7	3.1	2.7	2.3	1.9	1.6	1.4	1.2	1.0

C.4 Interest Schedule

USD Mn

Working Capital																	
Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
interest Rate	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%
Loan Amount	26.2	33.6	46.3	70.8	78.8	82.8	96.5	101.4	106.5	111.9	117.5	123.5	129.7	136.3	143.2	150.4	158.0
interest	2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.4	9.9	10.4	10.9	11.5	12.0	12.6

C.5 Cash Flow Statement

(USD Mn)

Description	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Source of funds																	
PAT	9.5	8.5	16.6	32.1	40.1	42.6	57.3	60.6	64.2	67.9	71.7	75.8	80.1	84.7	89.5	94.5	99.8
Depreciation	-	0.9	1.7	1.5	1.3	1.1	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.3	0.2	0.2	0.2
Equity Investment	12.5	-															
Increase in Term Loans	-	-															
Increase in Unsecured Loans																	
Expenses written off																	
Increase in Working capital loan	26.2	7.4	12.8	24.5	7.9	4.0	13.7	4.9	5.1	5.4	5.7	5.9	6.2	6.6	6.9	7.2	7.6
Total Sources	48.1	16.8	31.1	58.0	49.3	47.6	71.9	66.3	69.9	73.8	77.9	82.2	86.7	91.5	96.6	102.0	107.6
Use of Funds																	
Gross Fixed Asset	12.5	-															
Increase in working capital other current assets	34.9	9.8	17.0	32.7	10.6	5.3	18.3	6.5	6.8	7.2	7.5	7.9	8.3	8.7	9.2	9.7	10.1
Dividends	1.4	1.3	2.5	4.8	6.0	6.4	8.6	9.1	9.6	10.2	10.8	11.4	12.0	12.7	13.4	14.2	15.0
Increase in Capital WIP	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Repayment of loans	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Uses	48.8	11.1	19.5	37.5	16.6	11.7	26.9	15.6	16.5	17.4	18.3	19.3	20.3	21.5	22.6	23.8	25.1
Net Cash Flow	(0.6)	5.7	11.6	20.6	32.7	35.9	45.0	50.7	53.5	56.4	59.6	62.9	66.4	70.1	74.0	78.1	82.5
Opening Balance		(0.6)	5.0	16.6	37.1	69.8	105.8	150.8	201.5	254.9	311.4	370.9	433.8	500.2	570.3	644.3	722.4
Closing Balance	(0.6)	5.0	16.6	37.1	69.8	105.8	150.8	201.5	254.9	311.4	370.9	433.8	500.2	570.3	644.3	722.4	804.9

C.6 Working Capital Assessment

(USD Mn)

Sr.		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A. Current Assets																		
1	Raw Materials	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2	Stores & Spares	-																
3	Work - in - Progress	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
4	Finished Goods	2	2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.3	9.8	10.3	10.8	11.4	12.5
5	Debtors	30	32.8	42.0	58.0	88.8	98.7	103.7	120.9	127.1	133.5	140.3	147.4	154.8	162.6	170.9	179.5	188.6
Total Current Assets		34.9	44.7	61.8	94.5	105.0	110.3	128.6	135.1	142.0	149.2	156.7	164.6	173.0	181.7	190.9	200.5	210.7
B. Current Liabilities																		
1	Creditors	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
C. Working Capital Requirement		34.9	44.7	61.8	94.5	105.0	110.3	128.6	135.1	142.0	149.2	156.7	164.6	173.0	181.7	190.9	200.5	210.7
D. Margin Money	25%	8.7	11.2	15.4	23.6	26.3	27.6	32.2	33.8	35.5	37.3	39.2	41.2	43.2	45.4	47.7	50.1	52.7
E. Bank Borrowings		26.2	33.6	46.3	70.8	78.8	82.8	96.5	101.4	106.5	111.9	117.5	123.5	129.7	136.3	143.2	150.4	158.0

C.7 Project Evaluation

(USD Mn)

Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Outflows																	
Capex	3.8	0.0															
Margin Money	8.7	0.0															
Capex outflow	12.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Inflows																	
PAT	9.5	8.5	16.6	32.1	40.1	42.6	57.3	60.6	64.2	67.9	71.7	75.8	80.1	84.7	89.5	94.5	99.8
Depreciation	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Interest	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Others																	
Operating in-flow	9.5	8.6	16.7	32.2	40.2	42.7	57.4	60.8	64.3	68.0	71.9	75.9	80.3	84.8	89.6	94.6	99.9
Terminal cashflows																	53.7
Total Cash flows	-3.0	8.6	16.7	32.2	40.2	42.7	57.4	60.8	64.3	68.0	71.9	75.9	80.3	84.8	89.6	94.6	153.6
NPV (USD Mn)	238.3																
IRR (%)	373%																
payback (yrs from COD)	5.35																

C.8 Debt Service Coverage and other key ratios

Description		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
DSCR																		
Cash Accruals		9.5	8.6	16.7	32.2	40.2	42.7	57.4	60.8	64.3	68.0	71.9	75.9	80.3	84.8	89.6	94.6	99.9
Interest		2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.4	9.9	10.4	10.9	11.5	12.0	12.6
Total	242.00	11.6	11.2	20.4	37.8	46.5	49.3	65.1	68.9	72.8	76.9	81.3	85.8	90.6	95.7	101.0	106.6	112.5
Interest		2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.4	9.9	10.4	10.9	11.5	12.0	12.6
Repayment		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	34.79	2.1	2.7	3.7	5.7	6.3	6.6	7.7	8.1	8.5	8.9	9.4	9.9	10.4	10.9	11.5	12.0	12.6
AVG DSCR	6.96	5.5	4.2	5.5	6.7	7.4	7.4	8.4	8.5	8.5	8.6	8.6	8.7	8.7	8.8	8.8	8.9	8.9

Leverage Ratio	Average																	
Debt Equity (D:E Ratio)	0.00	0.0	0.0	0.0	0.0	0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Interest Coverage Ratio		6.0	4.5	5.9	7.2	8.0	8.1	9.2	9.2	9.3	9.3	9.4	9.4	9.5	9.5	9.6	9.6	9.7

Y-o-Y growth																		
Growth in revenues			28%	38%	53%	11%	5%	17%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%
Growth in Net profits			-11%	96%	93%	25%	6%	34%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%

Profitability Ratio																		
Gross Profit Margin		7%	5%	5%	5%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%	6%
Operating Profit Margin		3%	2%	3%	4%	4%	4%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%
EBIT Margin		3%	2%	3%	4%	4%	4%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%
Net Profit Margin		2%	2%	2%	3%	3%	3%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%	4%

Liquidity ratios	Avg																	
Current Ratio	3.75	1.3	1.5	1.7	1.9	2.2	2.6	2.9	3.32	3.73	4.12	4.49	4.85	5.19	5.52	5.83	6.14	6.43
Quick Ratio	3.67	1.2	1.4	1.6	1.8	2.1	2.5	2.8	3.24	3.65	4.04	4.41	4.77	5.11	5.44	5.75	6.06	6.35

Description		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Asset turn over ratio	Average																	
Inventory Turnover Ratio	12.17	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2	12.2
Asset Turnover Ratio	5.14	11.6	10.1	8.7	8.0	6.7	5.7	5.1	4.5	4.0	3.6	3.3	3.1	2.9	2.7	2.6	2.5	2.3
Return on Assets	14%	24%	12%	16%	19%	19%	16%	17%	15%	14%	12%	11%	11%	10%	10%	9%	9%	8%
	Average																	
Return on Net Worth (RONW)	74.1%	39%	55%	70%	82%	88%	91%	93%	95%	96%	96%	97%	97%	98%	98%	98%	98%	99%
Return on equity (ROE)	31.9%	39%	26%	34%	39%	33%	26%	26%	22%	19%	16%	15%	14%	12%	12%	11%	10%	10%
Return on Investments (ROI)	201.1%	65%	58%	113%	219%	273%	290%	390%	413%	437%	462%	489%	517%	546%	577%	610%	644%	680%

Appendix

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